

# The niche of Romik's ambidextrous sofa in convex-linear data, and a stability bound under a curvature condition

Vico Bonfioli  
vicobonfioli@gmail.com

August 2026

## Abstract

Baek proved that Gerver's sofa is the maximum-area moving sofa for a corridor with one corner, by writing a sofa as a convex cap minus a niche and obtaining optimality from the concavity of a quadratic upper bound. We carry that architecture to the ambidextrous moving sofa problem, which is open, and to Romik's candidate  $\Sigma$  of area  $A_R^* = 1.6449552184\dots$

The niche of an ambidextrous sofa is expressed exactly in convex-linear data: writing  $H(\theta) = h_K(\mu_\theta)$  for the cap's support function, the two arms  $\alpha_1, \alpha_2$  and the reach  $\sigma$  are affine in  $H$ , and  $|N| = \int_0^{\pi/2} [\frac{1}{2}(\alpha_2^+)^2 + \frac{1}{2}(\sigma - \alpha_1)^2 - \frac{1}{2}(\alpha_1^-)^2] dt$  whenever the associated sweep is injective. Injectivity follows from a single linear condition (RC), that the cap's radius of curvature be at most the corridor width; the proof reduces each of the three required statements to a forced harmonic oscillator whose kernel is non-negative because the rotation angle is  $\pi/2$ . (RC) is classical: by the dual of Blaschke's rolling theorem it says the cap slides freely inside a unit ball, up to the corridor facets. What is new is its use.

For the functional  $Q = |C_2| - 2V$  we prove  $Q(\Sigma) = A_R^*$ ; that  $\delta Q(\Sigma) = 0$ , so Romik's ODE families are exactly the critical-point equations of  $Q$ , identically in his constants; and that  $\delta^2 Q < 0$ , with an explicit constant. Two Cauchy–Schwarz steps decouple the two halves of  $[0, \pi]$  and reduce the second variation to a pair of Sturm–Liouville eigenvalue problems, certified from below by Sturm oscillation; this gives  $\frac{1}{2}\delta^2 Q \leq -\frac{2}{3}\|\eta\|_{L^2(0,\pi)}^2$ . Not splitting at all turns the second variation into a single  $2 \times 2$  system whose first eigenvalue is the sharp constant  $c^* = 0.7309566\dots$ , certified to exceed  $\frac{73}{100}$ . The same system, with the cell boundaries as parameters, proves concavity for every cell whose sign sets are ordered and anchored, by a one-line monotonicity reduction to a one-parameter family certified by an adaptive interval covering; on the mirror class the constant is uniform,  $\frac{3}{10}$ , and the resulting stability bound assumes no smallness of  $H - H_\Sigma$  beyond (RC). On the convex domain  $\mathcal{D}$  cut out by (RC), a separation inequality and  $\Sigma$ 's own sign pattern,

$$|T| \leq A_R^* - \frac{73}{100} \|H - H_\Sigma\|_{L^2(0,\pi)}^2$$

for every ambidextrous sofa  $T$  with cap data in  $\mathcal{D}$ , and  $\Sigma$  is the unique member attaining  $A_R^*$ . The domain is a  $C^2$ -neighbourhood of  $\Sigma$ : its radius along  $\sin(2k\theta)$  scales like  $k^{-2}$ .

Two limits are recorded rather than smoothed over. The separation inequality is independent of (RC) and cannot be removed. And  $V$  is the exact flux of the advancing wedge boundary, so  $V \geq |N|$  by Reynolds' transport theorem and  $Q$  bounds nothing where the sweep fails to be injective; in particular the construction does not recover Gerver's sofa, whose cap violates (RC). Separately, a two-hallway relaxation gives the first upper bound specific to the ambidextrous problem: the area of a left- and a right-turning hallway at angle  $\pi/4$  is bounded by translates of one explicit profile integral, and a four-case analysis of the corner offsets removes every hypothesis, giving

$$A_{\text{ambi}} \leq 2\sqrt{2} - 1 = 1.8284271\dots$$

unconditionally. The bound is attained by an explicit placement, so it is exactly the value of the two-hallway relaxation; it improves on the 2.2195 inherited from the one-corner theorem, and with Romik’s sofa it brackets the ambidextrous supremum within a factor of 1.1115. Optimality of  $\Sigma$  remains open.

## 1 Introduction

The moving sofa problem asks for the largest area of a connected shape that can be manoeuvred around a right-angled corner of a hallway of unit width. It was settled by Baek [1], who proved Gerver’s sofa [2] optimal. Romik [5] posed the *ambidextrous* variant, in which the shape must turn corners of both handedness, exhibited a candidate  $\Sigma$  of area

$$A_R^* = 1.6449552184254408 \dots,$$

and asked whether it is optimal. That question is open, and this note does not settle it. What follows is a structural analysis of the ambidextrous problem in the coordinates in which the constraint data is affine, and a local optimality statement for  $\Sigma$  on an explicitly described domain.

### The coordinates

Every ambidextrous sofa is  $C_2 \setminus (U \sqcup \rho U)$  for a convex *cap*  $C_2$  and a *niche*  $U$  swept by the hallway’s inner corner, with  $\rho$  the reflection in  $y = \frac{1}{2}$ . Writing  $H(\theta) = h_{C_2}(\mu_\theta)$  for the cap’s support function on  $[0, \pi]$ , the corner path is *affine* in  $H$  (Lemma 18), and so are the two arm lengths  $\alpha_1, \alpha_2$  and the reach  $\sigma$  that determine the sweep. In these coordinates the cap area is an exact quadratic form,

$$|C_2| = \int_0^\pi (H^2 - H'^2) d\theta - H(0) - H(\pi)$$

(Proposition 24), and the niche area has the closed form (20). The functional  $Q := |C_2| - 2V$ , with  $V$  the flux of the advancing sweep, is therefore an explicit quadratic polynomial in  $H$  on each cell of the sign pattern of  $(\alpha_1, \alpha_2)$ .

### The results

The analysis rests on one geometric hypothesis, that the cap is nowhere flatter than a circle of the corridor’s width:

$$(RC) \quad (H + H'')_{ac} \leq 1,$$

which is linear in  $H$  and hence convex. It is classical, being the sliding-ball condition dual to Blaschke’s rolling theorem (Remark 51); what is new is its use. Theorem 52 shows that (RC) implies all three injectivity conditions of the sweep, by reducing each to a forced harmonic oscillator whose kernel is non-negative precisely because the rotation angle is  $\pi/2$ ; consequently  $V = |N|$  and  $Q$  is tight.

At  $\Sigma$  the three hypotheses of the concavity method hold exactly:  $Q(\Sigma) = A_R^*$  structurally, with no integral evaluated (Theorem 64);  $\delta Q(\Sigma) = 0$  identically in Romik’s constants, so his ODE families *are* the critical-point equations of  $Q$  (Theorem 57); and  $\delta^2 Q < 0$ .

The second variation is the technical core. Setting  $p(t) = \eta(t)$  and  $q(t) = \eta(t + \pi/2)$  identifies it with a  $2 \times 2$  Sturm–Liouville system on  $[0, \pi/2]$  whose first eigenvalue *is* the best constant,

$$c^* = 0.7309566\dots,$$

certified above  $\frac{73}{100}$  in ball arithmetic (Theorem 88). On the overlap of the two sign sets the coupling is a total derivative, which is why every pointwise estimate loses and this one does not. The same system, with the cell boundaries as parameters, gives concavity on *every* ordered anchored cell (Theorem 91), and a uniform constant  $\frac{3}{10}$  on the mirror class (Theorem 93).

These combine into two optimality statements. Theorem 84 gives  $|T| \leq A_R^*$  for every ambidextrous sofa with cap data in the convex domain  $\mathcal{D}$  cut out by the gauge, (RC), a separation inequality and  $\Sigma$ ’s sign pattern, with the quantitative form

$$|T| \leq A_R^* - \frac{73}{100} \|H - H_\Sigma\|_{L^2(0,\pi)}^2$$

of Theorem 94 and uniqueness (Corollary 95). Proposition 103 exhibits an explicit  $C^2$ -ball of radius  $\frac{1}{20}$  inside  $\mathcal{D}$ . Theorem 98 is the non-perturbative version: on the mirror class, with no smallness of  $H - H_\Sigma$  assumed,  $|T| \leq A_R^* - \frac{3}{10} \|H - H_\Sigma\|^2$ .

## What is not proved, and why

What the local result addresses is Romik’s own first open problem [5, §7], which asks for a proof that Gerver’s sofa and  $\Sigma$  are local maxima of the area functional in the one-corner and ambidextrous problems respectively. Romik established the first-order conditions for  $\Sigma$ ; the second-order statement has not been proved, and Theorem 84 proves it only on the domain  $\mathcal{D}$ , so that problem stays open. For Gerver’s sofa the corresponding question is settled a fortiori by [1].

Optimality of  $\Sigma$  remains open, and the obstructions are recorded rather than smoothed over.  $V$  is the exact flux of the advancing sweep, so  $V \geq |N|$  always by Reynolds’ transport theorem (Proposition 40) and  $Q$  bounds nothing where the sweep fails to be injective; the domain cannot be widened by a concave majorant of the excess, since none exists (Proposition 96), though a semiconcave one does (Theorem 97). The separation hypothesis is independent of (RC). Gerver’s cap violates (RC), but that is not what excludes it: it is not the cap of any *connected* ambidextrous sofa (Remark 105). The sign hypothesis of Theorem 98 is permanent, not provisional: the incircle of the unit square is a connected ambidextrous sofa satisfying every other condition and violating it (Remark 101).

Claims are labelled throughout: VERIFIED means machine-checked in Lean 4 with no **sorry**, PROVED a complete human proof, HEURISTIC computational evidence only. Numerical values that carry weight are computed in ball or exact rational arithmetic with stated precision; Section 18 separates what is ours from what is Baek’s and Romik’s.

## 2 Setting

Write  $L$  for the closed L-shaped hallway of unit width with its corner at the origin,  $\mu_t = (\cos t, \sin t)$ ,  $\nu_t = (-\sin t, \cos t)$ , and for a rotation path  $c : [0, \pi/2] \rightarrow \mathbb{R}^2$  let

$$C_t = \{p : \langle p - c(t), \mu_t \rangle \leq 1\} \cap \{p : \langle p - c(t), \nu_t \rangle \leq 1\}, \quad Q_t = \{p : \langle p - c(t), \mu_t \rangle < 0\} \cap \{p : \langle p - c(t), \nu_t \rangle < 0\},$$

so that the hallway in position  $t$  is  $H_t = C_t \setminus Q_t$ . The one-sided sofa is  $S = \bigcap_t H_t$ , and writing  $C = \bigcap_t C_t$  and  $U = \bigcup_t Q_t$  we have  $S = C \setminus U$  with  $C$  convex.

Let  $\rho(x, y) = (x, 1 - y)$  be the reflection in the line  $y = \frac{1}{2}$ . Romik's ambidextrous sofa is

$$\Sigma = S \cap \rho(S) = C_2 \setminus (U \cup \rho U), \quad C_2 := C \cap \rho C \text{ convex}, \quad (1)$$

with  $|\Sigma| = A_R^* = 1.6449552184\dots$ , and  $c$  is Romik's piecewise path with breakpoints at  $\beta$  and  $\pi/2 - \beta$ , where

$$\beta = \arctan\left(\frac{1}{2}\left(\sqrt[3]{\sqrt{2} + 1} - \sqrt[3]{\sqrt{2} - 1}\right)\right) = 0.2896538208\dots \quad (2)$$

Baek's argument for the one-corner problem builds an overestimating region whose area is a quadratic functional on a convex domain of convex bodies, and proves it concave by exhibiting it as a linear functional minus Mamikon regions, whose areas are convex quadratics. Concavity reduces global optimality to a first-order condition. The construction is organised around  $S = K \setminus \mathcal{N}(K)$ : one cap, one niche.

By (1) the ambidextrous analogue has *two* niches, and

$$|U \cup \rho U| = |U| + |\rho U| - |U \cap \rho U|. \quad (3)$$

The last term appears in the upper bound with a  $+$  sign. Since concavity is obtained by *subtracting* convex quadratics, an added overlap term is not covered by the mechanism. The purpose of this note is Theorem 8: the term is zero.

### 3 The niches are disjoint

**Lemma 1** (Niche ceiling). *For every  $t \in [0, \pi/2]$  we have  $Q_t \subseteq \{(x, y) : y \leq c_y(t)\}$ .*

*Proof.*  $Q_t$  is the open convex cone with apex  $c(t)$  spanned by  $-\mu_t$  and  $-\nu_t$ , so  $q \in Q_t$  means  $q - c(t) = -a\mu_t - b\nu_t$  with  $a, b > 0$ . For  $t \in [0, \pi/2]$  both  $\sin t \geq 0$  and  $\cos t \geq 0$ , hence

$$q_y - c_y(t) = -a \sin t - b \cos t \leq 0. \quad \square$$

**Corollary 2.** *Put  $M := \max_{t \in [0, \pi/2]} c_y(t)$ . Then  $U \subseteq \{y \leq M\}$  and  $\rho U \subseteq \{y \geq 1 - M\}$ . If  $M < \frac{1}{2}$  then  $U \cap \rho U = \emptyset$ .*

*Proof.* The first inclusion is Lemma 1 applied to each  $t$ ; the second is its image under  $\rho$ . If  $M < \frac{1}{2}$  then  $1 - M > \frac{1}{2} > M$ , so the two half-planes are disjoint.  $\square$

It remains to evaluate  $M$  for Romik's path. On the middle piece  $[\beta, \pi/2 - \beta]$  Romik's path is

$$c(t) = R(t)v(t) + \kappa, \quad v(t) = \begin{pmatrix} f_1 \cos \frac{t}{2} + f_2 \sin \frac{t}{2} - 1 \\ -f_2 \cos \frac{t}{2} + f_1 \sin \frac{t}{2} - 1 \end{pmatrix}, \quad (4)$$

with  $R(t)$  the rotation by  $t$ , and the two facts we use are

$$\kappa_y = \frac{1}{2} \quad \text{exactly}, \quad f_2 = (1 - \sqrt{2}) f_1. \quad (5)$$

**Proposition 3.** For  $t \in [\beta, \pi/2 - \beta]$ ,

$$c_y(t) - \frac{1}{2} = f_1 \sin \frac{3t}{2} - f_2 \cos \frac{3t}{2} - (\sin t + \cos t) = f_1 \sqrt{4 - 2\sqrt{2}} \sin\left(\frac{3t}{2} + \frac{\pi}{8}\right) - \sqrt{2} \sin\left(t + \frac{\pi}{4}\right). \quad (6)$$

In particular  $c_y$  is symmetric about  $t = \pi/4$  on this interval, and

$$M = c_y(\pi/4) = \frac{1}{2} - \left(\sqrt{2} - f_1 \sqrt{4 - 2\sqrt{2}}\right). \quad (7)$$

*Proof.* By (4) and  $\kappa_y = \frac{1}{2}$ ,  $c_y(t) - \frac{1}{2} = \sin t v_x + \cos t v_y$ . Write  $S = \sin \frac{t}{2}$ ,  $C = \cos \frac{t}{2}$ , so  $\sin t = 2SC$  and  $\cos t = C^2 - S^2$ . Expanding and collecting the coefficients of  $f_1$  and  $f_2$  gives

$$\sin t v_x + \cos t v_y = f_1 S(3C^2 - S^2) + f_2 C(3S^2 - C^2) - (\sin t + \cos t),$$

and the triple-angle identities  $S(3C^2 - S^2) = \sin \frac{3t}{2}$ ,  $C(3S^2 - C^2) = -\cos \frac{3t}{2}$  give the first equality in (6). For the second, substitute  $f_2 = (1 - \sqrt{2})f_1$  from (5): the bracket becomes  $\sin \frac{3t}{2} + (\sqrt{2} - 1) \cos \frac{3t}{2}$ , of amplitude  $\sqrt{1 + (\sqrt{2} - 1)^2} = \sqrt{4 - 2\sqrt{2}}$  and phase  $\arctan(\sqrt{2} - 1) = \pi/8$ .

Both sinusoids in (6) are invariant under  $t \mapsto \pi/2 - t$ : indeed  $\sin(\frac{3}{2}(\pi/2 - t) + \pi/8) = \sin(7\pi/8 - \frac{3t}{2}) = \sin(\pi/8 + \frac{3t}{2})$  and  $\sin(3\pi/4 - t) = \sin(\pi/4 + t)$ . Hence  $c_y$  is symmetric about  $\pi/4$ , where both sinusoids attain the value 1 simultaneously; (7) follows.  $\square$

That  $\pi/4$  maximises  $c_y$  over all of  $[0, \pi/2]$ , and not merely over the middle piece, follows from the corresponding closed form on the outer pieces.

**Proposition 4.** For  $t \in [0, \beta]$ ,

$$c_y(t) = \frac{1}{2} + a_1 \sin 2t - \sin t - \frac{1}{2} \cos t, \quad (8)$$

and  $c_y$  is strictly increasing there, so  $\max_{[0, \beta]} c_y = c_y(\beta)$ . By the symmetry  $c_y(t) = c_y(\pi/2 - t)$  the same bound holds on  $[\pi/2 - \beta, \pi/2]$ . Consequently  $M = c_y(\pi/4)$  as claimed in (7).

*Proof.* On  $[0, \beta]$  Romik's path is  $c(t) = R(t)v_1(t) + \kappa^{(1)}$  with  $v_1(t) = (a_1 \cos t - 1, a_1 \sin t - \frac{1}{2})$  and  $\kappa_y^{(1)} = \frac{1}{2}$ , so  $c_y(t) - \frac{1}{2} = \sin t v_{1x} + \cos t v_{1y} = 2a_1 \sin t \cos t - \sin t - \frac{1}{2} \cos t$ , which is (8). Differentiating,

$$c'_y(t) = 2a_1 \cos 2t - \cos t + \frac{1}{2} \sin t \geq 2a_1 \cos 2\beta - 1 = 0.4650\dots > 0 \quad (t \in [0, \beta]),$$

using  $\cos 2t \geq \cos 2\beta$  and  $\cos t - \frac{1}{2} \sin t \leq 1$  for  $t \geq 0$ . Hence  $c_y$  is strictly increasing on  $[0, \beta]$ , with  $c_y(\beta) = 0.2143795161\dots$ , which is below  $c_y(\pi/4) = 0.3878381292\dots$   $\square$

**Remark 5.** Evaluating (8) and (6) at  $t = \beta$  gives the same value to  $4.4 \cdot 10^{-31}$  at 30-digit precision. Since the two expressions were derived from different pieces of Romik's path, this is an independent check on both, and on (10).

## 4 A closed form for $f_1$

Romik's constant  $f_1$  is recorded in the literature as the decimal 1.202938908156911389. The following determines it in closed form. Let

$$a_1 = \frac{1}{4} \sqrt{4 + \sqrt[3]{71 + 8\sqrt{2}} + \sqrt[3]{71 - 8\sqrt{2}}} = 0.8752873624 \dots \quad (9)$$

**Proposition 6.**

$$f_1 = \frac{\frac{4}{3} a_1 \cos \beta}{\cos \frac{\beta}{2} + (1 - \sqrt{2}) \sin \frac{\beta}{2}}. \quad (10)$$

*Proof.* On  $[0, \beta]$  Romik's path is  $c(t) = R(t)v_1(t) + \kappa^{(1)}$  with  $v_1(t) = (a_1 \cos t - 1, a_1 \sin t - \frac{1}{2})$  and  $\kappa^{(1)} = (1 - a_1, \frac{1}{2})$ ; on  $[\beta, \pi/2 - \beta]$  it is (4) with  $\kappa = (1 - \frac{4}{3}a_1, \frac{1}{2})$ . Both  $\kappa$ 's have second coordinate  $\frac{1}{2}$  and their first coordinates differ by exactly  $\frac{1}{3}a_1$ . Continuity at  $t = \beta$  therefore gives  $R(\beta)(v_1(\beta) - v(\beta)) = (-\frac{1}{3}a_1, 0)$ , so

$$v_1(\beta) - v(\beta) = R(-\beta)(-\frac{1}{3}a_1, 0) = (-\frac{1}{3}a_1 \cos \beta, \frac{1}{3}a_1 \sin \beta).$$

Reading the first coordinate and using  $f_2 = (1 - \sqrt{2})f_1$ ,

$$(a_1 \cos \beta - 1) - \left(f_1 \left(\cos \frac{\beta}{2} + (1 - \sqrt{2}) \sin \frac{\beta}{2}\right) - 1\right) = -\frac{1}{3}a_1 \cos \beta,$$

which rearranges to (10).  $\square$

**Remark 7.** Evaluating (10) at 50 digits gives  $f_1 = 1.202938908156911389070223 \dots$ , so the tabulated decimal is its rounding. As an independent check, the *second* coordinate of the junction relation, which was not used in the derivation, is satisfied to  $1.7 \cdot 10^{-51}$  at the same precision.

## 5 Disjointness of the niches for $\Sigma$

**Theorem 8.** *For Romik's ambidextrous sofa,  $U \cap \rho U = \emptyset$ , and consequently*

$$|\Sigma| = |C_2| - |U| - |\rho U| = |C_2| - 2|U|.$$

*Proof.* By Proposition 3,  $M < \frac{1}{2}$  is equivalent to  $f_1 \sqrt{4 - 2\sqrt{2}} < \sqrt{2}$ , i.e. to

$$f_1^2 < \frac{2 + \sqrt{2}}{2}. \quad (11)$$

With  $f_1$  given by (10) in terms of the algebraic numbers (2) and (9), both sides of (11) are explicit algebraic numbers, and

$$f_1^2 = 1.4470620167577420940 \dots, \quad \frac{2 + \sqrt{2}}{2} = 1.7071067811865475244 \dots,$$

so (11) holds with margin  $0.2600447644 \dots$ . Corollary 2 gives  $U \cap \rho U = \emptyset$ , and then (3) together with (1). Finally  $|\rho U| = |U|$  because  $\rho$  is an isometry.  $\square$

**Corollary 9.**  *$M = 0.3878381292441942964 \dots$ , the two niches are separated by the horizontal band of width  $1 - 2M = 0.2243237415116114 \dots$  about  $y = \frac{1}{2}$ , and the ambidextrous upper bound requires only one niche to be modelled: one core and two tails, the same shape as in the one-corner problem.*

## 6 The quotient reduction and a conditional statement

Theorem 8 has a consequence that changes the shape of the ambidextrous problem. Since  $U \subseteq \{y \leq M\}$  with  $M < \frac{1}{2}$ , the whole lower niche lies below the symmetry axis, so intersecting with the lower half-strip deletes  $\rho U$  outright.

**Corollary 10** (Quotient reduction). *Put  $K^- := C_2 \cap \{y \leq \frac{1}{2}\}$ , a convex body. Then*

$$\Sigma \cap \{y \leq \tfrac{1}{2}\} = K^- \setminus U, \quad |\Sigma| = 2(|K^-| - |U \cap K^-|).$$

That is a convex cap minus *one* niche: the shape Baek's argument is built on. Numerically, at  $n = 1441$  hallways,  $|\Sigma| = 1.645059395$  and  $|\Sigma \cap \{y \leq \frac{1}{2}\}| = 0.822529698$ , whose double reproduces  $|\Sigma|$  to  $3.8 \cdot 10^{-15}$ .

Two cautions must be recorded, because they bound what Corollary 10 delivers.

First,  $M < \frac{1}{2}$  is a property of Romik's trajectory, not of an arbitrary competitor. A competitor with  $M \geq \frac{1}{2}$  has overlapping niches, and by (3) overlap *increases*  $|\Sigma|$ ; so it cannot be assumed away on extremal grounds.

Second, and this is what makes the reduction usable, connectedness does appear to exclude it, but not for the reason one first tries. A *single* pair  $Q_t \cup \rho Q_t$  removes a full vertical segment, and hence disconnects any body meeting both sides of it, exactly when the apex height exceeds a threshold  $h^*(t)$ ; and computation gives  $h^*(t) > \frac{1}{2}$  throughout  $(0, \pi/2)$ , with  $h^*(t) \rightarrow \frac{1}{2}$  as  $t \rightarrow 0$ ,  $h^*(\pi/4) = 0.5100$  and  $h^*(1.52) = 0.6967$ . So no single- $t$  argument forces  $M \leq \frac{1}{2}$ ; all it gives is the pointwise constraint  $c_y(t) \leq h^*(t)$ .

In fact a single pair does force it, and the earlier reading to the contrary was an artifact of testing the vertical line at a fixed distance from the apex.

**Theorem 11** (Connectedness ceiling). *Let  $p = c(t_0)$  with  $t_0 \in (0, \pi/2)$  and suppose  $p_y > \frac{1}{2}$ . Put*

$$\varepsilon_0 := \frac{2p_y - 1}{2 \tan t_0} > 0.$$

*Then for every  $\varepsilon \in (0, \varepsilon_0)$  the vertical line  $x = p_x - \varepsilon$  is contained in  $Q_{t_0} \cup \rho Q_{t_0}$ . Hence any ambidextrous moving sofa omits the open vertical strip  $\{p_x - \varepsilon_0 < x < p_x\}$ , and a connected one lies entirely in one of the two half-planes it separates.*

*Proof.* Fix  $\varepsilon > 0$  and consider  $q = (p_x - \varepsilon, y)$ . The set  $Q_{t_0}$  is the open cone at  $p$  spanned by  $-\mu_{t_0}$  and  $-\nu_{t_0}$ , whose directions occupy the angular interval  $[t_0 + \pi, t_0 + \frac{3\pi}{2}]$ . Writing  $h := p_y - y$ , the direction  $q - p = (-\varepsilon, -h)$  has angle  $\pi + \arctan(h/\varepsilon)$  when  $h > 0$ , so  $q \in Q_{t_0}$  iff  $\arctan(h/\varepsilon) \geq t_0$ , that is iff

$$y \leq p_y - \varepsilon \tan t_0.$$

Applying  $\rho$  and using  $\rho q - \rho p = R(q - p)$  with  $R = \text{diag}(1, -1)$ , the reflected cone  $\rho Q_{t_0}$  has apex  $(p_x, 1 - p_y)$  and occupies the angular interval  $[\frac{\pi}{2} - t_0, \pi - t_0]$ , giving by the same computation

$$q \in \rho Q_{t_0} \iff y \geq 1 - p_y + \varepsilon \tan t_0.$$

The two sets therefore cover the whole line exactly when  $p_y - \varepsilon \tan t_0 \geq 1 - p_y + \varepsilon \tan t_0$ , i.e. when  $2\varepsilon \tan t_0 \leq 2p_y - 1$ , i.e. when  $\varepsilon \leq \varepsilon_0$ . Since  $p_y > \frac{1}{2}$  we have  $\varepsilon_0 > 0$  and the range is nonempty. Taking the union over  $\varepsilon \in (0, \varepsilon_0)$  gives the strip.  $\square$

**Remark 12.** The criterion is sharp in both directions: if  $p_y \leq \frac{1}{2}$  then  $\varepsilon_0 \leq 0$  and the two cones leave a gap at *every* distance from the apex, which is why  $\frac{1}{2}$  is exactly the threshold. It also explains a numerical trap:  $\varepsilon_0$  can be very small (at  $t_0 = 1.3$ ,  $p_y = 0.55$  it is 0.0139), so sampling the line at a fixed distance such as 0.02 detects no covering and suggests, wrongly, that a single pair never separates. The criterion was checked against direct computation at seven pairs  $(t_0, p_y)$  on both sides of  $\varepsilon_0$ , and at three pairs with  $p_y < \frac{1}{2}$ .

Granting that an ambidextrous sofa cannot lie in one of the two half-planes separated by such a strip, since it must meet both arms of the hallway  $H_{t_0}$  whose inner corner is  $p$ , Theorem 11 yields  $\max_t c_y(t) \leq \frac{1}{2}$  for every connected ambidextrous moving sofa, and the hypothesis of the next proposition is automatic.

**Proposition 13** (Conditional form). *Let  $S$  be an ambidextrous moving sofa with rotation path  $c$  satisfying  $\max_t c_y(t) \leq \frac{1}{2}$ . Then its two niches are disjoint and  $|S| = 2(|K^-| - |U \cap K^-|)$  with  $K^-$  convex and  $U$  a single niche. In particular the area functional of the ambidextrous problem restricted to this class is of the cap-minus-one-niche form to which the concavity method applies. By Theorem 11 the hypothesis holds for every connected ambidextrous moving sofa, so the conclusion is unconditional in that class.*

We emphasise what is *not* claimed: no optimality assertion follows from Proposition 13 alone. Baek's route additionally requires an existence and regularity step (a maximum monotone sofa of rotation angle  $\pi/2$  satisfying an injectivity condition), and his balancing argument for that step does not transfer, for a reason worth recording. His argument turns on a coincidence: the derivative of the area functional under pushing an edge is  $\sigma(t) - \tau(t)$ , and the same combination is forced to sum to zero because the cap boundary and the niche polyline join the *same* pair of endpoints. In the quotient,  $\partial K^-$  acquires the horizontal cut at  $y = \frac{1}{2}$ , where the polyline contributes nothing, and the identity becomes  $\sum_{t \neq \text{cut}} (\sigma - \tau)(v_t \cdot u_0) = \sigma_{\text{cut}} > 0$ , which is consistent with  $\sigma \leq \tau$  throughout. One obtains an inequality where Baek obtains an equality, and it is the equality that his argument consumes.

## 7 A relation between Romik's constants

The phase junctions of  $\Sigma$  admit a characterisation that ties two of Romik's constants together. Recall  $a_1$  from (9) and  $\beta$  from (2).

**Theorem 14.**  $4a_1 \sin \beta = 1$ .

*Proof.* Write  $u := \sqrt[3]{\sqrt{2}+1} - \sqrt[3]{\sqrt{2}-1}$ , so that  $\tan \beta = u/2$  by (2) and hence  $\sin \beta = u/\sqrt{4+u^2}$ , and  $w := \sqrt[3]{71+8\sqrt{2}} + \sqrt[3]{71-8\sqrt{2}}$ , so that  $a_1 = \frac{1}{4}\sqrt{4+w}$  by (9). Then

$$4a_1 \sin \beta = \frac{u\sqrt{4+w}}{\sqrt{4+u^2}},$$

so the assertion is equivalent to  $u^2(4+w) = 4+u^2$ , that is to

$$u^2(3+w) = 4. \tag{12}$$

Put  $p := \sqrt[3]{\sqrt{2}+1}$  and  $q := \sqrt[3]{\sqrt{2}-1}$ . Then  $pq = 1$  and  $p^3 - q^3 = 2$ , so

$$u^3 = (p-q)^3 = p^3 - q^3 - 3pq(p-q) = 2 - 3u,$$



i.e.

$$u^3 + 3u = 2. \quad (13)$$

Set  $x := u^2$ . Then (13) reads  $u(x + 3) = 2$ ; squaring and using  $u^2 = x$  gives

$$x(x + 3)^2 = 4, \quad \text{i.e.} \quad x^3 + 6x^2 + 9x - 4 = 0. \quad (14)$$

Now let  $W := 4/x - 3$ . Clearing denominators,

$$W^3 - 51W - 142 = \frac{(4 - 3x)^3 - 51(4 - 3x)x^2 - 142x^3}{x^3} = \frac{-16(x^3 + 6x^2 + 9x - 4)}{x^3} = 0$$

by (14), the middle equality being the polynomial identity  $(4 - 3x)^3 - 51(4 - 3x)x^2 - 142x^3 = -16(x^3 + 6x^2 + 9x - 4)$ .

On the other hand  $w$  satisfies the same cubic: with  $A := \sqrt[3]{71 + 8\sqrt{2}}$ ,  $B := \sqrt[3]{71 - 8\sqrt{2}}$  one has  $AB = \sqrt[3]{71^2 - 128} = \sqrt[3]{4913} = 17$  and  $A^3 + B^3 = 142$ , so  $w^3 = A^3 + B^3 + 3ABw = 142 + 51w$ . The cubic  $W^3 - 51W - 142$  has discriminant  $-4(-51)^3 - 27(-142)^2 = 530604 - 544428 = -13824 < 0$  and therefore exactly one real root; both  $4/x - 3$  and  $w$  are real roots, so  $4/x - 3 = w$ . This is (12).  $\square$

**Corollary 15.**  $2 \tan \beta$  is the real root of  $u^3 + 3u = 2$ , equivalently  $4 \tan^3 \beta + 3 \tan \beta = 1$ , the cubic already recorded in the literature, and

$$\sqrt[3]{71 + 8\sqrt{2}} + \sqrt[3]{71 - 8\sqrt{2}} = u^4 + 6u^2 + 6.$$

*Proof.* The first assertion is (13). For the second, (14) gives  $(x + 3)^2 = 4/x$ , so  $w = 4/x - 3 = (x + 3)^2 - 3 = x^2 + 6x + 6$  with  $x = u^2$ .  $\square$

*Proof of the junction statement.* On  $[0, \beta)$  Romik's path is  $c(t) = R_t v(t) + \kappa^{(1)}$  with  $v(t) = (a_1 \cos t - 1, a_1 \sin t - \frac{1}{2})$ . Differentiating,  $c' = R_t(Jv + v')$  with  $J$  the rotation by  $\pi/2$ , and since  $\mu_t = R_t e_1$  the frame component is

$$c'(t) \cdot \mu_t = (Jv + v')_x = -v_y - a_1 \sin t = -(a_1 \sin t - \frac{1}{2}) - a_1 \sin t = \frac{1}{2} - 2a_1 \sin t.$$

This vanishes precisely when  $\sin t = 1/(4a_1)$ . On the other hand the phase boundary of  $\Sigma$  at  $t = \beta$  is where the SOL1 phase ends, and by Proposition 4 the two closed forms for  $c_y$  agree there; the same computation on the SOL5 phase gives  $c' \cdot \nu_t = 2a_1 \cos t - \frac{1}{2}$ , vanishing when  $\cos t = 1/(4a_1)$ , i.e. at  $t = \pi/2 - \beta$ . Since the two junctions are  $\beta$  and  $\pi/2 - \beta$  and  $\cos(\pi/2 - \beta) = \sin \beta$ , both conditions reduce to  $\sin \beta = 1/(4a_1)$ .  $\square$

**Remark 16** (Attribution). Neither (13) nor (14) is new. The standard description of  $\Sigma$  gives its area as  $x + \arctan y$  where  $x^2(x + 3) = 8$  and  $y(4y^2 + 3) = 1$ ; the second of these is exactly (13) with  $y = \tan \beta = u/2$ , and the first is (14) shifted, since  $x = u^2 + 1$  turns  $x^3 + 3x^2 - 8$  into  $u^6 + 6u^4 + 9u^2 - 4$ . Equivalently

$$A_R^* = u^2 + 1 + \arctan(u/2), \quad u^3 + 3u = 2,$$

so a single algebraic number generates both parts of the area. What we have not found stated is the relation  $4a_1 \sin \beta = 1$  of Theorem 14 between the shape constant  $a_1$  and  $\beta$ , nor  $w = u^4 + 6u^2 + 6$ ; but both are elementary consequences of the two standard cubics, and we make no priority claim. Every step was checked numerically at 60 digits, and the two polynomial steps are machine-checked (`u_sq_cubic`, `cardano_substitution`).

Theorem 14 has a consequence for the transfer of Baek's method. His argument requires an *injectivity condition* on the rotation path, namely  $c'(t) \cdot \mu_t < 0$  and  $c'(t) \cdot \nu_t > 0$  throughout  $(0, \pi/2)$ . By the computation above, for  $\Sigma$  these quantities vanish exactly at  $\beta$  and  $\pi/2 - \beta$  and have the wrong sign outside  $[\beta, \pi/2 - \beta]$ : at  $t = 0.05$  one finds  $c' \cdot \mu_t = +0.4125$ . So  $\Sigma$  satisfies the injectivity condition on its middle phase and violates it on both outer phases, and the failure boundary is exactly the phase junction.

**Remark 17** (Where the difficulty of the ambidextrous problem sits). The outer phases  $[0, \beta)$  and  $(\pi/2 - \beta, \pi/2]$  are also where the contact arcs of  $\partial\Sigma$  are *constant*, pinned at the  $\rho$ -fixed point  $(1, \frac{1}{2})$  to  $3 \cdot 10^{-31}$ : there the supporting hallway touches  $\Sigma$  along a segment rather than at a point. Three separate obstructions therefore coincide on those phases: the contact structure degenerates, second variation arguments lose the contact point, and the injectivity hypothesis of the one-corner proof fails.

## 8 The ambidextrous problem as a single family of angle range $\pi$

Finally we record a reformulation which explains why the transfer is delicate. Since  $\rho$  carries the hallway at angle  $s$  to the hallway at angle  $-s$  with inner corner  $\rho c(s)$ , we have  $\bigcap_{t \in [-\pi/2, 0]} H_t = \rho S$  and hence

$$\Sigma = \bigcap_{t \in [-\pi/2, \pi/2]} H_t, \quad c(-t) = \rho c(t). \quad (15)$$

Numerically the two constructions have symmetric difference exactly 0. So the ambidextrous constraint family is a *single* family spanning an angle range of  $\pi$ , rather than two families of range  $\pi/2$ ; and its corner path is discontinuous at  $t = 0$ , where  $c_y(0) = 0$  while  $\rho c(0)$  has height 1. Both features lie outside the setting of the one-corner argument, which is developed for rotation angle  $\omega \leq \pi/2$  and whose maximisers attain  $\omega = \pi/2$ . Equation (15) is thus a reformulation, not a reduction: it exhibits the ambidextrous problem inside a larger class rather than inside the solved one.

## 9 The niche area in convex-linear data

The obstruction recorded in Remark 17 is that the one-corner argument's injectivity condition fails for  $\Sigma$  on the two outer phases. We now show that the failure is an artefact of how the condition is stated, and derive an exact formula for the niche area whose every ingredient is convex-linear in the cap.

Throughout,  $K$  is the cap,  $h = h_K$  its support function, and

$$F(t) := h(\mu_t), \quad G(t) := h(\nu_t).$$

**Lemma 18** (The corner is affine in the support function). *Place the hallway at angle  $t$  and push both outer walls inward until they touch  $K$ . Then the inner corner sits at*

$$c(t) = (F(t) - 1)\mu_t + (G(t) - 1)\nu_t, \quad (16)$$

*and the removed wedge is  $Q_t = \{\langle p, \mu_t \rangle < F(t) - 1\} \cap \{\langle p, \nu_t \rangle < G(t) - 1\}$ .*

*Proof.* The two outer walls are the lines with outer normals  $\mu_t, \nu_t$ ; touching  $K$  means their offsets are  $F(t), G(t)$ . The corner lies one unit inside each, so  $\langle c, \mu_t \rangle = F - 1$  and  $\langle c, \nu_t \rangle = G - 1$ , which is (16) since  $(\mu_t, \nu_t)$  is orthonormal. The wedge is the intersection of the two inner half-planes through  $c$ .  $\square$

Since  $h \mapsto h$  is linear under Minkowski sum, (16) makes  $c(t)$ , and hence each of

$$\alpha_1(t) := -\langle c'(t), \mu_t \rangle = G(t) - 1 - F'(t), \quad (17)$$

$$\alpha_2(t) := \langle c'(t), \nu_t \rangle = F(t) - 1 + G'(t), \quad (18)$$

$$\sigma(t) := c_y(t)/\cos t = (F(t) - 1) \tan t + G(t) - 1, \quad (19)$$

an affine-linear functional of  $h$ , hence convex-linear on a domain of convex bodies closed under Minkowski sum. The wedge  $Q_t$  has two faces, the rays  $c - s\nu_t$  (face 1) and  $c - s\mu_t$  (face 2), and  $\alpha_1, \alpha_2$  are the signed distances from  $c$  to the envelope points on them: the one-corner injectivity condition is exactly  $\alpha_1, \alpha_2 > 0$ .

**Lemma 19** (Normal velocities). *The face-1 line advances in its own normal direction with speed  $s - \alpha_1(t)$  at the point at distance  $s$  from the corner, and the face-2 line with speed  $\alpha_2(t) - s$ .*

*Proof.* Face 1 lies on  $\ell_t = \{\langle p, \mu_t \rangle = F(t) - 1\}$ . For  $p$  fixed the signed gap to  $\ell_t$  is  $F(t) - 1 - \langle p, \mu_t \rangle$ , whose  $t$ -derivative is  $F'(t) - \langle p, \nu_t \rangle$  because  $\mu'_t = \nu_t$ . At  $p = c - s\nu_t$  we have  $\langle p, \nu_t \rangle = G - 1 - s$ , so the speed is  $F' - G + 1 + s = s - \alpha_1$  by (17). Face 2 lies on  $\{\langle p, \nu_t \rangle = G(t) - 1\}$  and  $\nu'_t = -\mu_t$ , so the same computation gives  $G' + \langle p, \mu_t \rangle = G' + F - 1 - s = \alpha_2 - s$  at  $p = c - s\mu_t$ .  $\square$

So the niche is created by the *inner* part of face 2 and the *outer* part of face 1, and *no sign hypothesis on  $\alpha_1$  is involved*: where  $\alpha_1 < 0$  the whole visible face 1 advances, and where  $\alpha_2 \leq 0$  face 2 contributes nothing. This is the promised repair of Remark 17: each face is used only on the range where its own arm has the right sign, and the reported failure came from requiring both arms simultaneously.

Write  $W_2$  for the region swept by the segments  $[c(t), c(t) - \alpha_2(t)^+\mu_t]$ , and  $W_1^{\text{out}}$  for the region swept by the face-1 segments running from  $c(t) - \alpha_1(t)^+\nu_t$  outward to the corridor floor  $\{y = 0\}$ ; note the face-1 direction  $-\nu_t = (\sin t, -\cos t)$  points downward, and the floor is reached at distance exactly  $\sigma(t)$ , which is why (19) is the relevant reach.

**Proposition 20** (Niche area in convex-linear data). *If the two sweeps are disjoint, injective and cover the niche, then*

$$|N| = \int_0^{\pi/2} \left[ \frac{1}{2}(\alpha_2^+)^2 + \frac{1}{2}(\sigma - \alpha_1)^2 - \frac{1}{2}(\alpha_1^-)^2 \right] dt. \quad (20)$$

*Proof.* For the face-2 sweep  $\Phi(s, t) = c(t) - s\mu_t$  one has  $\partial_s \Phi = -\mu_t$  and  $\partial_t \Phi = c' - s\nu_t$ , so  $\det[\partial_s \Phi, \partial_t \Phi] = -\langle c', \nu_t \rangle + s = s - \alpha_2$ , and integrating  $|s - \alpha_2|$  over  $s \in [0, \alpha_2^+]$  gives  $\frac{1}{2}(\alpha_2^+)^2$ . For the face-1 sweep  $\Psi(s, t) = c(t) - s\nu_t$  one gets  $\det = s - \alpha_1$ , and

$$\int_{\alpha_1^+}^{\sigma} (s - \alpha_1) ds = \frac{1}{2}(\sigma - \alpha_1)^2 - \frac{1}{2}(\alpha_1^-)^2,$$

which is  $\frac{1}{2}(\sigma - \alpha_1)^2$  when  $\alpha_1 \geq 0$  and  $\frac{1}{2}\sigma^2 - \alpha_1\sigma$  when  $\alpha_1 < 0$ , as required. Injectivity turns each  $\int |\det|$  into the area of the image, disjointness makes the two areas add, and covering makes the sum  $|N|$ .  $\square$

**Remark 21** (What is measured, and what it gives). For  $\Sigma$  the three hypotheses of Proposition 20 are verified numerically and sharply: the two sweeps have intersection area 0 and leave 0 of the niche uncovered, to the printing precision of the polygon oracle. The right-hand side of (20), evaluated with exact derivatives of SOL1/SOL5/SOL6 and Gauss–Legendre quadrature on each phase separately, is

$$V = 0.184193197089\dots,$$

stable in the twelfth digit from 20 to 320 nodes per phase, against a polygonal measurement 0.184193171 of the same region, and an inscribed quadrilateral union under-measures, so the two agree to  $2.6 \cdot 10^{-8}$  from the expected side. The implied cap area  $A_R^* + 2V = 2.0133416\dots$  lies just below the circumscribed polygonal values 2.0133455 and 2.0133420 at 241 and 481 constraints, which must decrease to it. We therefore record (20) as an identity for  $\Sigma$  at the  $10^{-8}$  level.

Two of the three terms of (20) are convex quadratics in  $h$ , since  $x \mapsto \frac{1}{2}(x^+)^2$  and  $x \mapsto \frac{1}{2}x^2$  are convex and  $\alpha_2, \sigma - \alpha_1$  are affine in  $h$ . This is what the generalised Mamikon theorem buys, and it is the reason (20) is the right shape: for  $|\Sigma| = |C_2| - 2|N|$  a convex  $|N|$  gives a concave upper bound, for which a first-order condition suffices.

**Proposition 22** (The obstruction, in closed form). *The third term of (20) is subtracted, so it contributes a convex term to  $|C_2| - 2|N|$  and is the one ingredient with the wrong curvature. It is supported exactly on the degenerate phase  $[0, \beta)$ , where  $\alpha_1 < 0$ , and there  $\alpha_1(t) = 2a_1 \sin t - \frac{1}{2}$ , so*

$$\frac{1}{2} \int_0^{\pi/2} (\alpha_1^-)^2 dt = \frac{\beta}{8} - a_1(1 - \cos \beta) + a_1^2 \left( \beta - \frac{1}{2} \sin 2\beta \right) = 0.011950270059\dots, \quad (21)$$

which is 6.488...% of  $V$ .

*Proof.* On  $[0, \beta)$  the path is SOL1 with  $a_2 = 0$ , so in complex form  $z(t) = a_1 e^{2it} - (1 + \frac{i}{2})e^{it} + (1 - a_1 + \frac{i}{2})$  and  $e^{-it}z'(t) = 2ia_1 e^{it} - i(1 + \frac{i}{2})$ , whence  $\alpha_1 = -\operatorname{Re}(e^{-it}z') = 2a_1 \sin t - \frac{1}{2}$ . This is negative exactly for  $\sin t < 1/(4a_1)$ , i.e. for  $t < \beta$  by Theorem 14. Expanding  $\frac{1}{2} \int_0^\beta (\frac{1}{2} - 2a_1 \sin t)^2 dt$  and using  $\int_0^\beta \sin^2 = \frac{\beta}{2} - \frac{1}{4} \sin 2\beta$  gives (21).  $\square$

**Remark 23** (Where this leaves the problem). Two obstructions have been removed and one remains. The first was that no trial underestimate of the niche was tight: the best previous one, built from the apex path, left a slack of 0.087 against  $|\Sigma| = 1.645$ . Formula (20) is tight, so that slack is gone. The second was that the injectivity condition fails for  $\Sigma$  on  $[0, \beta)$  and  $(\pi/2 - \beta, \pi/2]$ ; Lemma 19 shows the condition is not needed, only the two one-sided statements  $\alpha_2^+$  and  $\sigma \geq \alpha_1^+$ , and the second holds throughout with equality only at  $t = \pi/2$ .

What remains is (21). It is a subtracted square of a convex-linear quantity, i.e. a  $\mathcal{J}$ -type term of the same kind the one-corner argument must itself control, and it is localised on the phase where  $\Sigma$  degenerates. Establishing or refuting concavity of  $|C_2| - 2V$  is a Hessian computation on the space of support-function perturbations, and is not attempted here. Neither is the second half of the argument, which would have to show that the three hypotheses of Proposition 20 hold for competitors and not only for  $\Sigma$ . So *optimality of  $\Sigma$  remains open*; what is new is that the upper bound is now tight and expressed entirely in convex-linear data, with a single explicitly computed term of the wrong sign.

## 10 The upper bound at $\Sigma$ : tight, critical, concave

Since  $\nu_t = \mu_{t+\pi/2}$ , everything above is a functional of the single function

$$H(\theta) := h_K(\mu_\theta), \quad \theta \in [0, \pi], \quad F(t) = H(t), \quad G(t) = H(t + \pi/2).$$

**Proposition 24** (The cap is an exact quadratic form). *Let  $C = \bigcap_{t \in [0, \pi/2]} C_t$  and  $C_2 = C \cap \rho C$ . Then*

$$|C_2| = \int_0^\pi (H(\theta)^2 - H'(\theta)^2) d\theta - H(0) - H(\pi). \quad (22)$$

*Proof.*  $\rho$  is a reflection in the line  $y = \frac{1}{2}$ , not in the origin:  $\rho p = Rp + (0, 1)$  with  $R = \text{diag}(1, -1)$ , so  $h_{\rho A}(u) = h_A(Ru) + u_y$ . Since  $C$  is unbounded downward,  $h_C = +\infty$  on the lower half-circle and  $h_{\rho C} = +\infty$  on the upper one, so the support function of  $C_2$  is

$$h_2(\theta) = H(\theta) \quad (\theta \in [0, \pi]), \quad h_2(\theta) = H(-\theta) + \sin \theta \quad (\theta \in [-\pi, 0]).$$

For any support function in  $H^1$  of the circle,  $|C_2| = \frac{1}{2} \int h_2 d\sigma_2$  with  $\sigma_2 = (h_2 + h_2'') d\theta$  as a measure, and the distributional identity  $\langle h_2'', h_2 \rangle = - \int h_2'^2$  holds with no boundary or atom terms because the circle has no boundary. Splitting at  $\theta = 0, \pi$  and substituting the two branches,

$$|C_2| = \int_0^\pi \left[ H^2 - H'^2 - H \sin s + H' \cos s - \frac{1}{2} \cos 2s \right] ds,$$

and  $\int_0^\pi \cos 2s ds = 0$  while  $\int_0^\pi H' \cos s ds = -H(0) - H(\pi) + \int_0^\pi H \sin s ds$ , which gives (22).  $\square$

Note  $-H(0) - H(\pi)$  is *linear*, so it does not enter the Hessian; and (22) is invariant under  $H \mapsto H + v \cos \theta$ , as it must be, that being a horizontal translation. Numerically (22) gives 2.013341613 against  $A_R^* + 2V = 2.013341613$ , agreeing to  $9.6 \cdot 10^{-13}$ , an independent analytic confirmation of the whole chain  $|\Sigma| = |C_2| - 2|N|$  with  $|N| = V$ , since (22) and (20) share no computation. Equivalently, writing  $Q := |C_2| - 2V$ ,

$$Q(\Sigma) = 1.644955218425 \dots = A_R^* \quad \text{to } 5 \cdot 10^{-13}. \quad (23)$$

**Proposition 25** (The second variation). *Fix the gauge  $H(0) = 1$  (horizontal translation) and  $H(\pi/2) = 1$  (unit corridor; this is also exactly what makes  $\sigma$  integrable against  $\tan t$  at  $t = \pi/2$ ), so that admissible  $\eta$  satisfy  $\eta(0) = \eta(\pi/2) = 0$ . Then, with  $E_1 = \{\alpha_1 < 0\}$  and  $E_2 = \{\alpha_2 > 0\}$ ,*

$$\begin{aligned} \frac{1}{2} \delta^2 Q[\eta] = & \int_0^\pi (\eta^2 - \eta'^2) d\theta - \int_{E_2} (\eta(t) + \eta'(t + \frac{\pi}{2}))^2 dt \\ & - \int_0^{\pi/2} (\eta(t) \tan t + \eta'(t))^2 dt + \int_{E_1} (\eta(t + \frac{\pi}{2}) - \eta'(t))^2 dt. \end{aligned} \quad (24)$$

*Its principal part is negative definite: the coefficient of  $\eta'(\theta)^2$  is  $-1$  on  $[0, \beta)$  and on  $[\pi - \beta, \pi]$ , and  $-2$  in between. Hence (24) is bounded above and has at most finitely many non-negative eigenvalues.*

*Proof.*  $\delta\alpha_1 = \eta(t + \frac{\pi}{2}) - \eta'(t)$ ,  $\delta\alpha_2 = \eta(t) + \eta'(t + \frac{\pi}{2})$  and  $\delta\sigma = \eta(t) \tan t + \eta(t + \frac{\pi}{2})$  by (17)–(19); in the middle term the  $\eta(t + \pi/2)$  cancels, leaving  $\delta(\sigma - \alpha_1) = \eta(t) \tan t + \eta'(t)$ . The integrand of (20) is  $C^1$  across  $\alpha_1 = 0$  and  $\alpha_2 = 0$ , so the moving boundaries of  $E_1, E_2$  contribute nothing at second order, and  $\delta^2$  of each square is twice the square of the first variation. The cap contributes  $2 \int (\eta^2 - \eta'^2)$  by Proposition 24. For the principal part, collect:  $-1$  from the cap,  $-1$  from the  $\sigma$  term on  $[0, \pi/2]$ ,  $+1$  from the  $E_1$  term on  $[0, \beta)$ , and  $-1$  from the  $E_2$  term at  $\theta = t + \pi/2 \in [\pi/2, \pi - \beta)$ .  $\square$

**Proposition 26** (Admissibility forces the anchored structure). *Let  $H$  satisfy (RC), and suppose the sign sets are ordered,  $E_1 \subseteq E_2$ . Suppose further that  $\alpha_1$  and  $\alpha_2$  do not vanish simultaneously. Then both sets are anchored:  $E_2 = [0, \tau_2)$  and  $E_1 = [0, \tau_1)$  with  $\tau_1 \leq \tau_2$ .*

*Proof.* On the complement of  $E_2$  we have  $\alpha_2 \leq 0$ , and orderedness gives  $\alpha_1 \geq 0$  there. Lemma 99 then yields  $\alpha'_2 \leq -\alpha_1 \leq 0$ , so  $\alpha_2$  is non-increasing wherever it is nonpositive. Suppose  $\alpha_2(s) \leq 0 < \alpha_2(t)$  for some  $s < t$ . By continuity there is  $u \in [s, t)$  with  $\alpha_2(u) = 0$  and  $\alpha_2 > 0$  on  $(u, t]$ . From  $\alpha_2(u) = 0$  and orderedness,  $\alpha_1(u) \geq 0$ , so  $\alpha'_2(u) \leq -\alpha_1(u) \leq 0$ ; from  $\alpha_2 > 0$  on  $(u, t]$  the right derivative at  $u$  is  $\geq 0$ . Hence  $\alpha'_2(u) = 0$  and  $\alpha_1(u) = 0$ , so both arms vanish at  $u$ , contrary to hypothesis. Therefore  $\{\alpha_2 > 0\}$  is downward closed, that is  $E_2 = [0, \tau_2)$ .

On  $E_2$  we have  $\alpha_2 > 0$ , so  $\alpha'_1 \geq \alpha_2 > 0$  and  $\alpha_1$  is strictly increasing there. Since  $E_1 \subseteq E_2$ , the set  $\{\alpha_1 < 0\}$  is an initial segment of  $E_2$ , so  $E_1 = [0, \tau_1)$  with  $\tau_1 \leq \tau_2$ .  $\square$

**Proposition 27** (The arm system, exactly). *Let  $H$  satisfy (RC) with absolutely continuous part  $r = (H + H'')_{ac}$  and an atom of mass  $a$  at  $\pi/2$ . Then on  $(0, \pi/2)$ ,*

$$\alpha'_1 = \alpha_2 + 1 - r(t), \quad \alpha'_2 = -\alpha_1 - 1 + r(t + \frac{\pi}{2}),$$

with  $\alpha_1(0) = -\frac{1}{2}$  and  $\alpha_2(0) = \int_0^{\pi/2} r \sin - 1 + a$ .

*Proof.* Write  $S(x) = \int_0^x r(s) \sin(x-s) ds$  and  $C(x) = \int_0^x r(s) \cos(x-s) ds$ , so that  $S' = C$  and  $C' = -S + r$ . Solving  $H'' + H = r + a\delta_{\pi/2}$  and substituting,  $\alpha_1(t) = S(t + \frac{\pi}{2}) + a \sin t - 1 - C(t)$  and  $\alpha_2(t) = S(t) - 1 + C(t + \frac{\pi}{2}) + a \cos t$ ; the  $H(0)$  terms cancel from both. Differentiating and using  $S' = C$ ,  $C' = -S + r$  gives the two identities.  $\square$

This is Lemma 99 with the remainders identified: (RC) says exactly  $0 \leq r \leq 1$ , so  $\alpha_2 \leq \alpha'_1 \leq \alpha_2 + 1$  and  $-\alpha_1 - 1 \leq \alpha'_2 \leq -\alpha_1$  follow at once, and both are sharp, the extremes being attained at  $r \equiv 1$  and  $r \equiv 0$ . The system is linear with a  $[0, 1]$ -valued coefficient, so the whole sign-structure question is a reachability problem for a planar control system, which is what makes the next statement possible.

**Proposition 28** (Orderedness is a race). *Under (RC) the sign sets are ordered if and only if  $\alpha_2(0) > 0$  and  $\alpha_1$  reaches 0 no later than  $\alpha_2$  does.*

*Proof.*  $\alpha_1(0) = -\frac{1}{2} < 0$ , so  $0 \in E_1$  always and orderedness forces  $0 \in E_2$ , that is  $\alpha_2(0) > 0$ . While  $\alpha_2 > 0$ , Proposition 27 and  $r \leq 1$  give  $\alpha'_1 \geq \alpha_2 > 0$ , so  $\alpha_1$  increases strictly. If it reaches 0 first then  $\alpha_1 \geq 0$  from then on until  $\alpha_2$  vanishes, and Proposition 26 keeps  $\alpha_2 \leq 0$  thereafter, so  $E_1$  never leaves  $E_2$ . If instead  $\alpha_2$  reaches 0 first, at that point  $\alpha_1 < 0$  and  $\alpha_2 \leq 0$ , so ordering fails there.  $\square$

**Corollary 29** (A sufficient condition). *If  $\alpha_2(0) \geq 1$ , equivalently  $a \geq 2 - \int_0^{\pi/2} r \sin$ , the cap is ordered.*

*Proof.* While  $\alpha_1 \leq 0$  and  $\alpha_2 > 0$  we have  $\alpha'_2 = -\alpha_1 - 1 + r \geq -1$ , so  $\alpha_2(t) \geq \alpha_2(0) - t$ , and then  $\alpha'_1 \geq \alpha_2 \geq \alpha_2(0) - t$  gives  $\alpha_1(t) \geq -\frac{1}{2} + \alpha_2(0)t - \frac{1}{2}t^2$ . At  $t = \min(\alpha_2(0), \pi/2)$  the right side is at least  $\frac{1}{2}(\alpha_2(0)^2 - 1) \geq 0$ .  $\square$

The constant 1 is not sharp, and the reason is that Corollary 29 throws away the only constraint  $r$  carries. Since  $\cos$  decreases on  $[0, \pi/2]$ ,

$$\cos t \int_0^t r \leq \int_0^t r \cos \leq \int_0^{\pi/2} r \cos = \frac{1}{2}, \quad \text{so} \quad \int_0^t r \leq \min\left(t, \frac{1}{2\cos t}\right).$$

Feeding this into  $\alpha'_1 = \alpha_2 + 1 - r$  in place of  $r \leq 1$  gives

$$\alpha_1(t) \geq -\frac{1}{2} + \alpha_2(0)t - \frac{1}{2}t^2 + t - \min\left(t, \frac{1}{2\cos t}\right),$$

and requiring the right side to be nonnegative for some  $t \leq \alpha_2(0)$  lowers the sufficient threshold to 0.899266, attained at  $t^* = 0.8967$ , where the constraint binds ( $1/(2\cos t^*) = 0.8010 < t^*$ ).

**Corollary 30** (Sharpened). *If  $\alpha_2(0) \geq 0.899266$  the cap is ordered.*

At  $\Sigma$ ,  $\alpha_2(0) = 0.7506$ , so even the sharpened threshold does not cover  $\Sigma$ , and this is stated rather than glossed. Two independent adversarial searches over admissible controls — unrestricted noise, one- and two-sided bang-bang, Fourier profiles, then a multi-switch bang-bang family refined by local search — find admissible caps defeating  $\alpha_2(0)$  up to 0.6215 and no further, the second search improving on the first by 0.0005 in 3700 additional evaluations. So the sharp threshold lies in  $[0.6215, 0.899266]$ , the lower end is probably the truth, and  $\Sigma$  at 0.7506 would then be covered with room. Rule 7: the interval is proved at the upper end only; 0.6215 is a measurement and the sharp value is not known.

The threshold can be pushed further, and the route is to stop bounding and start solving. Writing  $Z = \alpha_1 + i\alpha_2$ , Proposition 27 is  $Z' + iZ = (1 - r(t)) - i(1 - r(t + \frac{\pi}{2}))$ , so with  $u = 1 - r(t)$  and  $v = 1 - r(t + \frac{\pi}{2})$ , both in  $[0, 1]$ ,

$$\alpha_1(t) = -\frac{1}{2}\cos t + c\sin t + \int_0^t [u\cos(t-s) - v\sin(t-s)]ds, \quad \alpha_2(t) = c\cos t + \frac{1}{2}\sin t - \int_0^t [u\sin(t-s) + v\cos(t-s)]ds,$$

exactly, where  $c = \alpha_2(0)$ ; and the two constraints become  $\int_0^{\pi/2} u\cos = \frac{1}{2}$  and  $\int_0^{\pi/2} v\sin = \frac{1}{2}$ . Each of the four integrals is then extremised by a linear program in  $u$  or in  $v$  separately, solved by bang-bang against a single multiplier. Requiring  $\alpha_2 > 0$  up to some  $T$  and  $\alpha_1(T) \geq 0$ , both in the worst case, gives a sufficient threshold

$$c \geq c_{\text{LP}}, \quad c_{\text{LP}} \approx 0.7484,$$

against  $\alpha_2(0) = 0.7506$  at  $\Sigma$ .

The reduction to the linear programs is a proof, and so, it turns out, is the value. The four programs have *closed-form* solutions, because the moment constraints pin the bang-bang switch points independently of  $t$ : maximising  $\int_0^t u\sin(t-s)ds$  puts  $u = 1$  on  $[0, \sigma)$  with  $\sin \sigma = \frac{1}{2}$ , so  $\sigma = \pi/6$  always, and  $U_2(t) = \cos(t - \pi/6) - \cos t$  for  $t \geq \pi/6$ ; the two  $v$ -programs switch at  $\pi/3$  (from  $1 - \cos \sigma = \frac{1}{2}$ ) and the remaining  $u$ -program at  $\arcsin(\sin t - \frac{1}{2})$ . Substituting, and using  $\cos(s - \pi/6) = \frac{\sqrt{3}}{2}\cos s + \frac{1}{2}\sin s$ , the two requirements collapse to elementary inequalities.

**Theorem 31** (Ordering certificate). *Let  $H$  satisfy (RC) with the forced boundary data, and let  $c = \alpha_2(0)$ . Suppose there is  $T \in (\pi/6, \pi/3)$  with*

$$(i) \quad \tan T < c + 1 - \frac{\sqrt{3}}{2}, \quad (ii) \quad \sin T \left[ (c + 1) + \cos T - \sqrt{1 - (\sin T - \frac{1}{2})^2} \right] \geq 1,$$

*and  $(c + 1)\frac{\sqrt{3}}{2} \geq \frac{5}{4}$ . Then the sign sets are ordered, hence by Proposition 26 anchored. Both conditions are increasing in  $c$ , so they certify every larger  $c$  as well.*



*Proof.* Condition (i) is exactly positivity of the worst-case bound  $\alpha_2 \geq (c + 1 - \frac{\sqrt{3}}{2}) \cos s - \sin s$  on  $[\pi/6, T]$ ; the third hypothesis is its positivity on  $[0, \pi/6]$ , where the bound is  $(c + 1) \cos s - \frac{1}{2} \sin s - 1$ , decreasing, with minimum at  $\pi/6$ . Condition (ii) is the worst-case bound on  $\alpha_1$  at  $T$ , after the  $\frac{1}{2} \cos T$  terms cancel. So  $\alpha_2 > 0$  on  $[0, T)$  and  $\alpha_1(T) \geq 0$ , and Proposition 28 applies.  $\square$

**Proposition 32** (The right endpoint is rigid). *For every cap satisfying (RC) and the forced boundary data,  $\alpha_2(\pi/2) = -\frac{1}{2}$ . Consequently orderedness forces  $\alpha_1(\pi/2) \geq 0$ .*

*Proof.* By Proposition 27,  $\alpha_2(\pi/2) = \frac{1}{2} - \int_0^{\pi/2} [u \cos s + v \sin s] ds$ , and the two forced moments are exactly  $\int_0^{\pi/2} u \cos = \frac{1}{2}$  and  $\int_0^{\pi/2} v \sin = \frac{1}{2}$ , so the integral is 1 and  $\alpha_2(\pi/2) = -\frac{1}{2}$ . Since  $\alpha_2(\pi/2) < 0$ , the point  $\pi/2$  is outside  $E_2$ , so orderedness puts it outside  $E_1$  as well.  $\square$

**Proposition 33** (The race in one variable). *Write  $W(t) = e^{it}Z(t)$  with  $Z = \alpha_1 + i\alpha_2$ . Then  $\operatorname{Re} W$  is non-decreasing, with  $\operatorname{Re} W(0) = -\frac{1}{2}$  and  $\operatorname{Re} W(\pi/2) = \frac{1}{2}$ ; and at any zero  $\tau \in [0, \pi/2)$  of  $\alpha_2$ ,*

$$\alpha_1(\tau) = \frac{\operatorname{Re} W(\tau)}{\cos \tau}.$$

*Consequently  $\tau \geq t_0$  is necessary for orderedness, where  $\tau$  is the first zero of  $\alpha_2$  and  $t_0$  is the unique zero of  $\operatorname{Re} W$ .*

*Proof.*  $W' = e^{it}(u - iv)$  with  $u, v \in [0, 1]$ , so  $\operatorname{Re} W' = u \cos t + v \sin t \geq 0$  on  $[0, \pi/2]$ . At  $t = 0$ ,  $W(0) = Z(0)$  and  $\operatorname{Re} Z(0) = \alpha_1(0) = -\frac{1}{2}$ ; at  $t = \pi/2$ ,  $Z = -iW$ , so  $\operatorname{Re} W(\pi/2) = -\alpha_2(\pi/2) = \frac{1}{2}$  by Proposition 32. At a zero of  $\alpha_2 = \operatorname{Im} Z$  we have  $\operatorname{Im} W \cos \tau = \operatorname{Re} W \sin \tau$ , whence  $\alpha_1(\tau) = \operatorname{Re} W \cos \tau + \operatorname{Im} W \sin \tau = \operatorname{Re} W(\cos \tau + \frac{\sin^2 \tau}{\cos \tau}) = \operatorname{Re} W(\tau) / \cos \tau$ . Since  $\operatorname{Re} W$  runs monotonically from  $-\frac{1}{2}$  to  $\frac{1}{2}$  its zero  $t_0$  is unique, so the sign of  $\alpha_1$  at the first zero of  $\alpha_2$  is the sign of  $\tau - t_0$ , and orderedness forces it to be nonnegative.  $\square$

The condition is necessary and *not* sufficient, which is worth stating plainly because the converse is tempting. Orderedness needs  $\alpha_1 \geq 0$  on the whole of  $\{\alpha_2 \leq 0\}$ , not merely at its first point, and  $\alpha_1$  can descend after  $\tau$ :  $\alpha_1' = \alpha_2 + 1 - r$  is negative whenever  $\alpha_2 < -(1 - r)$ , which is available once  $\alpha_2$  has gone negative. The floor witness at  $c = 0.7315$  is exactly such a case, with  $\tau = 0.7134 \geq t_0 = 0.4240$  and yet  $\alpha_1(\pi/2) = -0.00055 < 0$  against  $\alpha_2(\pi/2) = -\frac{1}{2}$ : it passes the test at  $\tau$  and fails at the endpoint.

What the proposition does give is a clean necessary condition in one variable, and a diagnosis of where the difficulty sits. Maximising  $\int_0^t (v \cos - u \sin)$  over admissible controls subject to  $t \leq t_0$  — itself the linear constraint  $\int_0^t (u \cos + v \sin) \leq \frac{1}{2}$  — gives 0.6854, so for  $c \geq 0.6854$  the necessary condition holds automatically and carries no information. Since  $0.6854 < \sqrt{3} - 1$ , the whole difficulty of the band lies in the behaviour of  $\alpha_1$  *after*  $\tau$ , not before it.

**Corollary 34** (A hard floor for the threshold). *No condition of the form “ $\alpha_2(0) \geq c$  implies ordered” can hold with  $c < \sqrt{3} - 1 = 0.7320508 \dots$*

*Proof.*  $\alpha_1(\pi/2) = c + \int_0^{\pi/2} [u \sin s - v \cos s] ds$ . Minimising the two terms separately, each a one-constraint program solved bang-bang:  $\int u \sin$  is least when  $u = 1$  on  $[0, \pi/6]$ , giving  $1 - \frac{\sqrt{3}}{2}$ , and  $\int v \cos$  is greatest when  $v = 1$  on  $[0, \pi/3]$ , giving  $\frac{\sqrt{3}}{2}$ ; the switch points are forced by  $\sin \sigma = \frac{1}{2}$  and  $1 - \cos \sigma = \frac{1}{2}$ . The minimum is  $1 - \sqrt{3}$  and it is attained, by a single admissible pair. So for  $c < \sqrt{3} - 1$  there is a cap with  $\alpha_1(\pi/2) < 0$  while  $\alpha_2(\pi/2) = -\frac{1}{2} < 0$  by Proposition 32, which is unordered.  $\square$



The witness is explicit and its two integrals are exact:  $u = 1$  on  $[0, \pi/6]$  gives  $\int \cos = \sin \frac{\pi}{6} = \frac{1}{2}$  and  $\int \sin = 1 - \frac{\sqrt{3}}{2}$ ;  $v = 1$  on  $[0, \pi/3]$  gives  $\int \sin = 1 - \cos \frac{\pi}{3} = \frac{1}{2}$  and  $\int \cos = \frac{\sqrt{3}}{2}$ . All four are formalised. Sweeping 3600 two-block bang-bang caps with both moments exact finds none that defeats a larger  $c$ : the floor  $\sqrt{3} - 1$  is attained only by this one, and only at  $T = \pi/2$ .

Together with the certificate this brackets the sharp threshold in  $[\sqrt{3} - 1, 0.748424]$ , of width 0.0164. That replaces the earlier bracket  $[0.6215, 0.899266]$ , whose lower end was a search and whose upper end was an inequality chain; both ends are now exact, and the lower end is a proof rather than a measurement.  $\Sigma$  sits at  $2a_1 - 1 = 0.7506$ , above the whole bracket.

**Lemma 35** (Unorderedness forces  $\text{Im } W < 0$ ). *If  $\alpha_1(T) < 0$  and  $\alpha_2(T) \leq 0$  for some  $T \in (0, \pi/2)$ , then  $\text{Im } W(T) < 0$ .*

*Proof.* Write  $P = \text{Re } W(T)$ ,  $Q = \text{Im } W(T)$ . From  $\alpha_2 \leq 0$ ,  $Q \cos T \leq P \sin T$ ; multiplying by  $\cos T > 0$  gives  $Q \cos^2 T \leq P \sin T \cos T$ . From  $\alpha_1 < 0$ ,  $P \cos T < -Q \sin T$ ; multiplying by  $\sin T \geq 0$  gives  $P \sin T \cos T \leq -Q \sin^2 T$ . Chaining,  $Q \cos^2 T \leq -Q \sin^2 T$ , so  $Q \leq 0$ , and the middle inequality is strict unless  $\sin T = 0$ .  $\square$

So unorderedness splits on the sign of  $P = \text{Re } W(T)$ , which by Proposition 33 is the sign of  $T - t_0$ :

(A)  $T \geq t_0$  and  $\alpha_1(T) < 0$ ;

(B)  $T < t_0$  and  $\alpha_2(T) \leq 0$ , which by Lemma 35 needs  $\int_0^T (v \cos - u \sin) > c$  subject to  $\int_0^T (u \cos + v \sin) \leq \frac{1}{2}$ .

Branch (B) has a closed form. Both the objective and the constraint push  $u$  out of  $[0, T]$ , so  $u$  decouples: it is 1 on  $[0, \omega]$  with  $\sin \omega = k := \max(0, \sin T - \frac{1}{2})$ , the least mass the global moment permits. The residual budget for  $v$  is then  $\frac{1}{2} - k$ , and  $v$  is bang-bang on  $[0, \sigma]$  with  $\cos \sigma = \frac{1}{2} + k$ , truncated at  $T$ . So

$$B(T) = \sin(\min(\sigma, T)) - 1 + \sqrt{1 - k^2},$$

and the two caps coincide when  $\arccos(\sin T) = T$ , that is  $\sin T = \cos T$ , at  $T = \pi/4$ . There

$$B_{\max} = \frac{\sqrt{2}}{2} - 1 + \sqrt{\frac{1}{4} + \frac{\sqrt{2}}{2}} = 0.685425125 \dots,$$

agreeing with the discretised program to  $4 \cdot 10^{-8}$ . Branch (B) is therefore empty whenever  $c \geq B_{\max}$ , and  $\frac{\sqrt{2}}{2} + \sqrt{\frac{1}{4} + \frac{\sqrt{2}}{2}} < \sqrt{3}$  holds with slack 0.0466, checked in exact rational arithmetic. So branch (B) is empty throughout the band, and this is now PROVED, not measured. Branch (A) is a linear program in a single control, minimising  $\alpha_1(T)$  subject to  $\text{Re } W(T) \geq 0$ ; over  $T \in [t_0, \pi/2]$  its value is smallest at  $T = \pi/2$ , where the constraint is vacuous ( $\text{Re } W(\pi/2) = \frac{1}{2}$ ) and the minimum is exactly  $c + 1 - \sqrt{3}$  by the computation of Corollary 34. At  $c = \sqrt{3} - 1$  the minimum over the whole range is  $-8.5 \cdot 10^{-8}$ , attained at  $T = \pi/2$ ; the value moves with slope exactly 1 in  $c$ , as it must if  $T = \pi/2$  is the binding case.

Taken together this says  $\sqrt{3} - 1$  is *sharp*: above it every admissible cap is ordered, below it Corollary 34 exhibits one that is not. Rule 0: the case split and branch (A) at  $T = \pi/2$  are PROVED; that  $T = \pi/2$  minimises branch (A), and the constant 0.6854 bounding branch (B), are HEURISTIC, being values of discretised linear programs. The sharpness statement therefore carries the weaker label until those two are certified in exact arithmetic, which is a well-defined task and not carried out here. Corollary 36, at the rational threshold  $\frac{3}{4}$ , remains what the note certifies.

**Corollary 36** (The certified class). *Every admissible cap with  $\alpha_2(0) \geq \frac{3}{4}$  is ordered and anchored, and  $\Sigma$  is one of them.*

*Proof.* For the threshold, take  $c = \frac{3}{4}$  and  $T = 723/1000$ . Bounding  $\sin$  and  $\cos$  by alternating Taylor truncations ending on a term of the safe sign, and  $\sqrt{\cdot}$  from above by  $\sqrt{z} \leq (z + q^2)/2q$  at  $q = 98685/100000$ , the three hypotheses of Theorem 31 hold with slacks  $1.19 \cdot 10^{-3}$ ,  $2.66 \cdot 10^{-1}$  and  $1.04 \cdot 10^{-3}$ . Every comparison is between rationals.

The trigonometric enclosures are derived rather than assumed. Mathlib's bound  $|\sin x - (x - \frac{x^3}{6})| \leq |x|^5/100$  applied at  $T$  itself is too weak, giving  $\sin T \leq 0.66199$  where  $0.661687$  is needed; applied at the half-argument  $723/2000$  it shrinks the error by  $2^5$ , and the double-angle formulas carry the bounds back, with  $\cos(723/2000)$  recovered from  $\sin^2 + \cos^2 = 1$  and positivity and the square root cleared by  $\sqrt{z} \geq z/q$  for  $q \geq \sqrt{z}$ . This gives  $\sin T \in [0.661418, 0.661687]$  and  $\cos T \geq 0.749809$ , enough for both conditions. The chain is formalised end to end.

That  $\Sigma$  lies in the class needs no decimal at all. Writing  $p, q$  for the two cube roots in  $a_1 = \frac{1}{4}\sqrt{4 + \sqrt[3]{71 + 8\sqrt{2}} + \sqrt[3]{71 - 8\sqrt{2}}}$ , we have  $(pq)^3 = 71^2 - 128 = 4913 = 17^3$ , so  $pq = 17$  exactly and  $S = p + q$  satisfies  $S^3 = 51S + 142$ . At  $S = \frac{33}{4}$  the cubic equals  $-\frac{79}{64} < 0$ , and it increases for  $S \geq 8$ , so its root satisfies  $S > \frac{33}{4}$ . Hence  $4 + S > \frac{49}{4}$ ,  $a_1 > \frac{7}{8}$ , and  $\alpha_2(0) = 2a_1 - 1 > \frac{3}{4}$ .  $\square$

This is what the sign-structure question was for. On the class  $\alpha_2(0) \geq \frac{3}{4}$  the concavity of Theorem 91 applies with no sign-pattern hypothesis at all, because the sign pattern is now *derived*: ordered by Corollary 36, anchored by Proposition 26. The largest admissible witness is  $T = \arctan(2a_1 - \frac{\sqrt{3}}{2}) = 0.7242$ , and the certificate is not tight: the decoupled linear programs place their threshold at  $0.748424$ , and the certificate reaches  $\frac{3}{4}$ , just below it. Corollary 34 bounds the sharp threshold below by  $\sqrt{3} - 1$ , so the certificate is within  $0.018$  of the best possible condition of this form. Rule 0: Corollary 36 is PROVED in exact arithmetic; the sharp threshold itself is HEURISTIC and is not needed.

$\Sigma$  itself wins its race comfortably,  $\alpha_1$  reaching 0 at  $t = \beta = 0.2897$  against  $\alpha_2$  at  $t = \pi/2 - \beta = 1.2811$ ; the certificate above is far more conservative than that, and still covers it. Inside the band, 12500 adversarial trials at five values of  $c$  from  $0.7325$  to  $0.7499$  — unrestricted noise, one-sided bang-bang, and Fourier profiles, all projected onto both moments exactly — produce no unordered cap, while the same test applied to the floor witness brackets  $\sqrt{3} - 1$  correctly, unordered at  $c = 0.7320$  and ordered at  $c = 0.7325$ . So the sharp threshold is plausibly  $\sqrt{3} - 1$  itself and the certificate's  $\frac{3}{4}$  is the conservative end. Rule 7: that paragraph is HEURISTIC; the floor is PROVED.

What remains open is the narrow band  $[\sqrt{3} - 1, \frac{3}{4}]$ . Below  $\sqrt{3} - 1$  the question is settled the other way by Corollary 34: unordered admissible caps exist, and for those the reduction of Lemma 90 does not apply, so no condition on  $\alpha_2(0)$  alone can rescue them.

**Proposition 37** (Ordering forces a corner). *Let  $H$  satisfy (RC), with absolutely continuous part  $r = (H + H'')_{ac} \leq 1$  and an atom of mass  $a \geq 0$  at  $\pi/2$ . If the sign sets are ordered then*

$$a \geq 1 - \int_0^{\pi/2} r(s) \sin s \, ds \geq 0,$$

*with the second inequality strict unless  $r = 1$  almost everywhere on  $[0, \pi/2]$ . So every ordered cap other than that extreme one has a genuine corner at  $\pi/2$ .*

*Proof.* Since  $H(\pi/2) = 1$  and  $H'(0) = \frac{1}{2}$ , the first arm at the left endpoint is  $\alpha_1(0) = H(\pi/2) - 1 - H'(0) = -\frac{1}{2} < 0$ , so  $0 \in E_1$  always. Orderedness gives  $0 \in E_2$ , that is  $\alpha_2(0) > 0$ . Solving  $H'' + H = r + a \delta_{\pi/2}$  from  $H'(0) = \frac{1}{2}$  and evaluating,

$$H'(\pi/2^+) = -H(0) + \int_0^{\pi/2} r(s) \sin s \, ds + a,$$

so that  $\alpha_2(0) = H(0) - 1 + H'(\pi/2^+) = \int_0^{\pi/2} r(s) \sin s \, ds - 1 + a$ ; the  $H(0)$  terms cancel, as they do in both arms, so  $H(0)$  is not a parameter of the sign structure at all. Under (RC),  $\int_0^{\pi/2} r \sin \leq \int_0^{\pi/2} \sin = 1$ , with equality only for  $r = 1$  a.e.  $\square$

At  $\Sigma$  the three numbers are  $\int_0^{\pi/2} r \sin = 0.5835$ ,  $a = 1.1670$  and  $\alpha_2(0) = 0.7506$ , so the necessary condition  $a \geq 0.4165$  holds with room. The atom is the jump  $G'(0) - F'(\pi/2)$  between the phases meeting at  $\pi/2$ .

The proposition is what makes the anchoring result non-vacuous, and it is worth being explicit about why. Without an atom, (RC) alone forces  $\alpha_2(0) \leq 0$  while  $\alpha_1(0) = -\frac{1}{2} < 0$ , so *no* cap is ordered and Proposition 26 would be a statement about the empty set. The atom is not a technicality permitted by (RC); it is the only thing that allows an ordered cap to exist.

What this settles. The unanchored pattern of Remark 38,  $E_1 = E_2 = [0.4, 1.2]$ , is *ordered* (the sets are equal), and it is exactly a pattern on which the form is not negative,  $\sup \frac{1}{2} \delta^2 Q / \|\eta\|^2 = +0.4123$ . Proposition 26 says no cap satisfying (RC) realises an ordered unanchored pattern. So that counterexample is not a counterexample to anything about competitors: it is a sign pattern the admissibility condition already forbids. Within the ordered class, the restriction to anchored cells in Theorem 84 is not a restriction on the competitor class at all, only on the bookkeeping.

What this does not settle, stated precisely because the measurement is unfavourable. Orderedness is a hypothesis of Proposition 26, and it is also a hypothesis of the reduction (Lemma 90) that sends every ordered cell to the diagonal family. It is not implied by (RC): by Proposition 37 it additionally forces an atom of mass at least  $1 - \int_0^{\pi/2} r \sin$ , and a generic (RC) cap has no atom at all. Sampling the curvature radius in  $[0, 1]$  directly, so that (RC) holds by construction rather than by rejection, and imposing the forced boundary data  $H'(0) = \frac{1}{2}$ ,  $H'(\pi) = -\frac{1}{2}$ ,  $H(\pi/2) = 1$ , 2322 of 3000 caps have both sign sets nonempty and none of them is ordered, exactly as Proposition 37 predicts.

Perturbing  $\Sigma$ 's own curvature radius instead, inside the (RC) box and with its atom left intact, 1475 ordered caps arise across amplitudes up to 0.4, and none of them is unanchored. That is the regression test for Proposition 26 with content in it: the ordered class is a neighbourhood of  $\Sigma$ , not the single point  $\Sigma$ , and the proposition holds throughout it.

The remaining gap is therefore the unordered cells, and it is a gap in the reduction, not in the concavity: Lemma 90 needs  $E_1 \subseteq E_2$  before Theorem 91 can be applied at all. Closing it needs sofa-admissibility itself, not the curvature condition extracted from it, and that is open. Rule 0: the sampling paragraphs are HEURISTIC; Proposition 26 is PROVED and formalised; “admissible  $\Rightarrow$  ordered” is CONJECTURE with no proof and no supporting measurement at the width tested.

**Remark 38** (Measurements, and exactly what they do and do not give). In a hat basis with  $\eta(0) = \eta(\pi/2) = 0$ :

*Criticality.*  $\delta Q[\eta] = 0$  in every direction,  $\|\delta Q\|_\infty / h = 1.8 \cdot 10^{-8}$  by central differences on  $Q$  itself. So Romik's ODEs are the Euler–Lagrange equations of  $Q$ . (An analytic assembly of the

gradient first reported a nonzero value on  $[0, \beta)$  and  $[\pi/2, \pi/2 + \beta)$ ; that was a sign error on  $\delta(-\frac{1}{2}(\alpha_1^-)^2) = +\alpha_1^- \delta\alpha_1$ , caught by the finite-difference check.)

*Concavity, on  $\Sigma$ 's cell only.* With  $\Sigma$ 's own sign pattern  $E_1 = [0, \beta)$ ,  $E_2 = [0, \pi/2 - \beta)$ , the form (24) is negative definite:  $\sup \frac{1}{2} \delta^2 Q[\eta] / \|\eta\|_{L^2(0, \pi)}^2 = -0.7358, -0.7339, -0.7331$  at  $m = 32, 64, 128$ , against the exact mass matrix. Under Baek's injectivity condition, which is exactly  $E_1 = \emptyset$ ,  $E_2 = [0, \pi/2]$ , it is  $-0.8751$ . But (24) is *not* negative for every sign pattern: with  $E_1 = E_2 = [0.4, 1.2]$  it is  $+0.4123$ , and with  $E_1 = E_2$  a union of three intervals  $+0.4225$ . An earlier scan over intervals *anchored at 0* suggested that  $\tau_1 \leq \tau_2$  suffices; the unanchored cases refute that and the suggestion is **retracted**.

Since  $\alpha_1, \alpha_2$  are affine in  $H$ , each pointwise sign condition is a half-space and  $\Sigma$ 's cell is convex. Concavity plus criticality on a convex set gives that  $\Sigma$  maximises  $Q$  *on that cell*. Nothing global follows.

*What is missing, and it is worse than missing.* All of the above concerns  $Q$  alone. The inequality  $Q \geq |\text{sofa}|$ , which is the only reason  $Q$  is relevant, **fails**; see Proposition 40. All numbers in this remark are floating point.

**Remark 39** (The boundary of the cap, and where the sweep terminates). The density  $H + H''$  of the surface measure determines  $\partial C_2$  completely: it vanishes on  $(-\beta, \beta)$ , where  $\partial C_2$  is the single vertex  $P = (1, \frac{1}{2})$ ; equals  $\frac{1}{2}$  on  $(\pi/2 - \beta, \pi/2 + \beta)$ , where  $\partial C_2$  is a circular arc of radius exactly  $\frac{1}{2}$  centred at  $(1 - 2a_1, \frac{1}{2})$ ; has an atom of mass 1.167050 at  $\theta = \pi/2$ , the corridor ceiling  $y = 1$ , a facet; vanishes again on  $(\pi - \beta, \pi]$ , the second vertex; and is a smooth arc in between. The  $\rho$ -image of the ceiling facet is the corridor floor, the segment  $[-0.750575, 0.416475] \times \{0\}$ , whose left endpoint lies directly below the centre of the radius- $\frac{1}{2}$  arc.

This settles the geometry of the outer arm of Proposition 20. Because  $C_2$  is *convex*, the face-1 segment lies in  $C_2$  as soon as its far endpoint does, so the claim  $S_1 = \sigma$  reduces to a one-parameter containment: the endpoint  $x(t) = c_x(t) + \sigma(t) \sin t$  must lie in the floor facet. It runs monotonically from  $x(0) = 0$  to  $x(\pi/2) = 0.416475$ , the right end of the facet, and

$$x(\pi/2) = c_x(\pi/2) + \alpha_1(\pi/2) = (c_x(0) - \alpha_2(0)) + (\alpha_2(0) + \alpha_1(\pi/2) - (G(\pi/2) - 1))$$

is an algebraic identity, the right-hand side being the left end of the facet plus its length  $H'(\pi/2^+) - H'(\pi/2^-)$ . So the tangency at  $t = \pi/2$  is exact, not coincidental, and what remains of  $S_1 = \sigma$  is the monotonicity of the explicit function  $x$ .

**Proposition 40** ( $V$  over-estimates the niche, so  $Q$  is not an upper bound). *For every admissible  $H$ ,  $V(H) \geq |N(H)|$ . Consequently  $Q = |C_2| - 2V \leq |C_2| - 2|N|$ , and for the maximal sofa  $T_{\max} = C_2 \setminus (U \cup \rho U)$  with that cap,  $|T_{\max}| = |C_2| - 2|N| \geq Q$ . So  $Q$  is an upper bound for  $|T_{\max}|$  only where  $V = |N|$ .*

*Proof.*  $V$  is precisely the flux of the advancing part of  $\partial Q_t$ , by Lemma 19 and the Jacobian computation in Proposition 20. By Reynolds' transport theorem the area gained by the sweeping region  $N_T = \bigcup_{t \leq T} Q_t \cap C_2$  satisfies  $\frac{d}{dT} |N_T| \leq \int_{\text{advancing part of } \partial Q_T} v_n ds$ , because advance into already-swept territory contributes to the flux but gains no area. Integrating over  $T \in [0, \pi/2]$  gives  $|N| \leq V$ .  $\square$

**Remark 41** (The size of the gap, measured). Equality in Proposition 40 holds exactly when nothing is re-swept, which is what the three hypotheses of Proposition 20 assert, and which is why  $V(\Sigma) = |N(\Sigma)|$ . Perturbing  $H$  by a smooth bump  $\eta$  supported where  $H + H'' \geq \frac{1}{2}$  (so

convexity survives) and measuring  $|N|$  with a polygon oracle against  $V$  from the same  $H$ , the excess  $V - |N|$ , corrected for the oracle's known  $O(1/n)$  bias, is

| $\varepsilon$ | -0.20               | -0.10               | -0.05               | 0.05                | 0.10                | 0.20                |
|---------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| $V -  N $     | $5.5 \cdot 10^{-3}$ | $1.4 \cdot 10^{-4}$ | $4.2 \cdot 10^{-6}$ | $9.3 \cdot 10^{-5}$ | $8.4 \cdot 10^{-4}$ | $5.0 \cdot 10^{-3}$ |

strictly positive on *both* sides and growing faster than  $\varepsilon^2$ . So  $\Sigma$  is an isolated zero of  $V - |N|$ : the functional  $Q$  touches the true bound at  $\Sigma$  and falls strictly below it in every direction. Therefore  $Q$  cannot certify  $\Sigma$ , and the fact that  $Q$  is tight, critical and concave there (Remark 38) does not yield an optimality statement.

This locates the difficulty exactly. The repair is the one the one-corner argument performs: replace the exact flux  $V$  by the flux over a sub-family on which the sweep is *provably* injective, so that the integral is the area of a genuine subset of  $N$  and the inequality runs the right way, while the restriction is vacuous at  $\Sigma$ . Producing such a sub-family is the content of that argument's Chapters 3–6 and is not done here.

One reformulation that may help. A point  $p$  lies on the face-2 line at parameter  $t$  iff  $\phi(t) := \langle p, \nu_t \rangle - (G(t) - 1)$  vanishes, and with  $s(p, t) := F(t) - 1 - \langle p, \mu_t \rangle$  one computes  $\phi' = s - \alpha_2$  and  $s' = -\phi - \alpha_1$ , i.e. in complex form

$$z := s + i\phi, \quad z' = iz - (\alpha_1 + i\alpha_2).$$

So the whole face-2 sweep is governed by one linear ODE whose homogeneous part is an exact rotation,  $z(t) = \zeta(t) + e^{it}w_0$  with  $w_0$  an affine coordinate for  $p$ ; the sweep is injective iff no solution meets  $\{\phi = 0, 0 \leq s \leq \alpha_2\}$  twice, and at every such meeting  $\phi' = s - \alpha_2 \leq 0$ , so the crossing is always downward. A second meeting therefore forces an upward crossing in between, at  $s \geq \alpha_2$ . Since the total rotation over  $[0, \pi/2]$  is only  $\pi/2$ , the homogeneous part alone cannot supply that extra crossing; whether the forcing  $\zeta$  can is the open point.

**Proposition 42** (Monotonicity of the outer arm). *The face-1 segment's far endpoint has  $x$ -coordinate*

$$x(t) = c_x(t) + \sigma(t) \sin t = \frac{F(t) - 1}{\cos t}, \quad (25)$$

the  $x$ -intercept of the face-1 line, and  $\cos^2 t x'(t) = F'(t) \cos t + (F(t) - 1) \sin t$ . For  $\Sigma$ :

- on  $[0, \beta]$ , where  $A(t) \equiv P$  so  $F(t) = \cos t + \frac{1}{2} \sin t$ ,

$$x = 1 + \frac{1}{2} \tan t - \sec t, \quad \cos^2 t x' = \frac{1}{2} - \sin t > 0,$$

the last because  $\sin \beta = 1/(4a_1) = 0.2856 \dots < \frac{1}{2}$  by Theorem 14, i.e.  $\beta < \pi/6$ ;

- on  $(\pi/2 - \beta, \pi/2]$ , where  $F(t) - 1 = A \cos t + \frac{1}{2} \sin t - \frac{1}{2}$  with  $A = 1 - \frac{2}{3}a_1$ ,

$$x = A - \frac{1}{2} \tan\left(\frac{\pi}{4} - \frac{t}{2}\right), \quad \cos^2 t x' = \frac{1}{2}(1 - \sin t) > 0 \quad (t < \pi/2),$$

and in particular  $x(\pi/2) = A = 1 - \frac{2}{3}a_1 = 0.4164750917 \dots$ , which is exactly the right endpoint of the corridor floor facet of Remark 39;

- on  $[\beta, \pi/2 - \beta]$ , substituting  $u = t/2 - \pi/8$  and using  $-f_2/f_1 = \sqrt{2} - 1 = \tan(\pi/8)$ ,

$$\cos^2 t x' = W(u) := \frac{3K}{4} \sin u + \frac{K}{4} \cos 3u + \frac{1}{2} - \sin\left(2u + \frac{\pi}{4}\right), \quad K = f_1 \sqrt{4 - 2\sqrt{2}},$$

on  $|u| \leq \pi/8 - \beta/2$ .

Hence  $x$  is strictly increasing, given  $W > 0$ .

*Proof.* The face-1 line is  $\{\langle p, \mu_t \rangle = F(t) - 1\}$ , so its  $x$ -intercept is  $(F - 1)/\cos t$ ; that it agrees with  $c_x + \sigma \sin t$  is  $c_x + c_y \tan t = (F - 1)/\cos t$ , immediate from  $c = (F - 1)\mu_t + (G - 1)\nu_t$ . The three closed forms follow by substituting  $F(t) - 1 = \operatorname{Re}(e^{-it}z(t))$  with the complex forms of SOL1, SOL6, SOL5. For the middle phase the  $\kappa_6$  terms cancel and the triple-angle identities  $\sin(3t/2) = s(3c^2 - s^2)$ ,  $\cos(3t/2) = c(c^2 - 3s^2)$  with  $s = \sin \frac{t}{2}$ ,  $c = \cos \frac{t}{2}$  reduce the result to the stated three-term form.  $\square$

**Proposition 43** ( $W > 0$ , by a finite certificate).  $W > 0$  on  $|u| \leq u_0 := \pi/8 - \beta/2 = 0.2478721712\dots$ , so  $x$  is strictly increasing on all of  $[0, \pi/2]$  and  $S_1 = \sigma$ .

*Proof.*  $K = 1.3020516916172893452\dots$ , and  $3u_0 = 0.7436\dots < \pi$ , so  $\cos 3u$  is unimodal on  $[-u_0, u_0]$  with maximum at 0 and its minimum over any subinterval is attained at an endpoint. Also  $\sin u$  is increasing, and  $2u + \pi/4$  ranges over  $[0.2896\dots, 1.2812\dots] \subset [0, \pi/2]$ , so  $\sin(2u + \pi/4)$  is increasing. Therefore on  $[p, q] \subseteq [-u_0, u_0]$ ,

$$W \geq L(p, q) := \frac{3K}{4} \sin p + \frac{K}{4} \min(\cos 3p, \cos 3q) + \frac{1}{2} - \sin\left(2q + \frac{\pi}{4}\right).$$

Dividing  $[-u_0, u_0]$  into 32 equal pieces gives  $L > 0$  on every piece, with  $\min L = 6.13 \cdot 10^{-3}$ ; the computation is at 40 working digits, so the margin exceeds the arithmetic error by 37 orders of magnitude. (16 pieces do not suffice:  $\min L = -8.6 \cdot 10^{-3}$ .)  $\square$

**Remark 44.**  $W$  is in fact strictly decreasing, with  $W(u_0) = \frac{1}{2}(1 - \cos \beta) = 0.020828595599846555\dots$  at the right junction, matching the last phase's  $\frac{1}{2}(1 - \sin t)$  there to  $1.7 \cdot 10^{-41}$ , as  $C^1$  gluing requires. The three closed forms of Proposition 42 were also checked against numerical differentiation of the path to  $10^{-10}$ .

**Remark 45** (The identity at 61 digits). With all constants from their closed forms ( $u^3 + 3u = 2$ ,  $\beta = \arctan(u/2)$ ,  $a_1 = \sqrt{4 + u^2}/(4u)$ ,  $f_1$  from Proposition 6) and 60 working digits,

$$V = 0.184193197088768988251655564606\dots, \quad |C_2| = 2.013341612602978828172120476813\dots,$$

$$Q(\Sigma) - A_R^* = -1.56 \cdot 10^{-61}.$$

The two sides share no computation, so this rules out the possibility that the double-precision agreement of (23) was a rounding coincidence. It remains floating point, not an interval enclosure.

## 11 A convex subdomain on which the bound is valid

Proposition 40 says  $V \geq |N|$  with equality exactly when nothing is re-swept, i.e. when the combined sweep is injective. We now show that injectivity is governed by inequalities that are *linear* in  $H$ , so the set where the bound is valid is convex.

Each sweep lies on a line: face 2 at parameter  $t$  on  $\ell_2(t) = \{\langle p, \nu_t \rangle = G(t) - 1\}$ , face 1 on  $\ell_1(t) = \{\langle p, \mu_t \rangle = F(t) - 1\}$ . Two distinct lines meet in one point, so a double cover forces that point into both segments. Solving  $\ell_2(t) \cap \ell_2(t')$  for the coordinate  $s$  on  $\ell_2(t)$ , and using  $\langle \mu_t, \nu_{t'} \rangle = \sin(t - t')$ ,

$$s_{22}(t, t') = \frac{\langle c(t), \nu_{t'} \rangle - (G(t') - 1)}{\sin(t - t')}, \quad s_{11}(t, t') = \frac{\langle c(t), \mu_{t'} \rangle - (F(t') - 1)}{\sin(t' - t)}. \quad (26)$$



The numerator of  $s_{22}$  vanishes at  $t' = t$  with derivative  $-\alpha_2(t)$ , so  $s_{22}(t, t') \rightarrow \alpha_2(t)$ : nearby face-2 lines meet at the *far* end of the face-2 segment  $[0, \alpha_2]$ . Likewise  $s_{11}(t, t') \rightarrow \alpha_1(t)$ , the *near* end of the face-1 segment  $[\alpha_1^+, \sigma]$ .

**Definition 46.** Let  $\mathcal{K}'$  be the set of admissible  $H$  satisfying, for all  $0 \leq t' < t \leq \pi/2$ ,

$$(C22) \quad \langle c(t), \nu_{t'} \rangle - (G(t') - 1) \geq \alpha_2(t) \sin(t - t'), \quad (27)$$

$$(C11) \quad \langle c(t), \mu_{t'} \rangle - (F(t') - 1) \leq \alpha_1(t)^+ \sin(t' - t) \quad (\text{read with } \sin(t' - t) < 0), \quad (28)$$

together with the analogous condition for  $\ell_2(t) \cap \ell_1(t')$ .

**Proposition 47.**  $\mathcal{K}'$  is convex, and on  $\mathcal{K}'$  the combined sweep is injective, hence  $V = |N|$  and  $Q = |C_2| - 2|N| \geq |T|$  for every ambidextrous sofa  $T$  with cap data  $H$ .

*Proof.* By Lemma 18,  $c(t)$  is affine in  $H$ , hence so are  $\langle c(t), \nu_{t'} \rangle$ ,  $G(t') - 1$ ,  $F(t') - 1$ ,  $\alpha_1(t)$  and  $\alpha_2(t)$ ;  $\sin(t - t')$  does not depend on  $H$ . So (27) and (28) are linear inequalities in  $H$  and cut out a convex set. Given (27), for any pair  $t' < t$  the intersection point of  $\ell_2(t)$  and  $\ell_2(t')$  has  $s_{22}(t, t') \geq \alpha_2(t)$ , so it is not interior to the segment at  $t$ , and the pair contributes no double cover; (28) does the same for face 1 with the inequality reversed because the relevant endpoint is the near one. With the cross condition, no point is swept twice.

For  $V = |N|$ : every point of  $N$  has a first capture time  $\tau(p)$ , at which  $\partial Q_{\tau(p)}$  is advancing at  $p$  (otherwise  $p$  would have been captured earlier), so by Lemma 19 it lies on face 2 with  $s \leq \alpha_2$  or on face 1 with  $s \geq \alpha_1$ . Hence  $N$  is covered by the two sweeps, and the sweeps lie in  $N$  because an advancing face point belongs to  $Q_{t+\varepsilon}$ . Injectivity turns each  $\int |\det|$  into the area of its image, and the two images are disjoint, so the sum is  $|N|$ .  $\square$

**Remark 48** ( $\Sigma \in \mathcal{K}'$ , and  $\mathcal{K}'$  has nonempty interior). Measured on a grid of 401 parameters,  $\Sigma$  satisfies all three conditions: of the 158 802 ordered pairs with  $|t - t'|$  above the diagonal cutoff, *zero* give a point interior to both face-2 segments and *zero* interior to both face-1 segments, and of 58 081 cross pairs, *zero* land in both. The margins in (27), (28) vanish linearly as  $t' \rightarrow t$ , which is forced: that is the envelope limit. Away from the diagonal they are strictly positive.

$\mathcal{K}'$  is not a single point. For  $H = H_\Sigma + \varepsilon \sin(k\theta)$ , which respects the gauge for even  $k$ , all conditions still hold with zero violations at  $k = 2$  for  $|\varepsilon| \leq 3 \cdot 10^{-2}$  and at  $k = 4$  for  $|\varepsilon| \leq 10^{-2}$ , in both signs, and the two margins trade off smoothly. So  $\Sigma$  lies in  $\mathcal{K}'$  with low-frequency room around it.

The mechanism of failure is identified: for the compactly supported bump perturbations of Remark 41, where  $V - |N|$  was found positive, the face-2 condition continues to hold and it is the *face-1* condition that fails, by  $-4.7 \cdot 10^{-2}$  at  $\varepsilon = 0.05$  and  $-2.3 \cdot 10^{-1}$  at  $\varepsilon = 0.20$ . Those perturbations have large higher derivatives, which is what creates face-1 envelope self-intersections.

**Theorem 49** (Conditional). *Assume:*

- (i)  $Q(\Sigma) = A_R^*$ ,  $\delta Q(\Sigma) = 0$  in every admissible direction, and  $\delta^2 Q \preceq 0$  on the cell  $\mathcal{C}$  of support functions sharing  $\Sigma$ 's sign pattern;
- (ii) the three conditions of Definition 46 hold on  $\mathcal{K}' \cap \mathcal{C}$ ;
- (iii) competitors satisfy the niche separation  $M < \frac{1}{2}$  of Corollary 2.



Then  $|T| \leq A_R^*$  for every ambidextrous sofa  $T$  whose cap data lies in  $\mathcal{K}' \cap \mathcal{C}$ .

This statement is superseded by Theorem 84, which reaches the same conclusion on an explicitly described convex domain with all three hypotheses proved. It is retained only because its three hypotheses isolate what the later argument has to supply, and because Section 12 closes hypothesis (ii) for it. Its own label is HEURISTIC; Theorem 84 is PROVED.

*Proof.*  $\mathcal{C}$  is convex because  $\alpha_1, \alpha_2$  are affine in  $H$ , so each pointwise sign condition is a half-space;  $\mathcal{K}'$  is convex by Proposition 47; hence so is the intersection. By (i),  $Q$  is a concave function on that convex set with a critical point at  $\Sigma$ , so  $Q(H) \leq Q(\Sigma) = A_R^*$  throughout. By (ii) and Proposition 47,  $Q(H) \geq |T|$  there, using (iii) to get the factor 2 in  $|T| \leq |C_2| - 2|N|$ .  $\square$

## 12 One curvature condition implies injectivity

The conditions of Definition 46 were verified on a grid. We now prove them, for all pairs at once, from a single linear condition on the cap.

**Definition 50.** Say  $H$  satisfies (RC) if the absolutely continuous part of the surface measure obeys

$$H(\theta) + H''(\theta) \leq 1 \quad \text{for a.e. } \theta \in [0, \pi],$$

the only atoms being the corridor ceiling and floor facets at  $\theta = \pm\pi/2$ . Equivalently: the cap is nowhere flatter than a circle whose radius is the corridor width.

Since  $H + H'' \leq 1$  is linear in  $H$ , (RC) cuts out a convex set.

**Remark 51** ((RC) is classical). The bound  $h + h'' \leq R$  is not new. By the dual of Blaschke's rolling theorem, a convex body  $K$  slides freely inside a ball of radius  $R$  exactly when  $h_K + h_K'' \leq R$ , and that in turn says  $K$  is a Minkowski summand of  $RB$ ; see Schneider [4, §3.2]. Our (RC) is that condition at  $R = 1$ , the corridor width, *relaxed* to permit atoms at  $\theta = \pm\pi/2$ . Since Minkowski addition adds surface area measures, and a body whose surface measure is two equal atoms at  $\pm\pi/2$  is a horizontal segment,

$$(RC) \iff C_2 = [\text{a horizontal segment}] \oplus D, \quad D \text{ a Minkowski summand of the unit disc.}$$

For  $\Sigma$  the segment has length 1.1670498..., the corridor facet, and  $D$  has curvature radius at most 0.8388253.... What we claim is not the condition but its use: that it forces the niche sweep to be injective (Theorems 52, 56).

**Theorem 52.** *If  $H$  satisfies (RC), then (27) and (28) hold for every pair  $0 \leq t' < t \leq \pi/2$ . Hence each of the two sweeps is injective.*

*Proof.* Fix  $t$  and set  $\tau = t - t' \in (0, t]$ . For (27) put  $n(t') = \langle c(t), \nu_{t'} \rangle - (G(t') - 1)$  and

$$\Phi(\tau) = n(t - \tau) - \alpha_2(t) \sin \tau,$$

so that (27) reads  $\Phi \geq 0$ . Now  $n(t) = 0$  because  $\langle c(t), \nu_t \rangle = G(t) - 1$ , and  $n'(t') = -\langle c(t), \mu_{t'} \rangle - G'(t')$  equals  $-(F(t) - 1) - G'(t) = -\alpha_2(t)$  at  $t' = t$  by (18); hence  $\Phi(0) = \Phi'(0) = 0$ . Differentiating again and using  $\nu'' = -\nu$ ,

$$n''(t') = -\langle c(t), \nu_{t'} \rangle - G''(t') = -(n(t') + G(t') - 1) - G''(t'),$$

so with  $n(t - \tau) = \Phi(\tau) + \alpha_2(t) \sin \tau$  the  $\alpha_2$  terms cancel and

$$\Phi''(\tau) + \Phi(\tau) = 1 - (G + G'')(t - \tau) =: R(\tau). \quad (29)$$

With  $\Phi(0) = \Phi'(0) = 0$ , (29) gives  $\Phi(\tau) = \int_0^\tau \sin(\tau - u) R(u) du$ . Since  $\tau \leq t \leq \pi/2 < \pi$  we have  $\sin(\tau - u) \geq 0$  for  $u \in [0, \tau]$ , and  $R \geq 0$  by (RC) because  $G(s) = H(s + \pi/2)$ . Therefore  $\Phi \geq 0$ .

The ceiling atom of  $H$  sits at  $\theta = \pi/2$ , i.e. at  $t - u + \pi/2 = \pi/2$ , i.e.  $u = t$ . For  $\tau < t$  that is outside  $[0, \tau]$ , and for  $\tau = t$  it is the endpoint, where the kernel  $\sin(\tau - u)$  vanishes. So the atom contributes nothing.

For (28) put  $m(t') = \langle c(t), \mu_{t'} \rangle - (F(t') - 1)$  and  $\Psi(\tau) = m(t - \tau) + \alpha_1(t) \sin \tau$ . Then  $m(t) = 0$ ,  $m'(t) = \alpha_1(t)$  by (17), so  $\Psi(0) = \Psi'(0) = 0$ , and  $\mu'' = -\mu$  gives in the same way

$$\Psi''(\tau) + \Psi(\tau) = 1 - (F + F'')(t - \tau),$$

whence  $\Psi(\tau) = \int_0^\tau \sin(\tau - u) [1 - (H + H'')(t - u)] du \geq 0$ . Multiplying by  $-\sin \tau < 0$  turns  $\Psi \geq 0$  into  $s_{11} \leq \alpha_1$ , which is (28). Here the atom needs  $t - u = \pi/2$ , i.e.  $u = t - \pi/2 \leq 0$ , reachable only at  $t = \pi/2$ , where  $\sigma(\pi/2) = \alpha_1(\pi/2)$  makes the face-1 segment empty and the condition vacuous.  $\square$

**Remark 53** (Verification). For  $\Sigma$  the absolutely continuous part of  $H + H''$  attains its maximum at  $\theta = \beta$  and at  $\theta = \pi - \beta$ , where it equals

$$\frac{3}{4} f_1 \sqrt{4 - 2\sqrt{2} \cos(\frac{\beta}{2} + \frac{\pi}{8})} = 0.8388253 \dots$$

in closed form, the value used in Proposition 103, so (RC) holds for  $\Sigma$  with margin  $0.1611747 \dots$ ; second differences approach it from below and converge too slowly to confirm more than four digits, so the closed form is what is quoted; the atom at  $\theta = \pi/2$  has mass  $1.1670498 \dots$ , the ceiling facet length. The reduction (29) was checked against direct evaluation of  $n(t - \tau) - \alpha_2(t) \sin \tau$  at several  $(t, \tau)$ , agreeing to  $2.4 \cdot 10^{-5}$ , the finite-difference floor for  $H''$ .

Theorem 52 also explains the failure mode of Remark 48. Across twelve perturbations, (RC) holds exactly when (28) holds: the compactly supported bumps give  $\max(H + H'')_{ac} = 1.08, 1.45, 2.18, 2.67, 4.58, 8.41$  and all violate (28);  $H_\Sigma + \varepsilon \sin(2\theta)$  at  $|\varepsilon| = 0.03$  gives  $0.888$  and holds;  $H_\Sigma + \varepsilon \sin(4\theta)$  gives  $0.976$  at  $|\varepsilon| = 0.01$  and holds,  $1.267$  at  $|\varepsilon| = 0.03$  and fails. The agreement is exact on both sides of the transition.

**Remark 54** (What Theorem 52 does and does not give). It replaces hypothesis (ii) of Theorem 49, for the two self-intersection conditions, by (RC): a single explicit linear inequality with a geometric reading. The cross condition, that  $\ell_2(t) \cap \ell_1(t')$  avoid both segments, is not covered; it is measured to hold at  $\Sigma$  (0 of 58 081 pairs) and remains a hypothesis. So the hypothesis class of Theorem 49 may be replaced by  $\{(RC)\} \cap \{\text{cross}\} \cap \mathcal{C}$ , an intersection of two convex sets with one measured condition.

**Remark 55** (The first variation, pointwise). Hypothesis (i) of Theorem 49 contains  $\delta Q(\Sigma) = 0$ . Integrating every  $\eta'$  in  $\delta Q$  by parts and collecting the coefficient of  $\eta(\theta)$  gives the Euler–Lagrange equations of  $Q$ :

$$\begin{aligned} H + H'' &= \alpha_2^+ + (\sigma - \alpha_1) \tan \theta - (\sigma - \alpha_1)' + (\alpha_1^-)' & (0 < \theta < \pi/2), \\ H + H'' &= -(\alpha_2^+)'(s) + \alpha_1^-(s), & s = \theta - \pi/2 \quad (\pi/2 < \theta < \pi). \end{aligned}$$

Evaluated on  $\Sigma$  at eighteen values of  $\theta$  spanning all six regions, the residual is at most  $9.4 \cdot 10^{-6}$ , which is the finite-difference floor. So  $\delta Q(\Sigma) = 0$  has an explicit pointwise form; proving it amounts to substituting the closed forms of SOL1, SOL5, SOL6 phase by phase, and is not carried out here.

### 13 The cross condition, the first variation, and a Gårding bound

**Theorem 56.** *(RC) implies the cross condition: for all  $t, t'$  the point  $\ell_2(t) \cap \ell_1(t')$  is not interior to both sweep segments. Consequently, under (RC) the combined sweep is injective and  $V = |N|$ , so  $Q = |C_2| - 2|N| \geq |T|$ .*

*Proof.* Work in the frame at  $t$  and write  $p = c(t) + a\mu_t + b\nu_t$ . The equation  $\langle p, \nu_t \rangle = G(t) - 1$  forces  $b = 0$ , so  $p$  moves along  $\ell_2(t)$  with  $s_2 = -a$ ; the equation  $\langle p, \mu_{t'} \rangle = F(t') - 1$  then gives, for  $t' = t - \tau$ ,

$$s_2 = \frac{m(t - \tau)}{\cos \tau}, \quad s_1 = -\Phi(\tau) - (\alpha_2(t) - s_2) \sin \tau,$$

using  $\langle \mu_t, \nu_{t-\tau} \rangle = \sin \tau$  and  $n(t - \tau) = \Phi(\tau) + \alpha_2(t) \sin \tau$ . If  $\tau > 0$  and  $p$  is interior to the face-2 segment, then  $0 < s_2 < \alpha_2(t)$ , and  $\Phi \geq 0$  by Theorem 52 with  $\sin \tau \geq 0$  because  $\tau \leq \pi/2$ ; hence  $s_1 < 0 \leq \alpha_1(t')^+$  and  $p$  lies below the face-1 segment.

For  $t' > t$  work in the frame at  $t'$ : the same computation with the roles of the two families exchanged gives, for  $t = t' - \tau'$  with  $\tau' > 0$ ,

$$s_2 = -\Psi(\tau') - (s_1 - \alpha_1(t')) \sin \tau',$$

so if  $p$  is interior to the face-1 segment then  $s_1 > \alpha_1(t')^+ \geq \alpha_1(t')$  and  $s_2 < 0$ , outside  $[0, \alpha_2(t)]$ . At  $t' = t$  the two lines are perpendicular and meet at  $c(t)$ , where  $s_1 = s_2 = 0$ , interior to neither. Both displayed identities were checked against a direct linear solve, agreeing to  $10^{-16}$ .  $\square$

So hypothesis (ii) of Theorem 49 may be replaced throughout by (RC).

**Theorem 57.**  $\delta Q(\Sigma) = 0$ . *Two things are needed and both hold: the Euler–Lagrange equations of Remark 55 hold identically on each of the six regions, as trigonometric identities in  $a_1, f_1, f_2$  with no relation among the constants used; and the boundary term of  $\delta Q$  vanishes, because  $H'(\pi) = -\frac{1}{2}$ .*

*Proof.* On each phase,  $F(t) - 1 = \operatorname{Re}(e^{-it}z(t))$  and  $G(t) - 1 = \operatorname{Im}(e^{-it}z(t))$  evaluate to

$$\begin{array}{ll} \text{SOL1} & F - 1 = \cos t + \frac{1}{2} \sin t - 1, & G - 1 = (2a_1 - 1) \sin t + \frac{1}{2} \cos t - \frac{1}{2}, \\ \text{SOL6} & F - 1 = f_1 \cos \frac{t}{2} + f_2 \sin \frac{t}{2} - 1 + \kappa \cos t + \frac{1}{2} \sin t, & G - 1 = -f_2 \cos \frac{t}{2} + f_1 \sin \frac{t}{2} - 1 + \frac{1}{2} \cos t - \kappa \sin t, \\ \text{SOL5} & F - 1 = (1 - \frac{2}{3}a_1) \cos t + \frac{1}{2} \sin t - \frac{1}{2}, & G - 1 = (\frac{8}{3}a_1 - 1) \sin t + \frac{1}{2} \cos t - 1, \end{array}$$

with  $\kappa = 1 - \frac{4}{3}a_1$ . Substituting these into  $\alpha_1 = G - 1 - F'$ ,  $\alpha_2 = F - 1 + G'$ ,  $\sigma = (F - 1) \tan t + G - 1$  and into the two Euler–Lagrange equations, with the sign pattern of each region fixed ( $\alpha_2 > 0$  except on the last phase,  $\alpha_1 < 0$  only on the first), every residual reduces identically to 0.

For the boundary term, integrating the  $\eta'$  of  $\delta|C_2| = 2 \int_0^\pi (H\eta - H'\eta') d\theta - \eta(0) - \eta(\pi)$  by parts and using  $\eta(0) = 0$  leaves  $-(2H'(\pi) + 1)\eta(\pi)$  beyond the pointwise equations. The variation  $\delta V$  contributes nothing there: its only boundary term at  $\theta = \pi$  carries the factor  $\alpha_2^+(\pi/2)$ , and  $\alpha_2(\pi/2) < 0$ , so the clamp kills it; every other boundary sits at  $\theta = 0$  or  $\pi/2$ , where  $\eta$  vanishes. Finally the boundary point of  $C_2$  with outer normal  $\mu_\pi = (-1, 0)$  is  $H(\pi)\mu_\pi + H'(\pi)\nu_\pi = (-H(\pi), -H'(\pi))$ , so its height is  $-H'(\pi)$ ; since  $C_2$  is  $\rho$ -symmetric about  $y = \frac{1}{2}$  its leftmost point is  $\rho$ -fixed when unique, forcing  $H'(\pi) = -\frac{1}{2}$  and the boundary term to vanish. The same mechanism at  $\theta = 0$  gives  $H'(0) = \frac{1}{2}$ : both extreme points of the cap sit at height  $\frac{1}{2}$ .  $\square$

**Remark 58.** The computation is a computer algebra verification, six residuals, each returning 0 with  $a_1, f_1, f_2$  free. It was controlled against five deliberately perturbed variants of the middle-phase equation (sign flip on the tan term, each of the two terms dropped,  $\alpha_1$  and  $\alpha_2$  exchanged,  $H + H''$  replaced by  $H - H''$ ), all of which return nonzero; and the three pairs of phase formulas were checked against the reference path to  $2.2 \cdot 10^{-16}$ . The content of Theorem 57 is that Romik's ODE families are exactly the critical-point equations of  $Q$ .

**Remark 59** (Towards  $\delta^2 Q \leq 0$ ). The remaining part of hypothesis (i) is the sign of the second variation. The step that makes it accessible is an exact algebraic cancellation. On  $[0, \beta]$  write  $p = \eta'(t)$ ,  $q = \eta(t)$ ,  $r = \eta(t + \pi/2)$ ,  $T = \tan t$ ; the three terms of (24) that carry  $\eta'(t)$  combine as

$$-p^2 - (qT + p)^2 + (r - p)^2 = -(p + qT + r)^2 + 2r^2 + 2qTr, \quad (30)$$

so the added term is absorbed exactly and the remainder carries no derivative of  $\eta$ . What is left is a lower-order estimate. With  $\eta(0) = \eta(\pi/2) = 0$  the relevant Poincaré constants are 4 on  $[0, \pi/2]$ , 1 on  $[\pi/2, \pi]$  (Dirichlet–Neumann,  $\eta(\pi)$  free) and  $(\pi/2\beta)^2 = 29.41$  on  $[0, \beta]$ . Retaining a fraction  $\delta$  of  $-\int_0^\beta \eta'^2$  before applying (30) and splitting  $(qT + r)^2$  with a parameter  $\kappa$ , the coefficient of  $\int_0^\beta \eta^2$  is

$$\left(\frac{1+\kappa}{1-\delta} - 1\right) \tan^2 \beta + 1 - \delta \left(\frac{\pi}{2\beta}\right)^2 = -1.83 \dots < 0 \quad (\delta = \frac{1}{10}, \kappa = 1),$$

and the  $r^2$  remainder, of coefficient  $3.22 \dots$ , is absorbed by the  $E_2$  term of (24), which supplies  $-\frac{1}{2} \int_{\pi/2}^{\pi-\beta} \eta'^2$ : since  $\eta(\pi/2) = 0$  gives  $\int_{\pi/2}^{\pi-\beta} \eta^2 \leq (\beta^2/2) \int \eta'^2$ , the requirement is  $3.22 \cdot 0.0420 = 0.135 \leq \frac{1}{2}$ . So every ingredient of a Gårding inequality  $\delta^2 Q \leq -c \|\eta\|_{L^2}^2$  is in place; Theorem 60 assembles it and Theorem 87 sharpens the constant to  $\frac{2}{3}$ .

The sharp value is  $\frac{1}{2}c = 0.7309566 \dots$  by Theorem 88. A P1 finite-element computation approaches it from above, as Rayleigh–Ritz must:  $-0.735818, -0.733924, -0.733086, -0.732285, -0.731311, -0.7312$  at  $m = 32, \dots, 2048$ .

## 14 The area identity

The concavity of  $Q$  is established in Section 20; what follows here is the earlier and weaker route to the constant, retained for its elementary ingredients, and the closed-form evaluation of  $Q(\Sigma)$ .

**Remark 60** (An earlier route to the constant). Superseded by Theorem 88, which gives the exact constant, and by Theorem 91, which gives concavity on the whole ordered anchored family. It is retained because its ingredients are elementary and its failure locates the loss. On the cell  $\mathcal{C}$  of  $\Sigma$ 's sign pattern,  $\delta^2 Q \leq 0$ , with equality only at  $\eta = 0$ . Explicitly, writing  $J_1 = \int_0^\beta \eta^2$ ,  $J_2 = \int_\beta^{\pi/2} \eta^2$ ,  $J_3 = \int_{\pi/2}^\pi \eta^2$ ,

$$\frac{1}{2} \delta^2 Q[\eta] \leq -1.6013 J_1 - 0.3710 J_2 + 0 \cdot J_3 - 0.005 \int_{\pi/2}^{\pi-\beta} \eta'^2. \quad (31)$$

*Proof of the statement in Remark 60.* On  $[0, \beta]$  retain a fraction  $\delta$  of  $-\int_0^\beta \eta'^2$  and apply (30) with the remaining  $-(1 - \delta)\eta'^2$ ; completing the square in  $p = \eta'(t)$  and discarding the resulting negative square,

$$-(1 - \delta)p^2 - (qT + p)^2 + (r - p)^2 \leq \frac{(qT + r)^2}{1 - \delta} - q^2 T^2 + r^2 \leq A q^2 T^2 + B r^2,$$

with  $A = \frac{1+\kappa}{1-\delta} - 1$ ,  $B = \frac{1+1/\kappa}{1-\delta} + 1$  from  $(qT+r)^2 \leq (1+\kappa)q^2T^2 + (1+1/\kappa)r^2$ , and  $T = \tan t \leq \tan \beta$  on  $[0, \beta)$ . The three Poincaré constants for  $\eta(0) = \eta(\pi/2) = 0$  are  $P_1 = (\pi/2\beta)^2 = 29.4135$  on  $[0, \beta]$ ,  $P_2 = (\pi/2(\pi/2 - \beta))^2 = 1.5035$  on  $[\beta, \pi/2]$  and  $P_3 = 1$  on  $[\pi/2, \pi]$ , each with a Dirichlet endpoint and a free one. Finally  $-(\eta(t) + \eta'(t + \pi/2))^2 \leq -\lambda \eta'(t + \pi/2)^2 + \frac{\lambda}{1-\lambda} \eta(t)^2$  turns the  $E_2$  term into  $-\lambda \int_{\pi/2}^{\pi-\beta} \eta'^2 + \frac{\lambda}{1-\lambda} (J_1 + J_2)$ , and  $\eta(\pi/2) = 0$  gives  $\int_{\pi/2}^{\pi/2+\beta} \eta^2 \leq (\beta^2/2) \int_{\pi/2}^{\pi-\beta} \eta'^2$ , so the  $B r^2$  remainder is absorbed as soon as  $\lambda \geq B\beta^2/2$ . Collecting,

$$\begin{aligned} \text{coefficient of } J_1 &= A \tan^2 \beta - \delta P_1 + 1 + \frac{\lambda}{1-\lambda}, & \text{of } J_2 &= 1 - P_2 + \frac{\lambda}{1-\lambda}, \\ \text{of } J_3 &= 1 - P_3 = 0, & \text{of } \int_{\pi/2}^{\pi-\beta} \eta'^2 &= B\beta^2/2 - \lambda. \end{aligned}$$

At  $\delta = \frac{1}{10}$ ,  $\kappa = 2$ ,  $\lambda = 0.117$  these are  $-1.6013$ ,  $-0.3710$ ,  $0$  and  $-0.005$ , giving (31). For strictness, equality forces  $\eta \equiv 0$  on  $[0, \pi/2]$  and  $\eta$  the first Dirichlet–Neumann mode on  $[\pi/2, \pi]$ ; but with  $\eta \equiv 0$  on  $[0, \pi/2]$  the  $E_2$  term contributes  $-\int_0^{\pi/2-\beta} \eta'(t + \pi/2)^2 < 0$ .  $\square$

The bound  $\frac{1}{2} \delta^2 Q \leq -0.3710 \|\eta\|_{L^2(0, \pi/2)}^2$  that (31) yields is about half of the sharp  $-0.7309566 \dots$  for the full interval, the shortfall being the  $J_3$  coefficient that (31) leaves at zero. Theorems 87 and 88 recover it.

**Proposition 61.**  $\sigma - \alpha_1 = \cos t x'(t)$  on each phase, with  $x = (F - 1)/\cos t$  as in (25). Hence the middle term of (20) is  $\frac{1}{2} \int \cos^2 t x'(t)^2 dt$ , an energy of the intercept path.

*Proof.*  $\sigma - \alpha_1 = (F - 1) \tan t + F'$  by (17) and (19), and  $\cos^2 t x' = F' \cos t + (F - 1) \sin t$  by (25); dividing by  $\cos t$  gives the claim. Verified symbolically on SOL1, SOL5 and SOL6, residual 0 in each case, which also recovers the first and third closed forms of Proposition 42.  $\square$

**Remark 62** (The area identity). Since  $u = 2 \tan \beta$ , Romik’s area is

$$A_R^* = u^2 + 1 + \arctan(u/2) = 1 + 4 \tan^2 \beta + \beta.$$

Using Proposition 61 to make every integrand elementary, the six blocks of  $|C_2|$  and the three blocks of  $V$  evaluate at 50 working digits to  $|C_2| = 2.0133416126029788281721 \dots$ ,  $V = 0.1841931970887689882517 \dots$ , and

$$Q(\Sigma) - (1 + 4 \tan^2 \beta + \beta) = -2.7 \cdot 10^{-51}.$$

A symbolic evaluation of the same integrals was attempted and did not terminate; the identity therefore remains HEURISTIC, now against a closed-form target and with a phase-by-phase decomposition.

## 15 $\Sigma$ satisfies (RC), and $Q(\Sigma) = A_R^*$

**Proposition 63.**  $\Sigma$  satisfies (RC). The surface-measure density is, block by block,

$$H + H'' = \begin{cases} 0 & \theta \in [0, \beta) \text{ and } \theta \in (\pi - \beta, \pi], \\ \frac{1}{2} & \theta \in (\pi/2 - \beta, \pi/2) \text{ and } \theta \in (\pi/2, \pi/2 + \beta), \\ \frac{3K}{4} \cos\left(\frac{\theta}{2} + \frac{\pi}{8}\right) & \theta \in [\beta, \pi/2 - \beta], \\ \frac{3K}{4} \sin\left(\frac{\theta - \pi/2}{2} + \frac{\pi}{8}\right) & \theta \in [\pi/2 + \beta, \pi - \beta], \end{cases} \quad K = f_1 \sqrt{4 - 2\sqrt{2}},$$

plus the ceiling atom at  $\theta = \pi/2$ . Hence

$$\max(H + H'')_{\text{ac}} = \frac{3K}{4} \cos\left(\frac{\beta}{2} + \frac{\pi}{8}\right) = 0.8388253494 \dots < 1,$$

and already the amplitude satisfies  $\frac{3}{4}K = 0.9765387687 \dots < 1$ , so (RC) holds as soon as

$$f_1^2 \leq \frac{16}{9(4 - 2\sqrt{2})} = 1.5174 \dots, \quad \text{against } f_1^2 = 1.4470 \dots \quad (32)$$

*Proof.* Each block of  $H$  is read off the three closed forms of Theorem 57. On the first phase  $H = \cos \theta + \frac{1}{2} \sin \theta$ , so  $H + H'' = 0$ ; the shifted last phase is the same. On the last phase  $H = (1 - \frac{2}{3}a_1) \cos \theta + \frac{1}{2} \sin \theta + \frac{1}{2}$  and on the shifted first phase  $H = (2a_1 - 1) \sin \theta + \frac{1}{2} \cos \theta + \frac{1}{2}$ , each giving  $H + H'' = \frac{1}{2}$ . On the middle phase only the half-angle terms survive differentiation twice, contributing a factor  $1 - \frac{1}{4}$ , so  $H + H'' = \frac{3}{4}(f_1 \cos \frac{\theta}{2} + f_2 \sin \frac{\theta}{2})$  and, on the shifted middle phase,  $\frac{3}{4}(f_1 \sin \frac{\theta}{2} - f_2 \cos \frac{\theta}{2})$ . Both have amplitude  $\frac{3}{4}\sqrt{f_1^2 + f_2^2} = \frac{3}{4}f_1\sqrt{4 - 2\sqrt{2}}$  by  $f_2 = (1 - \sqrt{2})f_1$ , and phase  $\pi/8$  because  $-f_2/f_1 = \sqrt{2} - 1 = \tan(\pi/8)$ . On  $[\beta, \pi/2 - \beta]$  the argument  $\frac{\theta}{2} + \frac{\pi}{8}$  runs over  $[0.5375, 1.0333] \subset [0, \pi/2]$ , where cosine decreases, so the maximum is at  $\theta = \beta$ .  $\square$

Criterion (32) is strictly stronger than the separation criterion  $f_1^2 < (2 + \sqrt{2})/2 = 1.7071 \dots$  of Theorem 8: within the family in which  $f_1$  is the free parameter, (RC) implies  $M < \frac{1}{2}$ .

**Theorem 64.**  $Q(\Sigma) = A_R^*$ .

*Proof.*  $\Sigma$  satisfies (RC) by Proposition 63, so  $V(\Sigma) = |N(\Sigma)|$  by Theorems 52 and 56. The niches are disjoint by Theorem 8, and  $C_2$  is  $\rho$ -invariant, so  $|\Sigma| = |C_2| - |N| - |\rho N| = |C_2| - 2|N|$ . Hence  $Q(\Sigma) = |C_2| - 2V(\Sigma) = |C_2| - 2|N| = |\Sigma| = A_R^*$ .  $\square$

No integral is evaluated. Independently, with  $A_R^* = 1 + 4 \tan^2 \beta + \beta$  and the integrands made elementary by Proposition 61, the two sides agree to  $2.7 \cdot 10^{-51}$  at 50 working digits, which is a consistency check on the whole chain rather than a proof of it.

**Remark 65** (Regression against the one-corner problem). The construction does *not* recover Gerver's sofa, and this should be stated plainly. Applying the same machinery to Gerver's rotation path:

- Gerver satisfies the one-corner injectivity condition everywhere:  $\alpha_1 > 0$  and  $\alpha_2 > 0$  on all of  $[0, \pi/2]$ , as expected.
- (RC) *fails* for Gerver's cap:  $\max(H + H'')_{\text{ac}} \geq 1.399$  near  $\theta = \pi$ .
- Gerver's face-2 sweep is genuinely not injective in this decomposition: 33 002 of 158 802 tested pairs meet interior to both segments, with  $\min(s_{22} - \alpha_2) = -0.0944$ . The face-1 sweep is injective.
- Consistently,  $V = 0.6426146$  against  $|N| = |C| - |S| = 0.6406$  to  $0.6412$  at 481 and 721 constraints, i.e.  $V > |N|$  as Proposition 40 requires.

So (RC) and the one-corner injectivity condition are independent:  $\Sigma$  satisfies the first and not the second, Gerver the second and not the first. By Remark 51 the difference has a classical reading:  $\Sigma$ 's cap is a corridor facet plus a body that slides freely inside the unit disc, whereas part of Gerver's cap boundary is *flatter than the corridor is wide*, its curvature radius reaching at least 1.399. The decomposition  $N = W_2 \sqcup W_1^{\text{out}}$  is therefore not the one the one-corner argument uses, and the fact that it fails on the solved case is a standing caution about the framework, recorded rather than explained.

**Remark 66** (The separation hypothesis for competitors). Hypothesis (iii) remains open. Over 250 random perturbations  $H_\Sigma + \sum_{k=1}^3 c_k \sin(2k\theta)$  with  $|c_k| \leq a/k^2$  for  $a = 0.02, 0.05, 0.10$ , every  $H$  satisfying (RC) had  $M \leq 0.3976$ , against the required  $M < \frac{1}{2}$ ; the (RC)-satisfying fraction fell from 250/250 to 34/250 as  $a$  grew. No counterexample and no implication.

## 16 Two negative findings

**Remark 67** (The mirror decomposition does not rescue Gerver). The decomposition of Proposition 20 has a mirror image under  $t \mapsto \pi/2 - t$ ,  $\mu \leftrightarrow \nu$ : sweep the inner part of face 1 and the outer part of face 2, the latter reaching the floor at distance  $\sigma_2 = c_y/\sin t$ . For Gerver both give the same value,  $0.6426144\dots$ , because Gerver's rotation path is itself symmetric under  $t \mapsto \pi/2 - t$ ; both exceed  $|N| \approx 0.641$ . So the failure of Remark 65 is not an artefact of orientation.

The structural reason is visible in Proposition 63.  $\Sigma$ 's cap has  $H + H'' = 0$  on  $[0, \beta)$  and  $(\pi - \beta, \pi]$ , i.e. *vertices*, because  $\Sigma$ 's outer phases have constant contact arcs pinned at  $(1, \frac{1}{2})$ . Gerver's cap has no such vertices: there  $H + H''$  reaches at least 1.399 near  $\theta = \pi$ . The degeneracy that makes  $\Sigma$  hard for local analysis is exactly what makes (RC) hold for it.

**Remark 68** (A false bound, retracted). The following was attempted for hypothesis (iii) and is **false**. Solving  $H'' + H = W$  with  $H(0) = 1$  gives  $H(\theta) = \cos \theta + H'(0) \sin \theta + \int_0^\theta \sin(\theta - s)W(s) ds$ , and substituting into  $c_y(t) = (H(t) - 1) \sin t + (H(t + \pi/2) - 1) \cos t$  yields

$$c_y(t) = H'(0) - \sin t - \cos t + \sin t \int_0^t \sin(t - s)W ds + \cos t \int_0^{t+\pi/2} \cos(t - s)W ds.$$

Both kernels are non-negative on their ranges, so “ $W \leq 1$ ” would give  $c_y \leq H'(0)$ , and for a  $\rho$ -symmetric cap  $H'(0) = \frac{1}{2} + \frac{1}{2}$  (vertical right edge), apparently yielding  $M \leq \frac{1}{2}$ .

The error is that  $W$  is *not*  $\leq 1$  as a measure: it carries the ceiling atom of mass 1.1670... at  $s = \pi/2$ , and here that atom meets the kernel  $\cos(t - \pi/2) = \sin t$ , which does not vanish. The corrected bound is  $M \leq H'(0) + \frac{1}{2} \cdot 1.1670 = 1.083\dots$ , useless against the required  $\frac{1}{2}$ .

This does *not* affect Theorem 52: there the same atom lands at  $u = t$ , which is outside the integration range when  $\tau < t$  and is the endpoint where the kernel  $\sin(\tau - u)$  vanishes when  $\tau = t$ ; and the face-1 atom needs  $u = t - \pi/2 \leq 0$ . The placement of the atom relative to the kernel is what distinguishes the two arguments, and it was checked in both.

Hypothesis (iii) therefore remains open.

## 17 A finite-angle relaxation, and what it can give

Everything above is local. We record here the one construction in this note that bears on the *global* problem, together with an honest account of how far it has been carried.

**Proposition 69** (Finite-angle relaxation). *Let  $S$  be an ambidextrous moving sofa. For any finite  $F_1, F_2 \subset [0, \pi/2]$ ,*

$$|S| \leq \sup \left| \bigcap_{t \in F_1} L(t, c_t) \cap \bigcap_{s \in F_2} R(s, d_s) \right|,$$



the supremum being over all placements  $c_t, d_s \in \mathbb{R}^2$ , where  $L(t, c)$  is the left-turning hallway rotated by  $t$  with inner corner  $c$ , and  $R(s, d)$  the right-turning one, that is, the mirror image rotated by  $s$  and placed at  $d$ .

*Proof.*  $S$  lies in the hallway at every instant of each of its two motions, so it lies in the intersection over any finite subset of instants, for the placements those instants realise. Dropping the constraints linking the placements to one another enlarges the right-hand side.  $\square$

The bound this yields depends on  $F_1, F_2$  and requires the supremum, which is a global maximisation over  $2(|F_1| + |F_2|) - 2$  real parameters. Two facts are recorded.

**Proposition 70** (A limitation at three angles). *For  $F_1 = F_2 = \{0, \pi/4, \pi/2\}$  the supremum in Proposition 69 is at least  $2\sqrt{2} - 1 = 1.8284271\dots$ , so no bound below that value is obtainable at those angles.*

*Proof.* An explicit placement attaining it is recorded in the accompanying computation; exhibiting one placement suffices.  $\square$

Three objects already force this value, the surrounding structure is classical, and the constant has an exact geometric meaning.

A moving sofa lies in the incoming straight corridor before it reaches the corner, so  $S \subseteq \mathcal{S}$  for a strip  $\mathcal{S}$  of width 1; an ambidextrous one meets a  $\pi/4$  hallway of each handedness. Hence

$$|S| \leq \sup_{c,d} |\mathcal{S} \cap L_{\pi/4}(c) \cap R_{\pi/4}(d)|. \quad (33)$$

With only  $L_{\pi/4}$  this is Hammersley's bound: each arm of that hallway is a strip of width 1 at  $\pi/4$  to  $\mathcal{S}$ , meeting it in a parallelogram of area  $1/\sin(\pi/4) = \sqrt{2}$ , so  $|S| \leq 2\sqrt{2}$ . The second handedness changes the geometry, and the following is what the constant  $2\sqrt{2} - 1$  measures.

**Lemma 71** (Cap lemma). *Let  $T$  be a unit square whose diagonals are horizontal and vertical, and  $\mathcal{S}$  a horizontal strip of width 1. Then  $|T \cap \mathcal{S}| \leq \sqrt{2} - \frac{1}{2}$ , with equality when the strip is centred on  $T$ .*

*Proof.*  $T$  has vertical extent  $\sqrt{2} > 1$ , and its horizontal cross-section at height  $y$  from the centre has length  $\sqrt{2} - 2|y|$  for  $|y| \leq \sqrt{2}/2$ . The cap above height  $h$  has area  $\int_h^{\sqrt{2}/2} (\sqrt{2} - 2y) dy = \frac{1}{2} - \sqrt{2}h + h^2$ , decreasing in  $h$  on  $[0, \sqrt{2}/2]$ , so a strip of width 1 removes least when centred, at  $h = \frac{1}{2}$ :

$$|T \cap \mathcal{S}| \leq 1 - 2\left(\frac{1}{2} - \frac{\sqrt{2}}{2} + \frac{1}{4}\right) = \sqrt{2} - \frac{1}{2}. \quad \square$$

All four arms of the two  $\pi/4$  hallways are strips of width 1 with normal  $(1,1)/\sqrt{2}$  or  $(-1,1)/\sqrt{2}$ . Two arms from different hallways with perpendicular normals meet in exactly such a square, so each perpendicular pair contributes at most  $\sqrt{2} - \frac{1}{2}$  by Lemma 71, and

$$2(\sqrt{2} - \frac{1}{2}) = 2\sqrt{2} - 1.$$

This is not merely numerology: the maximising configuration found for (33) splits into exactly two connected components, each of area  $0.914213562 = \sqrt{2} - \frac{1}{2}$ , one for each perpendicular pair, with the two parallel pairs empty. The corners sit at heights 0.995169 and 0.004831, symmetric about the midline.

The conjecture below reduces to a one-dimensional problem. Let  $\mu = (1,1)/\sqrt{2}$ ,  $\nu = (-1,1)/\sqrt{2}$  and use the orthonormal coordinates  $u = \langle p, \mu \rangle$ ,  $v = \langle p, \nu \rangle$ ; write  $a = \langle c, \mu \rangle$ ,  $a' = \langle c, \nu \rangle$ ,  $b = \langle d, \mu \rangle$ ,  $b' = \langle d, \nu \rangle$ .

**Lemma 72** (Rotated coordinates). *In these coordinates both hallways are axis-aligned and*

$$L_{\pi/4}(c) \cap R_{\pi/4}(d) = ([b-1, a+1] \times [b'-1, a'+1]) \setminus (\{u < a, v < a'\} \cup \{u > b, v > b'\}),$$

*a rectangle less two opposite quadrants, while  $\mathcal{S} = \{0 \leq u+v \leq \sqrt{2}\}$ . Consequently, with  $s = u+v$  and  $t = u-v$ ,*

$$|\mathcal{S} \cap L_{\pi/4}(c) \cap R_{\pi/4}(d)| = \frac{1}{2} \int_0^{\sqrt{2}} \ell(s) ds, \quad (34)$$

*where  $\ell(s)$  is the length of  $[\max(p, r) - 2, \min(q, w) + 2] \setminus ((p, q) \cup (r, w))$  with  $p = s - 2a'$ ,  $q = 2a - s$ ,  $r = 2b - s$ ,  $w = s - 2b'$ . In particular  $\ell$  is piecewise linear in  $s$ , and the two removed intervals are nonempty exactly for  $s < a + a'$  and  $s > b + b'$  respectively.*

*Proof.*  $L_{\pi/4}(c) = \{u \leq a+1, v \leq a'+1\} \setminus \{u < a, v < a'\}$  and  $R_{\pi/4}(d) = \{u \geq b-1, v \geq b'-1\} \setminus \{u > b, v > b'\}$  directly from the definitions, since reflection carries  $\mu, \nu$  to  $-\nu, -\mu$ . The cross-section of the rectangle at level  $s$  is  $[\max(p, r) - 2, \min(q, w) + 2]$  in  $t$ , the two quadrants cut out  $(p, q)$  and  $(r, w)$ , and  $du dv = \frac{1}{2} ds dt$ .  $\square$

Formula (34) agrees with direct polygon computation to  $2.5 \cdot 10^{-11}$  on test placements. The crude bound  $\ell \leq 4$  recovers Hammersley: it gives  $\frac{1}{2} \cdot 4\sqrt{2} = 2\sqrt{2}$ . The conjectured value is attained, and can be computed exactly.

**Proposition 73** (The extremal configuration). *At  $a = a' = b = b' = \sqrt{2}/4$ , that is with both corners at the same point at mid-corridor height,  $\ell(s) = 4 - 4|s - \frac{\sqrt{2}}{2}|$  and (34) equals exactly  $2\sqrt{2} - 1$ .*

*Proof.* Put  $\delta = s - \sqrt{2}/2$ . Then  $p = w = \delta$  and  $q = r = -\delta$ , so the outer interval is  $[|\delta| - 2, 2 - |\delta|]$ , of length  $4 - 2|\delta|$ . Exactly one of  $(p, q) = (\delta, -\delta)$  and  $(r, w) = (-\delta, \delta)$  is nonempty, according to the sign of  $\delta$ , and it has length  $2|\delta|$  and lies inside the outer interval because  $|\delta| \leq \sqrt{2}/2 < 1$ . Hence  $\ell = 4 - 4|\delta|$  and  $\frac{1}{2} \int_0^{\sqrt{2}} (4 - 4|\delta|) ds = \frac{1}{2} (4\sqrt{2} - 4 \cdot \frac{1}{2}) = 2\sqrt{2} - 1$ .  $\square$

A search of 600 multistarts over  $[-3, 3]^4$  in exact piecewise-linear arithmetic returns 1.828427124746, agreeing with  $2\sqrt{2} - 1$  to  $2.7 \cdot 10^{-15}$ , and no configuration exceeding it was found.

The supremum in (33) can be evaluated in closed form on most of the parameter space, and this gives the first upper bound specific to the ambidextrous problem. The key is that the set left after the two quadrants are removed is confined to two short end pieces, whose lengths are controlled by two gauge-invariant combinations of  $c$  and  $d$ .

**Lemma 74** (Four pieces). *With the notation of Lemma 72, set*

$$m_1 = \langle c, \nu \rangle + \langle d, \mu \rangle = a' + b, \quad m_2 = \langle c, \mu \rangle + \langle d, \nu \rangle = a + b',$$

$$\delta_\mu = \langle d - c, \mu \rangle = b - a, \quad \delta_\nu = \langle d - c, \nu \rangle = b' - a'.$$

*All four are unchanged by the translation  $(a, a', b, b') \mapsto (a + \varepsilon, a' - \varepsilon, b + \varepsilon, b' - \varepsilon)$ , and for every  $s$*

$$\ell(s) \leq (2 - 2|s - m_1|)^+ + (2 - 2|s - m_2|)^+ + \min(2(s - m_1)^+, 2\delta_\mu^+) + \min(2(s - m_2)^+, 2\delta_\nu^+). \quad (35)$$

*Proof.* Write  $I = [\max(p, r) - 2, \min(q, w) + 2]$ . Since  $I \setminus (p, q) = \{x \in I : x \leq p\} \cup \{x \in I : x \geq q\}$  and likewise for  $(r, w)$ , intersecting the two gives

$$I \setminus ((p, q) \cup (r, w)) = (A \cup B \cup C \cup D) \cap I,$$

$$A = (-\infty, \min(p, r)], \quad B = [q, r], \quad C = [w, p], \quad D = [\max(q, w), \infty).$$

Now  $|A \cap I| \leq \min(p, r) - \max(p, r) + 2 = (2 - |p - r|)^+$  and  $p - r = 2s - 2(a' + b)$ , giving the first term;  $|D \cap I| \leq \min(q, w) + 2 - \max(q, w) = (2 - |q - w|)^+$  and  $q - w = 2(a + b') - 2s$ , giving the second. For the cross piece  $C = [w, p]$ : it is empty unless  $p > w$ , i.e.  $\delta_\nu > 0$ , and  $C \setminus A \subseteq [\min(p, r), p]$ , so  $|C \setminus A| \leq \min((p - r)^+, (p - w)^+) = \min(2(s - m_1)^+, 2\delta_\nu^+)$ . Symmetrically  $B$  is empty unless  $\delta_\mu > 0$  and  $|B \setminus D| \leq \min((w - q)^+, (r - q)^+) = \min(2(s - m_2)^+, 2\delta_\mu^+)$ . Adding the four bounds can only over-count overlaps.  $\square$

**Proposition 75** (The profile integral). *Let  $g(x) = \int_0^{\sqrt{2}} (2 - 2|s - x|)^+ ds$ . Then  $g(\sqrt{2} - x) = g(x)$ ,  $g$  vanishes off  $(-1, \sqrt{2} + 1)$ , on  $[\sqrt{2} - 1, 1]$  it equals  $2\sqrt{2} - x^2 - (\sqrt{2} - x)^2$ , and*

$$\max_{x \in \mathbb{R}} g(x) = g\left(\frac{1}{\sqrt{2}}\right) = 2\sqrt{2} - 1.$$

*Proof.* The substitution  $s \mapsto \sqrt{2} - s$  gives the symmetry. Differentiating under the integral,  $g'(x) = 2|[0, \sqrt{2}] \cap (x - 1, x)| - 2|[0, \sqrt{2}] \cap (x, x + 1)|$ , which is  $\leq 0$  for  $x \geq \sqrt{2}/2$  because the second window then contains at least as much of  $[0, \sqrt{2}]$  as the first; with the symmetry,  $g$  increases up to  $\sqrt{2}/2$  and decreases after. For  $\sqrt{2} - 1 \leq x \leq 1$  the integrand is nonnegative throughout  $[0, \sqrt{2}]$ , so  $g(x) = 2\sqrt{2} - 2 \int_0^{\sqrt{2}} |s - x| ds = 2\sqrt{2} - x^2 - (\sqrt{2} - x)^2$ , and at  $x = \sqrt{2}/2$  this is  $2\sqrt{2} - \frac{1}{2} - \frac{1}{2}$ .  $\square$

Combining (35) with  $\ell \geq 0$  and integrating over the band gives a bound in closed form, with no computation of any kind.

**Theorem 76** (The sharp constant, off the cross region). *For all  $c, d$ ,*

$$|\mathcal{C}_2 \cap L_{\pi/4}(c) \cap R_{\pi/4}(d)| \leq \frac{1}{2}[g(m_1) + g(m_2)] + \sqrt{2}[\min(1, \delta_\mu^+) + \min(1, \delta_\nu^+)]. \quad (36)$$

*In particular, if  $\langle d - c, \mu \rangle \leq 0$  and  $\langle d - c, \nu \rangle \leq 0$  then*

$$|\mathcal{C}_2 \cap L_{\pi/4}(c) \cap R_{\pi/4}(d)| \leq 2\sqrt{2} - 1 = 1.8284271 \dots,$$

*with equality exactly when  $m_1 = m_2 = 1/\sqrt{2}$  and  $\delta_\mu = \delta_\nu = 0$ . Consequently every ambidextrous moving sofa whose two corner positions at angle  $\pi/4$  satisfy those two inequalities has area at most  $2\sqrt{2} - 1$ .*

*Proof.* Integrate (35) over  $s \in [0, \sqrt{2}]$  and halve, as in (34). The first two terms give  $\frac{1}{2}g(m_1) + \frac{1}{2}g(m_2)$  by definition of  $g$ ; the third is at most  $\frac{1}{2} \int_0^{\sqrt{2}} 2 \min(1, \delta_\mu^+) ds = \sqrt{2} \min(1, \delta_\mu^+)$  and likewise the fourth. If  $\delta_\mu \leq 0$  and  $\delta_\nu \leq 0$  the cross terms are absent and Proposition 75 bounds the rest by  $\frac{1}{2} \cdot 2(2\sqrt{2} - 1)$ . Equality forces both  $g$ 's to be maximal, hence  $m_1 = m_2 = 1/\sqrt{2}$ ; the configuration  $a = a' = b = b' = \sqrt{2}/4$  of Proposition 73 realises it, and (36) is therefore sharp.  $\square$

The bound (36) was tested against the objective computed independently from the definition at  $4 \cdot 10^4$  random parameter triples over scales from 0.1 to 160: the minimum slack is  $-1.0 \cdot 10^{-14}$ , i.e. zero to rounding, so the bound is both correct and attained.

When  $\delta_\mu > 0$  or  $\delta_\nu > 0$  the cross terms in (36) are genuinely present and the estimate exceeds  $2\sqrt{2} - 1$ . The next three lemmas remove the sign hypothesis. Throughout, the reflection across the diagonal  $u = v$  preserves the band and exchanges  $(a, a')$  with  $(a', a)$  and  $(b, b')$  with  $(b', b)$ , hence  $\delta_\mu \leftrightarrow \delta_\nu$  and  $m_1 \leftrightarrow m_2$ ; so assume without loss of generality  $t := \delta_\nu = \max(\delta_\mu, \delta_\nu) > 0$ .

**Lemma 77** (Wide offsets). *If  $t \geq 1$  then  $|\mathcal{C}_2 \cap L_{\pi/4}(c) \cap R_{\pi/4}(d)| \leq \sqrt{2}(2-t)^+ \leq \sqrt{2}$ .*

*Proof.*  $w - p = (s - 2b') - (s - 2a') = -2t$  identically in  $s$ , so  $\ell(s) \leq |I| \leq 4 + w - p = 4 - 2t$  for every  $s$ , and the band has length  $\sqrt{2}$ :  $\frac{1}{2} \int_0^{\sqrt{2}} \ell \leq \frac{1}{2} \sqrt{2} (4 - 2t)^+$ .  $\square$

**Lemma 78** (Both offsets positive). *If  $0 < \delta_\mu \leq \delta_\nu = t \leq 1$  then  $\ell(s) \leq (2 - 2|s - m_1|)^+ + (2 - 2|s - m_2|)^+$  for every  $s$ , so the area is at most  $\frac{1}{2}[g(m_1) + g(m_2)] \leq 2\sqrt{2} - 1$ .*

*Proof.* Write, as in Lemma 74, the kept set as  $(A \cup B \cup C \cup D) \cap I$  with  $A = (-\infty, \min(p, r)]$ ,  $B = [q, r]$ ,  $C = [w, p]$ ,  $D = [\max(q, w), \infty)$ , and note  $m_1 \leq m_2$  since  $\delta_\mu \leq \delta_\nu$ . For  $s \leq m_1$  we have  $p \leq r$  and  $w \leq q$ , so  $C \subseteq A$  and  $B \subseteq D$ , and  $\ell \leq |A \cap I| + |D \cap I| \leq (2 - 2(m_1 - s))^+ + (2 - 2(m_2 - s))^+$  exactly as in the proof of Theorem 76; symmetrically for  $s \geq m_2$ . On the middle window  $m_1 < s < m_2$ :  $A = (-\infty, r]$ ,  $D = [q, \infty)$ , and since  $\delta_\mu > 0$  gives  $q \leq r$ ,

$$A \cap D = [q, r] = B, \quad A \cap C \cap D = [q, r] \cap [w, p] = [q, r] = B,$$

the latter because  $w \leq q$  and  $r \leq p$  on the window. In the inclusion-exclusion for  $|A \cup C \cup D|$  the terms  $-|A \cap D|$  and  $+|A \cap C \cap D|$  therefore cancel, leaving

$$\ell = |A \cap I| + |D \cap I| + [|C \cap I| - |A \cap C \cap I| - |C \cap D \cap I|].$$

Since  $t \leq 1$ ,  $C = [w, p] \subseteq [p - 2, w + 2] = I$ , so  $|C \cap I| = 2t$ ; and  $A \cap C = [w, r]$ ,  $C \cap D = [q, p]$  are subsets of  $C$ , of lengths  $2(n_2 - s)$  and  $2(s - n_1)$  with  $n_1 = a + a'$ ,  $n_2 = b + b'$ , both nonnegative on the window. The bracket equals  $2t - 2(n_2 - n_1) = 2t - 2(\delta_\mu + \delta_\nu) = -2\delta_\mu < 0$ , and the first two terms are the two tents.  $\square$

**Lemma 79** (Mixed offsets, pointwise). *If  $\delta_\mu \leq 0 < \delta_\nu = t \leq 1$ , write  $u = -\delta_\mu$ , so  $m_2 - m_1 = t + u$ . Then for every  $s$*

$$\ell(s) \leq (2 - 2|s - m_1|)^+ + (2 - 2|s - m_2|)^+ + \min(2(s - m_1)^+, 2(m_2 - s)^+, 2\min(t, u)).$$

*Proof.* Outside the middle window the absorption of the previous proof applies verbatim (here  $B = \emptyset$ ). On the window,  $A \cap D = [q, r] = \emptyset$  since  $\delta_\mu < 0$ , and

$$\ell = |A \cap I| + |D \cap I| + c(s), \quad c(s) = |C \cap I| - |A \cap C \cap I| - |C \cap D \cap I| = 2t - 2(n_2 - s)^+ - 2(s - n_1)^+,$$

by the same computations ( $C \subseteq I$  from  $t \leq 1$ ;  $A \cap C = [w, r]$  when  $s \leq n_2$  and empty otherwise, likewise  $C \cap D$ ). Now  $n_2 = m_1 + t$  and  $n_1 = m_2 - t$  give  $c \leq 2t - 2(n_2 - s) = 2(s - m_1)$  and  $c \leq 2(m_2 - s)$ ; and  $(n_2 - s)^+ + (s - n_1)^+ \geq (n_2 - n_1)^+ = (t - u)^+$  gives  $c \leq 2t - 2(t - u)^+ = 2\min(t, u)$ . The first two terms are at most the two tents, as before.  $\square$

Integrating Lemma 79 over the band reduces everything to one two-variable inequality about the profile  $g$ .

**Lemma 80** (The profile inequality). *For  $m_1 \leq m_2$  set  $D = m_2 - m_1$ ,  $\tau^* = \min(1, D/2)$  and*

$$E(m_1, m_2) = \int_0^{\sqrt{2}} \min(2(s - m_1)^+, 2(m_2 - s)^+, 2\tau^*) ds.$$

*Then  $g(m_1) + g(m_2) + E(m_1, m_2) \leq 2(2\sqrt{2} - 1)$ , with equality exactly at  $m_1 = m_2 = 1/\sqrt{2}$ .*

Before the four regions, the statement simplifies. Write  $C(x) = \int_0^{\sqrt{2}} \min(|s - x|, 1) ds$ . The identity  $\max(0, y) = y + \max(0, -y)$  gives  $(2 - 2|s - x|)^+ = 2 - 2\min(|s - x|, 1)$  pointwise, hence  $g(x) = 2\sqrt{2} - 2C(x)$ ; and  $E = 2 \int_0^{\sqrt{2}} F$  with  $F(s) = \min((s - m_1)^+, (m_2 - s)^+, \tau^*)$ . Substituting, the inequality  $g(m_1) + g(m_2) + E \leq 2(2\sqrt{2} - 1)$  is equivalent to

$$\int_0^{\sqrt{2}} [\min(|s - m_1|, 1) + \min(|s - m_2|, 1) - F(s)] ds \geq 1. \quad (37)$$

Neither  $g$  nor  $E$  appears. Since  $\int_0^{\sqrt{2}} \min(|s - x|, 1) ds \geq \frac{1}{2}$  for every  $x$ , which is the pairing argument of the formalisation section, the two positive terms already contribute 1, and (37) says exactly that  $\int F$  does not exceed the excess of those two terms over  $\frac{1}{2}$  each. On the central square that excess is  $x_1^2 + x_2^2$  with  $x_i = m_i - \sqrt{2}/2$ , and  $\int F \leq D^2/4 = (x_2 - x_1)^2/4$ , so region (i) reduces to  $(x_2 - x_1)^2 \leq 4(x_1^2 + x_2^2)$ , which is  $(x_1 + x_2)^2 \geq 0$  together with  $(x_2 - x_1)^2 \leq 2(x_1^2 + x_2^2)$ .

**Remark 81** (A tempting simplification that is false).  $F \leq \min(\min(|s - m_1|, 1), \min(|s - m_2|, 1), \tau^*) =: M$  pointwise, so replacing  $F$  by  $M$  in (37) would give a symmetric statement with no one-sided parts. It is false: on a  $241 \times 241$  grid of  $[-1.6, 3.2]^2$  the left side with  $M$  has minimum 0.8751, at  $(m_1, m_2) = (0.520, 0.880)$ . The loss is at  $m_1 = m_2$ , where  $F \equiv 0$  but  $M = \min(|s - m_1|, 1)$  does not vanish. The one-sided form is not a convenience.

*Proof.* Write  $C(x) = \int_0^{\sqrt{2}} \min(|s - x|, 1) ds$  and  $e(x) = C(x) - \frac{1}{2}$ , which is nonnegative by the pairing argument. By (37) the claim is  $\int_0^{\sqrt{2}} F \leq e(m_1) + e(m_2)$ . Set  $D = m_2 - m_1 \geq 0$  and  $u_i = m_i - \sqrt{2}/2$ .

(L1)  $\int F \leq D^2/4$ .  $F$  is supported on  $[m_1, m_2]$  and dominated by  $\min(s - m_1, m_2 - s)$ , a triangle of base  $D$  and height  $D/2$ , of area  $D^2/4$ ; when  $\tau^* = 1 < D/2$  the profile is instead a trapezoid of area  $D - 1$ , and  $D - 1 \leq D^2/4$  because  $(D - 2)^2 \geq 0$ . Clipping the domain to  $[0, \sqrt{2}]$  only decreases the integral.

(L2)  $\int F \leq \min(C(m_1), C(m_2))$ . Pointwise  $F \leq \tau^* \leq 1$  and  $F \leq (s - m_1)^+ \leq |s - m_1|$ , so  $F \leq \min(|s - m_1|, 1)$ ; likewise for  $m_2$ .

(L3)  $C(x) < 1$  forces  $|x - \sqrt{2}/2| \leq 1$ , and (L4) on that range  $e(x) \geq (x - \sqrt{2}/2)^2/2$ . Both are finite piecewise-polynomial verifications: with no clipping the pairing identity gives  $e(x) = \int_0^{\sqrt{2}/2} 2 \operatorname{dist}(x, [s, s + \sqrt{2}/2]) ds$ , equal to  $u^2$  for  $|u| \leq 1 - \sqrt{2}/2$ , and clipping contributes the remaining two polynomial pieces.

*Assembly.* If  $\max(C(m_1), C(m_2)) \geq 1$  then by (L2)

$$\int F \leq \min(C_1, C_2) = C_1 + C_2 - \max(C_1, C_2) \leq C_1 + C_2 - 1 = e_1 + e_2.$$

Otherwise both  $C(m_i) < 1$ , so (L3) gives  $|u_i| \leq 1$  and (L4) gives  $e(m_i) \geq u_i^2/2$ ; then

$$e_1 + e_2 \geq \frac{1}{2}(u_1^2 + u_2^2) \geq \frac{1}{4}(u_2 - u_1)^2 = \frac{D^2}{4} \geq \int F$$

by (L1), the middle step being  $(u_1 + u_2)^2 \geq 0$ . Equality throughout requires  $u_1 = u_2 = 0$ , that is  $m_1 = m_2 = 1/\sqrt{2}$ .  $\square$

The case split is on a comparison of real numbers, not on a box: no covering, and no interval arithmetic, appears in the proof. An earlier version certified this lemma by an adaptive covering of 28 512 boxes in ball arithmetic. Numerically the assembled inequality has minimum slack  $6.6 \cdot 10^{-7}$  over  $6 \cdot 10^4$  sampled pairs, attained at  $m_1 = m_2 = \sqrt{2}/2$ , which is the extremal placement of Proposition 73.

**Theorem 82** (The unconditional bound). *Every ambidextrous moving sofa satisfies*

$$|S| \leq 2\sqrt{2} - 1 = 1.8284271\dots$$

*Proof.* By (33) it suffices to bound the two-hallway area for every placement. If  $\delta_\mu \leq 0$  and  $\delta_\nu \leq 0$  this is Theorem 76. Otherwise, WLOG  $t = \delta_\nu = \max > 0$ . If  $t \geq 1$ , Lemma 77 gives  $\sqrt{2} < 2\sqrt{2} - 1$ . If  $t < 1$  and  $\delta_\mu > 0$ , Lemma 78 and Proposition 75 give  $2\sqrt{2} - 1$ . If  $t < 1$  and  $\delta_\mu \leq 0$ , integrate Lemma 79 over the band: the tents give  $\frac{1}{2}[g(m_1) + g(m_2)]$  and the flat-top term gives  $\frac{1}{2}$  of an integral that is monotone in its height, so with  $\min(t, u) \leq \min(1, D/2) = \tau^*$  it is at most  $\frac{1}{2}E(m_1, m_2)$ ; Lemma 80 finishes.  $\square$

Since Romik’s sofa is ambidextrous, Theorem 82 brackets the ambidextrous supremum:

$$1.6449552\dots = A_R^* \leq A_{\text{ambi}} \leq 2\sqrt{2} - 1 = 1.8284271\dots,$$

a factor of 1.1115. The previously available bound was 2.2195...: an ambidextrous sofa is in particular a one-corner sofa, so it inherits whatever the one-corner problem gives, and since [1] that is Gerver’s area itself. Before [1] the inherited value was the 2.37 of [3].

Improving on the inherited bound is a question Kallus and Romik put in print. Their concluding section observes that any one-corner upper bound applies to the ambidextrous variant, calls the gap between the bounds an appealing opportunity, and asks for their techniques to be extended to the ambidextrous setting so as to obtain better upper bounds on the ambidextrous moving sofa constant [3, §6]. Theorem 82 answers that question, though by evaluating a small relaxation in closed form rather than by extending their branch and bound. As of the most recent survey-level treatment [7, Problem 6.63], which records the lower bound 1.6449... and states that whether it is sharp remains open, no upper bound specific to the problem had been given.

The constant  $2\sqrt{2} - 1$  cannot be improved by adding angles to this pair: it is attained, by Proposition 73, so it is exactly the value of the two-hallway relaxation. The route to a smaller constant is more constraints, not a better estimate of these.

The construction is not new, and its lineage should be stated. Relaxing a sofa to the intersection of finitely many placed hallways is Hammersley’s idea [6] and is the framework of Kallus and Romik [3]; the worked example following their Proposition 4 evaluates the one-angle case in closed form, finds the optimum at  $\alpha = \pi/4$ , and recovers exactly Hammersley’s  $2\sqrt{2}$ , which those authors describe as Hammersley’s proof rewritten in their notation. What follows is the two-hallway ambidextrous analogue of that example. The novelty claimed here is the resulting bound, not the method that produces it: we know of no closed-form evaluation of a relaxation with more than one hallway, in either handedness, and Romik has described the multi-hallway maximum as very hard to obtain by hand, which is why [3] passes to interval-arithmetic branch and bound.

Four of the six constraints of Proposition 69 also force this value.

**Proposition 83** (Four constraints suffice, and they recover Hammersley). *Write  $L_\theta, R_\theta$  for a left- and a right-turning hallway at angle  $\theta$ , with free placements. Then*

$$\sup |L_0 \cap L_{\pi/4} \cap L_{\pi/2}| = 2\sqrt{2}, \quad \sup |L_0 \cap R_0 \cap L_{\pi/4} \cap R_{\pi/4}| \geq 2\sqrt{2} - 1,$$

*the first being Hammersley’s classical bound for the one-corner problem.*

The first equality is a regression check rather than a new result: it says the construction reproduces the known elementary bound  $2\sqrt{2}$  for one corner, from three left-turning hallways.

That  $2\sqrt{2} - 1$  is Hammersley’s constant less exactly 1 is structural, not a coincidence, and Lemma 71 says where the 1 goes. In Hammersley’s configuration the strip meets the  $\pi/4$ -rotated corridor in two unit squares set on their diagonals, of total area  $2\sqrt{2}$ . Adding the oppositely-handed hallway at the same angle cuts a quadrant from each square, and Lemma 71 evaluates what survives in one of them as  $\sqrt{2} - \frac{1}{2}$ . Two of those give  $2\sqrt{2} - 1$ : each square loses exactly  $\frac{1}{2}$ . Since dropping constraints only enlarges the supremum, the four-constraint value already bounds the six-constraint one, so a proof of

$$|L_0 \cap R_0 \cap L_{\pi/4} \cap R_{\pi/4}| \leq 2\sqrt{2} - 1 \quad \text{for all placements} \quad (38)$$

gives  $A_{\text{ambi}} \leq 2\sqrt{2} - 1$  outright, with no branch and bound and no appeal to the remaining angles. That is what Theorem 76 and Lemmas 77–80 establish: (38) *holds*, since the corridor pair  $L_0 \cap R_0$  is the strip whose rotated coordinates give the band, and the four sign cases of the offsets exhaust the placements. Theorem 82 is its restatement for sofas. Two structurally different placements attain the value: one splits the region into two congruent pieces of area  $\sqrt{2} - \frac{1}{2}$ , one leaves it connected. So  $2\sqrt{2} - 1$  is the supremum at those angles, attained by Proposition 73 and bounded above by Theorem 82, with no interval arithmetic anywhere in the two-hallway argument.

The construction was validated before use: intersecting along  $\Sigma$ ’s own corner path must give an area decreasing to  $|\Sigma|$ , and it does, through 2.050458, 2.000690, 1.991986, 1.990120, 1.645249 at 3, 9, 33, 129, 513 angles per side for the two-family intersection. Two errors were caught by that check, one in the side of the inner corner quadrant and one in the identification of the mirrored family, before any bound was computed.

## 18 Relation to prior work

The one-corner problem was settled by Baek [1], who proved Gerver’s sofa optimal. Our reading of that paper, for the purpose of separating what is his from what is not:

- The *arm lengths*  $f, g$  are his (Definition 6.2.1), defined through the contact points  $A$  and  $C$ ; our  $\alpha_1, \alpha_2$  are these shifted by the corridor width.
- His injectivity condition (Definition 6.1.2) has three parts. Part (1) requires the surface measure  $\sigma_K$  to be absolutely continuous on  $[0, \pi/2)$  and on  $(\pi/2, \pi]$ , so an atom is permitted only at  $\theta = \pi/2$ ; part (3) is  $x'_K(t) \cdot u_t < 0$  and  $x'_K(t) \cdot v_t > 0$  on  $(0, \pi/2)$ .
- Our (RC) shares part (1) verbatim and *adds* a bound on the density, while dropping part (3) entirely. No curvature bound appears in [1]; its only mentions of curvature are the standard identification of the surface-measure density with the radius of curvature, which we also use. The bound itself is classical, however: it is the sliding-ball condition of Remark 51, so (RC) is a known condition put to a new use rather than a new condition.



- His concavity argument attaches Mamikon regions to the overestimating region so that the total is linear in the cap; ours subtracts them from the cap. He works on a space of triples  $(K, B, D)$  of convex bodies; we work with the single support function  $H$  on  $[0, \pi]$ . Whether the two bookkeepings are equivalent we do not determine.
- The ambidextrous problem is not treated in [1], and we have found no upper bound for it in the literature.

The constants  $a_1, \beta, f_1$  and the ODE families are Romik's [5]; the cubics of Section 7 are the standard ones and we claim no priority for them. This audit was made against [1] and the Kallus–Romik line [3] only, not against the wider literature.

## 19 The main theorem

**Theorem 84.** *Let  $\mathcal{D}$  be the set of support functions  $H$  on  $[0, \pi]$  of admissible caps such that*

- (a)  $H(0) = H(\pi/2) = 1$  (the gauge: horizontal translation, unit corridor);
- (b) (RC)  $\sigma_K$  is absolutely continuous on  $[0, \pi/2)$  and  $(\pi/2, \pi]$  with density at most 1, the only atom being at  $\theta = \pi/2$ ; equivalently, on  $(0, \pi)$   $0 \leq (H + H'')_{ac} \leq 1$  with no atom, together with  $H'(0) = \frac{1}{2}$  and  $H'(\pi) = -\frac{1}{2}$ , the last two being the vanishing of the junction atoms of the  $\rho$ -extension (Lemma 100);
- (c)  $c_y(t) = (H(t) - 1) \sin t + (H(t + \pi/2) - 1) \cos t \leq \frac{1}{2}$  for  $t \in [0, \pi/2]$  (note  $\leq$ , not  $<$ : this gives  $|U \cap \rho U| = 0$ , which is all an area bound needs);
- (d)  $\alpha_1 < 0$  exactly on  $[0, \beta)$  and  $\alpha_2 > 0$  exactly on  $[0, \pi/2 - \beta)$ .

Then  $\mathcal{D}$  is convex,  $\Sigma \in \mathcal{D}$ , and

$$|T| \leq A_R^*$$

for every ambidextrous moving sofa  $T$  whose cap data lies in  $\mathcal{D}$ .

*Proof.* Each of (a)–(d) is a family of affine inequalities or equalities in  $H$ , by Lemma 18 and (17)–(19), so  $\mathcal{D}$  is convex.  $\Sigma$  satisfies (b) by Proposition 63, (c) with margin  $\frac{1}{2} - M = 0.1121\dots$  by Theorem 8, and (d) by definition of  $\beta$ .

By (b) and Theorems 52 and 56 the combined sweep is injective, so  $V = |N|$  by Proposition 47, whose proof also supplies the covering and containment that Proposition 20 assumes. By (c) and Lemma 1,  $U \subseteq \{y \leq \frac{1}{2}\}$  and  $\rho U \subseteq \{y \geq \frac{1}{2}\}$ , so  $|U \cap \rho U| = 0$ ; with  $C_2$   $\rho$ -invariant this gives  $|T| \leq |C_2| - |N| - |\rho N| = |C_2| - 2|N| = Q$ . (Corollary 2 states the strict form  $M < \frac{1}{2}$ , which is what  $\Sigma$  satisfies; the null-set form used here is what the hypothesis  $\leq$  delivers.) By (d) and Theorem 60,  $\delta^2 Q \leq 0$  on  $\mathcal{D}$ ; by Theorem 57,  $\delta Q(\Sigma) = 0$ . A concave function with a critical point on a convex set attains its maximum there, so  $Q(H) \leq Q(\Sigma)$ , and  $Q(\Sigma) = A_R^*$  by Theorem 64. Combining,  $|T| \leq Q(H) \leq A_R^*$ .  $\square$

## 20 The second variation

The bound above needs  $\delta^2 Q \leq 0$  on the cells the segment meets. This section establishes that, with the best constant, and on a family of cells rather than one.

Two routes to the constant are recorded and only the second is load-bearing. The first splits  $[0, \pi/2]$  into  $[0, \beta]$  and  $[\beta, \pi/2]$ , applies a separate Poincaré inequality on each, and discards the  $E_2$  term, giving the weak constants of Remark 60. Treating each half as a *single* weighted eigenvalue problem removes both losses; not splitting at all is better still and gives the exact value.

**Proposition 85** (An elementary constant, no computer). *On the cell  $\mathcal{C}$ , for every  $\eta$  with  $\eta(0) = \eta(\pi/2) = 0$ ,*

$$\frac{1}{2} \delta^2 Q[\eta] \leq -\frac{1}{12} \|\eta\|_{L^2(0, \pi)}^2,$$

*by elementary inequalities alone: no transfer matrix, no oscillation argument, and no ball arithmetic anywhere in the proof.*

*Proof.* Run the decoupling of Theorem 87 at  $\lambda = \frac{9}{20}$ ,  $\kappa = \frac{1}{4}$ , so  $r = \frac{9}{11}$  and  $q = 5$ , and bound the two halves by hand, using only  $\beta \leq \frac{3}{10}$  and  $\frac{\pi}{2} \leq 1.5708$ .

*Left half.*  $w_L \geq 1 - \kappa = \frac{3}{4}$  and  $m_L \leq 2 + r$ , and the Dirichlet–Dirichlet Poincaré inequality on  $[0, \frac{\pi}{2}]$  gives  $\int p'^2 \geq 4 \int p^2$ , so

$$\int w_L p'^2 - \int m_L p^2 \geq [3 - 2 - \frac{9}{11}] \int p^2 = \frac{2}{11} \int p^2.$$

*Right half.* Write  $c = \frac{\pi}{2} - \beta$  and  $s = \lambda - q\beta^2/2$ . Three elementary steps. The heavy mass sits on  $[0, \beta)$ , next to the Dirichlet end, so  $q \int_0^\beta \eta'^2 \leq q \frac{\beta^2}{2} \int_0^\beta \eta'^2 \leq \frac{9}{40} \int_0^c \eta'^2$  by the half-Poincaré inequality  $\eta(t)^2 \leq t \int_0^t \eta'^2$ ; this absorbs the mass into the weight surplus and leaves  $s = \frac{9}{20} - \frac{9}{40} = \frac{9}{40}$ . The surplus converts to mass through the cut at  $c$ : by the same half-Poincaré inequality and  $\eta(t)^2 \leq 2\eta(c)^2 + 2\beta \int_c^{\pi/2} \eta'^2$  for  $t \geq c$ ,

$$\int_0^{\pi/2} \eta'^2 \leq \left(\frac{c^2}{2} + 2\beta c\right) \int_0^c \eta'^2 + 2\beta^2 \int_c^{\pi/2} \eta'^2 \leq \frac{11}{5} \int_0^c \eta'^2 + \frac{9}{50} \int_c^{\pi/2} \eta'^2,$$

so  $\int_0^c \eta'^2 \geq \frac{5}{11} \int \eta'^2 - \frac{9}{110} \int_c^{\pi/2} \eta'^2$ . With the Dirichlet–Neumann Poincaré inequality  $\int \eta'^2 \geq \int \eta^2$  on  $[0, \frac{\pi}{2}]$ ,

$$\int w_R \eta'^2 - \int m_R \eta^2 \geq \left(1 - \frac{9}{40} \cdot \frac{9}{110}\right) \int \eta'^2 - \left(1 - \frac{9}{40} \cdot \frac{5}{11}\right) \int \eta^2 \geq \frac{369}{4400} \int \eta^2.$$

Since  $\frac{369}{4400} > \frac{1}{12}$  and  $\frac{2}{11} > \frac{1}{12}$ , the two halves give the claim. Every constant is rational and every inequality is Cauchy–Schwarz, a Poincaré inequality on an interval, or arithmetic.  $\square$

**Remark 86** (A rational certificate for the decoupled half). The gap between  $\frac{1}{12}$  and the decoupled truth is closed almost entirely by a certificate that is still exact arithmetic. In Prüfer form the half-problem  $-(w\eta')' - m\eta = c\eta$  becomes  $\theta' = \cos^2 \theta/w + (m+c)\sin^2 \theta$  with  $\theta(0) = 0$ , and  $c_1 > c$  exactly when  $\theta$  stays below  $\pi/2$ ; a piecewise-linear  $B$  with  $B(0) = 0$  and  $B' \geq$  the right side dominates  $\theta$ , so a barrier reaching less than  $\pi/2$  is a proof and every check is an inequality between rationals. The identity

$$\frac{\cos^2 \theta}{w} + (m+c)\sin^2 \theta = \frac{1}{w} + \left(m+c - \frac{1}{w}\right)\sin^2 \theta$$

is what makes it work: the coefficient is positive on every piece, so an upper bound on  $\sin^2 \theta$  gives one on the whole, which is exactly what an upper barrier supplies. Bounding  $\cos^2 \leq 1$

instead is fatal, since the slope then never decays and the barrier crosses  $\pi/2$  prematurely. With  $\sin$  bounded above by  $x - x^3/6 + x^5/120$ , every constant rounded in the safe direction, and slopes rounded upward to denominator  $10^7$ , 1200 segments give

$$c_1 > \frac{37}{100}$$

for the right half. The left half is Dirichlet–Dirichlet, so the phase runs to  $\pi$ , and there the method has a genuine weakness: once an upper barrier passes  $\pi/2$  it knows only  $\sin^2 \leq 1$ , because an upper bound on  $\theta$  is a *lower* bound on  $\sin$  beyond the maximum. At the note’s  $(\lambda, \kappa) = (\frac{9}{20}, \frac{1}{4})$  that made the left half, not the right, the limit. Raising  $\kappa$  shrinks the left weight until the left half is slack; scanning both halves,  $(\lambda, \kappa) = (\frac{11}{20}, \frac{1}{5})$  gives

$$\frac{1}{2} \delta^2 Q[\eta] \leq -\frac{9}{20} \|\eta\|_{L^2(0,\pi)}^2$$

on the cell, by exact rational arithmetic: 5.4 times the  $\frac{1}{12}$  of Proposition 85 and 61.6% of the sharp  $c^*$ . The true decoupled values there are  $c_R = 0.4748$  and  $c_L = 2.4414$ , so the certificate captures 95% of what is available and the right half binds. Both negative controls behave: the right half at  $c = 1$  crosses  $\pi/2$  and the left at  $c = 3$  crosses  $\pi$  (`ambi_barrier.py`). A two-sided barrier, tracking a lower phase as well as an upper one, does *not* remove the limitation, and the reason is worth recording. It does recover information past  $\pi/2$ : with a bracket  $[L, B]$  the bound  $\sin^2 \leq 1$  is replaced by  $\max(\sin^2 L, \sin^2 B)$ , which the trace shows falling back to 0.55 and then 0.26 once the bracket clears the maximum. But beyond  $\pi/2$  the upper barrier’s slope must be driven by  $\sin^2 L$ , the value at the *lower* endpoint, because  $\sin$  decreases there; the lagging endpoint therefore sets the pace and the bracket widens. At  $(\frac{9}{20}, \frac{1}{4})$  the left half still fails at  $c = 1$ , the barrier reaching  $\pi$  at  $t = 1.5624$  against  $T = 1.5708$ , and refining the mesh to 20 000 segments does not close that gap, so the loss is structural rather than a discretisation artefact. At  $(\frac{7}{10}, \frac{2}{5})$  it fails at every  $c$  tested. Reaching  $\frac{2}{3}$  by a barrier would need a genuinely different device, not a finer one. Both the monotone bound and the comparison principle that turns a barrier into a bound on the phase are machine-checked (`prufer_rhs_eq`, `prufer_rhs_le`, `sin_sq_le_of_le`, `coef_pos_right`, `coef_pos_left`, `prufer_comparison`, `barrier_certifies`). The comparison is not an ODE-theoretic fact requiring a Lipschitz constant or a Grönwall bound: it is the fencing lemma of the mean value theory, whose hypothesis is checked only where the two functions touch, and at a touching point the barrier’s own strictness gives the conclusion.

The constant is 11% of the sharp  $c^*$ , and that is the price of dispensing with the transfer matrices: Theorem 87 reaches  $\frac{2}{3}$  and Theorem 88 the sharp  $\frac{73}{100}$ , both through certified evaluations. What Proposition 85 buys is qualitative: strict concavity of the form on the cell, hence the local maximality statements at  $\Sigma$ , no longer depends on any computation.

**Theorem 87** (Concavity, sharp order). *On the cell  $\mathcal{C}$ , for every  $\eta$  with  $\eta(0) = \eta(\pi/2) = 0$ ,*

$$\frac{1}{2} \delta^2 Q[\eta] \leq -\frac{2}{3} \|\eta\|_{L^2(0,\pi)}^2. \quad (39)$$

*Proof.* Fix  $\lambda \in (0, 1)$  and  $\kappa \in (0, 1)$ , and put  $r = \lambda/(1 - \lambda)$  and  $q = 1 + 1/\kappa$ . Two elementary inequalities dispose of the two coupling terms of (24). Since  $(1 + r)(1 - \lambda) = 1$ ,

$$(1+r)a^2 + 2ab + (1-\lambda)b^2 = (\sqrt{1+r}a + \sqrt{1-\lambda}b)^2 \geq 0, \quad \text{that is} \quad -(a+b)^2 \leq -\lambda b^2 + r a^2,$$

applied with  $a = \eta(t)$ ,  $b = \eta'(t + \pi/2)$  on  $E_2 = [0, \pi/2 - \beta]$ ; it buys gradient weight  $\lambda$  on  $[\pi/2, \pi - \beta]$  at the price of mass  $r$  on  $[0, \pi/2 - \beta]$ . And  $(x - y)^2 \leq (1 + 1/\kappa)x^2 + (1 + \kappa)y^2$  with

$x = \eta(t + \pi/2)$ ,  $y = \eta'(t)$  on  $E_1 = [0, \beta)$  pays mass  $q$  on  $[\pi/2, \pi/2 + \beta]$  and gradient weight  $1 + \kappa$  on  $[0, \beta)$ . Substituting  $s = t + \pi/2$  in both, the two halves decouple:

$$\frac{1}{2}\delta^2 Q[\eta] \leq \int_0^{\pi/2} (m_L \eta^2 - w_L \eta'^2) + \int_{\pi/2}^{\pi} (m_R \eta^2 - w_R \eta'^2) =: B[\eta],$$

$$\begin{aligned} w_L &= 1 - \kappa \text{ on } [0, \beta), \text{ 2 after,} & m_L &= 2 + r \text{ on } [0, \frac{\pi}{2} - \beta), \text{ 2 after,} \\ w_R &= 1 + \lambda \text{ on } [\frac{\pi}{2}, \pi - \beta), \text{ 1 after,} & m_R &= 1 + q \text{ on } [\frac{\pi}{2}, \frac{\pi}{2} + \beta), \text{ 1 after.} \end{aligned}$$

Each half is a Rayleigh quotient, so  $B[\eta] \leq -\min(c_L, c_R) \|\eta\|_{L^2(0, \pi)}^2$ , where  $c_L$  and  $c_R$  are the first eigenvalues of

$$-(w_L \eta')' - m_L \eta = c \eta \text{ on } (0, \frac{\pi}{2}), \quad \eta(0) = \eta(\frac{\pi}{2}) = 0; \quad -(w_R \eta')' - m_R \eta = c \eta \text{ on } (\frac{\pi}{2}, \pi), \quad \eta(\frac{\pi}{2}) = 0, \eta'(\pi) = 0.$$

The Dirichlet conditions at 0 and  $\pi/2$  are the gauge, not a restriction, and nothing pins  $\eta(\pi)$ .

Both coefficients are piecewise constant with three pieces, so the solution of the initial value problem  $\eta = 0$ ,  $w\eta' = 1$  at the left endpoint is a product of three transfer matrices acting on  $(\eta, w\eta')^\top$ ,

$$T(\ell, w, m; c) = \begin{pmatrix} \cos k\ell & (wk)^{-1} \sin k\ell \\ -wk \sin k\ell & \cos k\ell \end{pmatrix}, \quad k = \sqrt{(m + c)/w}$$

(hyperbolic when  $m + c < 0$ ). Write  $\Phi_L(c)$  for the resulting  $\eta(\pi/2)$  and  $\Phi_R(c)$  for  $(w\eta')(\pi)$ . Its zeros are exactly the eigenvalues, and by Sturm oscillation  $\Phi > 0$  on  $(-\infty, c_1)$ , so a single sign evaluation bounds  $c_1$  from *below*, which is the direction a certificate needs and the one a Rayleigh–Ritz or finite-element computation cannot supply.

Positivity alone does not locate  $c$ , because  $\Phi > 0$  again on  $(c_2, c_3)$ ; one also needs  $c < c_2$ . That comes from min–max, crudely. Since  $\int w\eta'^2 \geq w_{\min} \int \eta'^2$  and  $-\int m\eta^2 \geq -m_{\max} \int \eta^2$ , the form dominates the constant-coefficient one, so  $c_k \geq w_{\min} \lambda_k - m_{\max}$  with  $\lambda_k$  the eigenvalues on an interval of length  $\pi/2$ , namely  $(2k)^2$  for Dirichlet–Dirichlet and  $(2k - 1)^2$  for Dirichlet–Neumann. Hence

$$c_2^L \geq \frac{3}{4} \cdot 16 - \frac{63}{13} = 7.153\dots, \quad c_2^R \geq 1 \cdot 9 - 6 = 3,$$

both above  $\frac{2}{3}$ . (At  $k = 1$  the same bound gives  $-1.85$  and  $-5$ , useless; the second eigenvalue is far enough away to be reached crudely while the first is not, which is exactly why the transfer matrix is needed.)

It remains to evaluate two signs. At  $\kappa = \frac{1}{4}$ ,  $\lambda = \frac{37}{50}$  and  $c = \frac{2}{3}$  every weight and mass is rational ( $w_L = \frac{3}{4}, 2, 2$  and  $m_L = \frac{63}{13}, \frac{63}{13}, 2$ ;  $w_R = \frac{87}{50}, \frac{87}{50}, 1$  and  $m_R = 6, 1, 1$ ), so each  $k^2 = (m + c)/w$  is exact, and the only inexact inputs are the three lengths  $\beta$ ,  $\frac{\pi}{2} - 2\beta$ ,  $\beta$  and the trigonometric functions of  $k\ell$ . In ball arithmetic at 300 bits, with  $\beta = \arctan(u/2)$  enclosed from  $u^3 + 3u = 2$  by exact rational bisection,

$$\Phi_L(\frac{2}{3}) \in 0.0046739486452026048\dots \pm 6.4 \cdot 10^{-73}, \quad \Phi_R(\frac{2}{3}) \in 0.0026447861413929500\dots \pm 8.3 \cdot 10^{-73},$$

both certified strictly positive. Hence  $c_L, c_R > \frac{2}{3}$ , which is (39).  $\square$

Optimising the two parameters gives  $\min(c_L, c_R) = 0.674944$  at  $(\kappa, \lambda) = (0.26, 0.74)$ . Since  $\|\eta\|_{L^2(0, \pi/2)} \leq \|\eta\|_{L^2(0, \pi)}$ , (39) also improves the  $[0, \pi/2]$  constant of Theorem 60 from 0.371 to  $\frac{2}{3}$ .

The splitting into two scalar problems is itself avoidable, and avoiding it gives the sharp constant.

**Theorem 88** (The sharp constant). *The best constant in (39) is the first eigenvalue  $c^*$  of an explicit  $2 \times 2$  Sturm–Liouville system, and*

$$\frac{1}{2} \delta^2 Q[\eta] \leq -\frac{73}{100} \|\eta\|_{L^2(0,\pi)}^2, \quad c^* = 0.7309566 \dots \quad (40)$$

*Proof.* The two halves of  $[0, \pi]$  are the same interval re-indexed. Put  $p(t) = \eta(t)$  and  $q(t) = \eta(t + \pi/2)$  for  $t \in [0, \pi/2]$ ; then (24) reads  $\frac{1}{2} \delta^2 Q = \int_0^{\pi/2} L$  with

$$L = 2p^2 - 2p'^2 + q^2 - q'^2 - \mathbf{1}_{E_2}(p + q')^2 + \mathbf{1}_{E_1}(q - p')^2,$$

and every boundary condition is forced:  $p(0) = p(\pi/2) = 0$  is the gauge,  $q(0) = \eta(\pi/2) = 0$  is the same gauge seen from the other half, and  $q(\pi/2) = \eta(\pi)$  is free. Nothing has been discarded, so the first eigenvalue of this system is the best constant. Writing  $M := -L$  as  $ap'^2 + bq'^2 + gp^2 + dq^2 + 2epq' + 2fq'p'$ ,

| piece                                    | $a$ | $b$ | $g$ | $d$ | $e$ | $f$ |
|------------------------------------------|-----|-----|-----|-----|-----|-----|
| $[0, \beta)$                             | 1   | 2   | -1  | -2  | 1   | 1   |
| $[\beta, \frac{\pi}{2} - \beta)$         | 2   | 2   | -1  | -1  | 1   | 0   |
| $[\frac{\pi}{2} - \beta, \frac{\pi}{2})$ | 2   | 1   | -2  | -1  | 0   | 0   |

On  $[0, \beta)$ , exactly where the obstruction sits,  $e = f = 1$  and  $2pq' + 2qp' = 2(pq)'$  is a total derivative: the  $E_1$  obstruction and the  $E_2$  resource cancel there up to the single boundary value  $2p(\beta)q(\beta)$ . The coupling is not pointwise, which is why every pointwise splitting loses and this one does not.

Eliminating  $p', q'$  in favour of the momenta  $P = ap' + fq$  and  $Q = bq' + ep$ , the Euler–Lagrange system is first order in  $(p, P, q, Q)$  with piecewise constant matrix

$$A(c) = \begin{pmatrix} 0 & 1/a & -f/a & 0 \\ g - c - e^2/b & 0 & 0 & e/b \\ -e/b & 0 & 0 & 1/b \\ 0 & f/a & d - c - f^2/a & 0 \end{pmatrix},$$

so the transfer across a piece of length  $\ell$  is  $\exp(A(c)\ell)$ . The left conditions  $p(0) = q(0) = 0$  leave a two-dimensional solution space; propagating a basis through the three pieces and imposing  $p(\pi/2) = 0$  and  $Q(\pi/2) = 0$  gives a  $2 \times 2$  determinant  $\Phi(c)$  whose zeros are exactly the eigenvalues. A sign change brackets  $c^* \in (0.7309, 0.7310)$ .

For the certificate the oscillation argument is unavailable, since for a system the count is a Maslov index rather than a zero count, and it is not needed. Two facts compose:  $c^* \geq \frac{2}{3}$  by Theorem 87, and  $\Phi$  has no zero on  $[\frac{2}{3}, \frac{73}{100}]$ . The latter is checked by covering that interval with 60 balls and verifying in 400-bit ball arithmetic that no enclosure of  $\Phi$  contains 0, with  $\beta$  enclosed from  $u^3 + 3u = 2$  by exact rational bisection. Hence no eigenvalue lies at or below  $\frac{73}{100}$ .  $\square$

The certified  $\frac{73}{100}$  is 99.9% of  $c^*$ ; the decoupled  $\frac{2}{3}$  is 91.2%. Both are lower bounds, and no step in either is floating point. The covering route is also self-contained: with the crude spectral bound  $-4$  of Theorem 91(c), covering  $[-4, \frac{73}{100}]$  at the fixed lengths  $(\beta, \frac{\pi}{2} - 2\beta, \beta)$  with 3074 boxes certifies  $c_1 > \frac{73}{100}$  directly, without the decoupled chain or the oscillation argument; the negative control at  $\frac{74}{100}$  is refused with the failing box at  $c \in [0.7310, 0.7313]$ , which lies just above  $c^* = 0.7309566 \dots$  and within  $4 \cdot 10^{-5}$  of it: the enclosure of  $\Phi$  there is still wide enough to

contain 0, which is how a target past  $c^*$  fails. The decoupled chain remains the hand-checkable route to  $\frac{2}{3}$ . Nothing in the argument stops at  $\frac{73}{100}$ : certifying a target  $c$  needs the enclosures to separate from 0 on every subinterval of  $[\frac{2}{3}, c]$ , and those must shrink as  $c$  approaches the root, so the number of balls grows: 60, 60, 800, 12000 for targets 0.73, 0.7305, 0.7309, 0.73095. The last is 99.9991% of  $c^*$ . We quote  $\frac{73}{100}$  because a reader can check it with 60 balls; the limit is arithmetic cost, not a gap.

**Remark 89** (The  $[\pi/2, \pi]$  eigenvalue in closed form). When only the weight is raised and the mass is left at 1 – that is,  $w = 1 + \lambda_J$  on  $[\pi/2, \pi - \beta]$  and 1 on  $[\pi - \beta, \pi]$  – the transfer product collapses to a transcendental equation for the first eigenvalue  $\Lambda(\lambda_J)$  of  $-(w\eta')' = \Lambda\eta$  with  $\eta(\pi/2) = 0$ ,  $\eta'(\pi) = 0$ :

$$\sqrt{w_1} \cot(L_1 \sqrt{\Lambda/w_1}) = \tan(L_2 \sqrt{\Lambda}), \quad w_1 = 1 + \lambda_J, \quad L_1 = \frac{\pi}{2} - \beta, \quad L_2 = \beta. \quad (41)$$

At  $\lambda_J = 0$  this reads  $\cot(L_1 \sqrt{\Lambda}) = \cot(\pi/2 - L_2 \sqrt{\Lambda})$ , so  $\sqrt{\Lambda}(L_1 + L_2) = \pi/2$ ; since  $L_1 + L_2 = \pi/2$  *exactly*,  $\Lambda(0) = 1$  exactly. That is the marginal case  $P_3 = 1$  of Theorem 60, and it explains why no gain is available on  $[\pi/2, \pi]$  until the weight is raised. For  $\lambda_J > 0$  put  $G(w_1) = \sqrt{w_1} \cot(L_1/\sqrt{w_1}) - \tan \beta$ . Then  $G(1) = 0$ , and  $G$  increases: the prefactor increases, while  $L_1/\sqrt{w_1}$  decreases and  $\cot$  decreases on  $(0, \pi)$ . So  $G > 0$  for  $w_1 > 1$ . The left side of (41) decreases in  $\Lambda$  and the right side increases, so the root moves right and  $\Lambda(\lambda_J) > 1$  for every  $\lambda_J > 0$ :

|             |   |          |          |          |          |          |
|-------------|---|----------|----------|----------|----------|----------|
| $\lambda_J$ | 0 | 0.05     | 0.10     | 0.16483  | 0.25     | 0.35     |
| $\Lambda$   | 1 | 1.049467 | 1.098883 | 1.162878 | 1.246819 | 1.345183 |

Theorem 87 runs this mechanism on both halves at once, raising the mass as well as the weight.

### Concavity on every ordered anchored cell

The concavity results so far concern single cells:  $\Sigma$ 's own (Theorems 60, 87, 88) and the eleven measured entries of Proposition 107 below. This subsection proves concavity for the whole ordered anchored family at once. In the variables  $p(t) = \eta(t)$ ,  $q(t) = \eta(t + \pi/2)$  on  $[0, \pi/2]$ , with  $p(0) = p(\pi/2) = q(0) = 0$  and  $q(\pi/2)$  free, the cell form is

$$B_{E_1, E_2}[\eta] = \int_0^{\pi/2} [2p^2 - 2p'^2 + q^2 - q'^2] - \int_{E_2} (p + q')^2 + \int_{E_1} (q - p')^2, \quad (42)$$

where  $E_1 = \{\alpha_1 < 0\}$  and  $E_2 = \{\alpha_2 > 0\}$ .

**Lemma 90** (Monotonicity in the sign sets). *If  $E_1 \subseteq E_2$  then  $B_{E_1, E_2}[\eta] = B_{E_2, E_2}[\eta] - \int_{E_2 \setminus E_1} (q - p')^2 \leq B_{E_2, E_2}[\eta]$ .*

*Proof.* The two forms differ only in the set carrying the  $(q - p')^2$  term. □

So every cell with  $E_1 \subseteq E_2$  and  $E_2$  *anchored* is dominated by a member of the one-parameter diagonal family  $D_\tau := B_{[0, \tau], [0, \tau]}$ , and it suffices to treat that family.

**Theorem 91** (Diagonal concavity).  *$D_\tau[\eta] \leq 0$  for every  $\tau \in [0, \pi/2]$  and every admissible  $\eta$ .*

*Proof.* On  $[0, \tau]$  the integrand of (42) combines, by the algebra of Theorem 88, into  $(p^2 - p'^2) + 2(q^2 - q'^2) - 2(pq)'$ , so with  $p(0)q(0) = 0$ ,

$$D_\tau = \int_0^\tau [(p^2 - p'^2) + 2(q^2 - q'^2)] - 2p(\tau)q(\tau) + \int_\tau^{\pi/2} [2(p^2 - p'^2) + (q^2 - q'^2)]. \quad (43)$$

Three overlapping ranges of  $\tau$  are treated by three arguments.

*Range (a):*  $\tau \leq 1/\sqrt{3}$ . Write (43) as  $B_{\text{bare}} + \int_0^\tau [(p'^2 - p^2) + (q^2 - q'^2)] - 2p(\tau)q(\tau)$  with  $B_{\text{bare}} := \int_0^{\pi/2} [2(p^2 - p'^2) + (q^2 - q'^2)]$ . The Poincaré constants for the boundary conditions are 4 (Dirichlet–Dirichlet, for  $p$ ) and 1 (Dirichlet–Neumann, for  $q$ ), so  $B_{\text{bare}} \leq -\frac{3}{2} \int_0^{\pi/2} p'^2$ . The added terms are bounded by  $\int_0^\tau p'^2 \leq \int_0^{\pi/2} p'^2$ , by  $\int_0^\tau q^2 \leq \tau^2 \int_0^\tau q'^2$  (from  $q(t)^2 \leq t \int_0^t q'^2$ ), and, using Young's inequality at weight  $s = 1/(2\tau)$  together with  $p(\tau)^2 \leq \tau \int_0^\tau p'^2$  and  $q(\tau)^2 \leq \tau \int_0^\tau q'^2$ ,

$$2|p(\tau)q(\tau)| \leq \frac{p(\tau)^2}{2\tau} + 2\tau q(\tau)^2 \leq \frac{1}{2} \int_0^{\pi/2} p'^2 + 2\tau^2 \int_0^\tau q'^2.$$

Collecting,  $D_\tau \leq (-\frac{3}{2} + 1 + \frac{1}{2}) \int_0^{\pi/2} p'^2 + (3\tau^2 - 1) \int_0^\tau q'^2 \leq 0$  for  $\tau^2 \leq \frac{1}{3}$ .

*Range (b):*  $\sigma := \pi/2 - \tau \leq \frac{1}{18}$ . The obstruction here is that the bound must degenerate:  $D_{\pi/2}$  annihilates the mode  $(p, q) = (0, \sin t)$ , so no estimate with a uniform negative constant exists near  $\tau = \pi/2$ , and the argument must see the mode. Expand  $q = a\varphi_1 + r$  in the Dirichlet–Neumann eigenbasis  $\varphi_k = \sin((2k-1)t)$  of  $[0, \pi/2]$ , with  $r \perp \varphi_1$  in both inner products, and set  $D := \int q'^2 - \int q^2 \geq 0$ . From the eigenvalues  $(2k-1)^2$ ,

$$\int r'^2 \leq \frac{9}{8}D, \quad \|r\|_{L^2}^2 \leq \frac{D}{8}, \quad r(t)^2 \leq t \int_0^t r'^2 \leq \frac{9\pi}{16}D. \quad (44)$$

Since  $\varphi_1'' = -\varphi_1$  and  $r(0) = 0$ , integration by parts gives  $\int_0^\tau (\varphi_1 r - \varphi_1' r') = -\varphi_1'(\tau)r(\tau) = -\sin \sigma r(\tau)$ , and  $\int_0^\tau (\varphi_1^2 - \varphi_1'^2) = -\frac{1}{2} \sin 2\sigma$ . Hence, using (44) and Young's inequality  $2XY \leq 3X^2 + \frac{1}{3}Y^2$  on the middle term,

$$2 \int_0^\tau (q^2 - q'^2) + \int_\tau^{\pi/2} (q^2 - q'^2) \leq -\left(\frac{7}{8} - \frac{3\pi}{16}\right)D - a^2 \sin \sigma (\cos \sigma - 3 \sin \sigma) \leq -\frac{2}{7}D - \frac{4}{5}\sigma a^2,$$

the last step using  $\frac{7}{8} - \frac{3\pi}{16} \geq \frac{2}{7}$  and  $\sin \sigma (\cos \sigma - 3 \sin \sigma) \geq \sigma(1 - 3\sigma - \sigma^2) \geq \frac{4}{5}\sigma$  for  $\sigma \leq \frac{1}{18}$ . For  $p$ , with  $P^2 := \int_\tau^{\pi/2} p'^2$  and the Dirichlet Poincaré constant  $(2\sigma/\pi)^2 \leq \frac{1}{400}$  on  $[\tau, \pi/2]$ ,

$$\int_0^\tau (p^2 - p'^2) + 2 \int_\tau^{\pi/2} (p^2 - p'^2) \leq -\frac{3}{4} \int_0^{\pi/2} p'^2 - \left(1 - \frac{1}{400}\right)P^2 \leq -\frac{174}{100}P^2.$$

The boundary term couples the pieces through  $p(\tau)^2 \leq \sigma P^2$  and  $q(\tau) = a \cos \sigma + r(\tau)$ :

$$2|p(\tau)||a| \cos \sigma \leq \frac{5}{4}P^2 + \frac{4}{5}\sigma \cos^2 \sigma a^2, \quad 2|p(\tau)||r(\tau)| \leq \frac{9}{25}P^2 + \frac{25}{9}\sigma \cdot \frac{9\pi}{16}D,$$

and  $\frac{25}{9} \cdot \frac{9\pi}{16} \sigma = \frac{25\pi}{16} \sigma \leq \frac{25\pi}{288} \leq \frac{2}{7}$  for  $\sigma \leq \frac{1}{18}$ . Summing the three displays, the coefficients of  $a^2$ ,  $D$  and  $P^2$  are respectively  $\frac{4}{5}\sigma(\cos^2 \sigma - 1) \leq 0$ , at most  $-\frac{2}{7} + \frac{25\pi}{288} \leq 0$ , and  $-\frac{174}{100} + \frac{5}{4} + \frac{9}{25} = -\frac{13}{100} < 0$ . Hence  $D_\tau \leq 0$ .

*Range (c):*  $\tau \in [0.55, 1.5153]$ . The form  $D_\tau$  is the  $2 \times 2$  Sturm–Liouville system of Theorem 88 with pieces of lengths  $\tau$ ,  $\pi/2 - \tau$  and coefficient rows  $(1, 2, -1, -2, 1, 1)$ ,  $(2, 1, -2, -1, 0, 0)$ , so



its eigenvalues are the zeros of the same shooting determinant  $\Phi(c; \tau)$ . Two facts combine. First, the spectrum is bounded below by  $-4$ : pointwise,  $2pq' \geq -(2p^2 + \frac{1}{2}q'^2)$  and  $2qp' \geq -(2q^2 + \frac{1}{2}p'^2)$  absorb the cross terms of  $-D_\tau$  leaving non-negative derivative coefficients, so  $-D_\tau[\eta] \geq -\int(3p^2 + 4q^2) \geq -4\|\eta\|^2$ . Second,  $\Phi(c; \tau) \neq 0$  for every  $(c, \tau) \in [-4, 0] \times [0.55, 1.5153]$ : the rectangle is covered adaptively by 305 boxes on which the ball-arithmetic enclosure of  $\Phi$  excludes zero, with the lengths carried as balls containing  $[\tau_{lo}, \tau_{hi}]$ . Hence no eigenvalue lies in  $[-4, 0]$  and none lies below  $-4$ , so the first eigenvalue is positive and  $D_\tau \leq 0$  (indeed  $< 0$ ) on this range.

The ranges  $[0, 1/\sqrt{3}]$ ,  $[0.55, 1.5153]$  and  $[\pi/2 - \frac{1}{18}, \pi/2]$  overlap and cover  $[0, \pi/2]$ .  $\square$

The covering was validated by negative controls: enlarged  $c$ -ranges crossing the measured first eigenvalue are refused, with the failing boxes localising the eigenvalue curve to  $(\tau, c) = (0.785, 0.377)$  for the mirror family below and  $(1.514, 0.041)$  for the diagonal, matching the measured values.

**Corollary 92** (Ordered anchored concavity).  *$\delta^2 Q \leq 0$  on every cell whose sign sets satisfy  $\{\alpha_1 < 0\} \subseteq \{\alpha_2 > 0\}$  with  $\{\alpha_2 > 0\} = [0, \tau]$  anchored. The set  $\{\alpha_1 < 0\}$  need not be an interval.*

**Theorem 93** (Uniform constant on the mirror family). *For every  $\tau \in [0, \pi/4]$  the mirror-anchored cell  $(\tau, \pi/2 - \tau)$  satisfies*

$$\frac{1}{2} \delta^2 Q[\eta] \leq -\frac{3}{10} \|\eta\|_{L^2(0, \pi)}^2.$$

*Proof.* The cell form is the system of Theorem 88 with lengths  $\tau$ ,  $\pi/2 - 2\tau$ ,  $\tau$ ;  $\Sigma$ 's cell is  $\tau = \beta$ . The lower bound  $-4$  on the spectrum holds as in range (c), and the covering of  $[-4, \frac{3}{10}] \times [0, 0.7854]$  by 249 boxes shows  $\Phi \neq 0$  there; since  $0.7854 > \pi/4$ , every  $\tau \in [0, \pi/4]$  is included. (For  $\tau$  slightly past  $\pi/4$  the middle length crosses zero and the transfer formula is an analytic continuation of no further relevance; the enclosure covers every real  $\tau$  in each ball, in particular the range where the formula is the shooting determinant of the cell.) Hence the first eigenvalue exceeds  $\frac{3}{10}$  on the whole family. Measured, it decreases from 0.875 at  $\tau = 0$  to 0.370 at  $\tau = \pi/4$ , with 0.7331 at  $\Sigma$ 's cell.  $\square$

## 21 Stability and uniqueness

Because  $Q$  is exactly quadratic on the cell and  $\delta Q(\Sigma) = 0$ , the same ingredients give more than a bound.

**Theorem 94** (Stability). *For every ambidextrous moving sofa  $T$  whose cap data  $H = H_\Sigma + \eta$  lies in  $\mathcal{D}$ ,*

$$|T| \leq A_R^* - \frac{73}{100} \|\eta\|_{L^2(0, \pi)}^2. \quad (45)$$

*Proof.* On the cell  $\mathcal{C}$  the functional  $Q$  is a quadratic polynomial in  $H$ , so  $Q(H_\Sigma + \eta) = Q(\Sigma) + \delta Q(\Sigma)[\eta] + \frac{1}{2} \delta^2 Q[\eta]$  exactly. The first variation vanishes by Theorem 57 and  $Q(\Sigma) = A_R^*$  by Theorem 64, while Theorem 88 bounds the second variation. Combining with  $|T| \leq Q(H)$  from Theorem 84 gives (45).  $\square$

**Corollary 95** (Uniqueness).  *$\Sigma$  is the unique ambidextrous moving sofa of area  $A_R^*$  whose cap data lies in  $\mathcal{D}$ ; every other one has strictly smaller area, by the amount (45).*

*Proof.* Equality in (45) forces  $\|\eta\|_{L^2(0,\pi/2)} = 0$ , and then  $\eta \equiv 0$  on  $[0, \pi/2]$ ; Theorem 60 shows  $\delta^2 Q < 0$  strictly unless  $\eta \equiv 0$  throughout, since with  $\eta \equiv 0$  on  $[0, \pi/2]$  the  $E_2$  term of (24) contributes  $-\int_0^{\pi/2-\beta} \eta'(t + \pi/2)^2 dt < 0$  unless  $\eta$  is constant on  $[\pi/2, \pi]$ , and  $\eta(\pi/2) = 0$  makes that constant zero.  $\square$

The domain  $\mathcal{D}$  is cut out in part by (RC), and (RC) enters at exactly one point: it forces the flux excess  $E := V - |N|$  to vanish, so that  $Q$  is tight. Separation and Reynolds give, with no curvature hypothesis,

$$|T| = |C_2| - 2|N| = Q + 2E, \quad E \geq 0. \quad (46)$$

It is natural to ask for a concave majorant of  $E$ , which would replace  $\{E = 0\}$  by something larger. There is none.

**Proposition 96** (No concave majorant). *Let  $\mathcal{D}'$  be convex with  $\Sigma$  in its algebraic interior, and let  $\tilde{E} : \mathcal{D}' \rightarrow \mathbb{R}$  be concave with  $\tilde{E} \geq 0$  and  $\tilde{E}(\Sigma) = 0$ . Then  $\tilde{E} \equiv 0$ , and hence  $E \equiv 0$  on  $\mathcal{D}'$ .*

*Proof.* Let  $a, b \in \mathcal{D}'$  with  $\Sigma = \frac{1}{2}(a + b)$ . Concavity gives  $0 = \tilde{E}(\Sigma) \geq \frac{1}{2}\tilde{E}(a) + \frac{1}{2}\tilde{E}(b) \geq 0$ , so  $\tilde{E}(a) = \tilde{E}(b) = 0$ . As  $\Sigma$  is interior, every point of  $\mathcal{D}'$  arises this way.  $\square$

So any architecture with a concave, tight majorant is confined to the injectivity locus. The hypothesis to weaken is concavity of  $\tilde{E}$ : what the argument needs is that  $Q + 2\tilde{E}$  be concave, and the strict concavity of  $Q$  leaves room.

**Theorem 97** (Excess budget). *Let  $\mathcal{D}'$  be convex with  $\Sigma \in \mathcal{D}'$  and  $\theta \in [0, 1]$ . Suppose  $M < \frac{1}{2}$  on  $\mathcal{D}'$ , that  $\frac{1}{2}\delta^2 Q \leq -c\|\eta\|_{L^2(0,\pi)}^2$  there, and that*

$$E(H) \leq \frac{1}{2} \theta c \|H - H_\Sigma\|_{L^2(0,\pi)}^2 \quad \text{on } \mathcal{D}'. \quad (47)$$

*Then  $|T| \leq A_R^* - (1 - \theta)c \|H - H_\Sigma\|_{L^2(0,\pi)}^2$  for every ambidextrous moving sofa  $T$  with cap data in  $\mathcal{D}'$ .*

*Proof.* Concavity of  $Q$  with  $\delta Q(\Sigma) = 0$  gives  $Q(H) \leq A_R^* - c\|\eta\|^2$ . Adding (46) and (47),  $|T| = Q + 2E \leq A_R^* - c\|\eta\|^2 + \theta c\|\eta\|^2$ .  $\square$

At  $\theta = 0$  this is Theorem 94. For  $\theta > 0$  it admits  $E > 0$ , so Proposition 96 is evaded, and the room available is proportional to  $c$ : the constant  $c^* = 0.7309566$  of Theorem 88 is what makes (47) usable. Measured along  $H_\Sigma + \varepsilon\phi$  with  $\phi$  the bump at  $\theta_0 = 1$  of half-width 0.45, the least admissible  $\theta$  is

|               |        |        |        |        |        |        |
|---------------|--------|--------|--------|--------|--------|--------|
| $\varepsilon$ | 0.0141 | 0.0200 | 0.0283 | 0.0400 | 0.0566 | 0.0800 |
| $\theta$      | 0.100  | 0.301  | 0.683  | 1.276  | 2.090  | 3.106  |

so  $\theta < 1$  holds out to  $\varepsilon \approx 0.031$ , against  $\varepsilon_c = 0.006492$  for (RC): the admissible radius grows by a factor of about 4.8 in that direction. This is a measurement, not a proof.

**Theorem 98** (Stability on the mirror class). *Let  $T$  be a connected ambidextrous moving sofa whose cap data  $H$  satisfies the gauge, the mirror condition (49), and (RC), and suppose that along the segment  $H_s = (1 - s)H_\Sigma + sH$  the set  $\{\alpha_1(\cdot; H_s) < 0\}$  is an interval  $[0, \tau_1(s))$  with  $\tau_1(s) \leq \pi/4$  for every  $s$ . Then*

$$|T| \leq A_R^* - \frac{3}{10} \|H - H_\Sigma\|_{L^2(0,\pi)}^2,$$

*and equality forces  $H = H_\Sigma$ .*

*Proof.* The mirror condition is linear and holds at  $H_\Sigma$ , so every  $H_s$  satisfies it, and by Lemma 104 each cell met by the segment is mirror-anchored with parameter  $\tau_1(s) \leq \pi/4$ . By Theorem 93,  $g(s) := Q(H_s)$  has  $g''(s) \leq -\frac{3}{5}\|H - H_\Sigma\|^2$  on each cell,  $g$  is  $C^1$  across cell boundaries as in Proposition 107,  $g(0) = A_R^*$  and  $g'(0) = 0$ ; integrating,  $Q(H) \leq A_R^* - \frac{3}{10}\|H - H_\Sigma\|^2$ . Connectedness gives  $M \leq \frac{1}{2}$  (Theorem 11), and (RC) gives  $V = |N|$ , so  $|T| = |C_2| - 2|N| = Q(H)$ .  $\square$

**Lemma 99** (The arm sandwich). *Suppose  $0 \leq (H + H'')_{\text{ac}} \leq 1$  on  $(0, \pi)$ , with atoms only at  $\pi/2$ . Then on  $(0, \pi/2)$ , in the absolutely continuous sense,*

$$\alpha_2 \leq \alpha'_1 \leq \alpha_2 + 1, \quad -\alpha_1 - 1 \leq \alpha'_2 \leq -\alpha_1.$$

*The upper bounds use convexity ( $H + H'' \geq 0$ ), the lower bounds use (RC).*

*Proof.*  $\alpha'_1 = G' - F''$  and  $-F'' \in [F - 1, F]$  by the two-sided bound on  $F + F''$ , so  $\alpha'_1 \in [G' + F - 1, G' + F] = [\alpha_2, \alpha_2 + 1]$ . Likewise  $\alpha'_2 = F' + G''$  with  $G'' \in [-G, 1 - G]$  gives  $\alpha'_2 \in [-\alpha_1 - 1, -\alpha_1]$ .  $\square$

The pair  $(\alpha_1, \alpha_2)$  therefore rotates: where  $\alpha_2 > 0$  the first arm strictly increases, and where  $\alpha_1 > 0$  the second strictly decreases. This is the differential form of the oscillator mechanism behind Theorem 52, and it makes the sign hypothesis of Theorem 98 checkable from the arms alone. It does not make it removable:

**Lemma 100** (Forced boundary derivatives). *Under (RC),  $H'(0) = \frac{1}{2}$  and  $H'(\pi) = -\frac{1}{2}$ .*

*Proof.*  $C_2$  is  $\rho$ -symmetric, so its support function satisfies  $h(\theta) = h(-\theta) + \sin \theta$  on the whole circle; this determines  $h$  on  $[\pi, 2\pi]$  from  $H$  on  $[0, \pi]$ . On  $(\pi, 2\pi)$ , writing  $\psi = 2\pi - \theta$ , one finds  $h + h'' = (H + H'')(\psi)$ , so no condition arises there. At the two junctions the extension has derivative jumps

$$h'(0^+) - h'(0^-) = 2H'(0) - 1, \quad h'(\pi^+) - h'(\pi^-) = -2H'(\pi) - 1,$$

which are the atoms of the surface area measure at  $\theta = 0$  and  $\theta = \pi$ . Convexity forces both to be non-negative; (RC) permits atoms only at  $\pm\pi/2$ , so both vanish.  $\square$

For  $\Sigma$  these read 0.4999999990 and  $-0.4999999987$  in the computed data, the deviation being the finite-difference error. Lemma 100 is what makes the class of admissible caps rigid at the endpoints, and it is what selects a single member of the following family.

**Remark 101** (The sign hypothesis is not removable). Consider  $H_c(\theta) = 1 - c + c(\sin \theta + \cos \theta)$ , the support data of the disc of radius  $1 - c$  centred at  $(c, c)$ . It satisfies the gauge and the mirror condition (49) with  $A = 2c$ , and  $H_c + H_c'' \equiv 1 - c$ , so the absolutely continuous part of (RC) holds for every  $c \in (0, 1)$ . By Lemma 100 only  $c = \frac{1}{2}$  is admissible:  $H'_c(0) = c$ , and  $c < \frac{1}{2}$  makes the junction atoms negative, so the  $\rho$ -extension is not convex at all, while  $c > \frac{1}{2}$  creates atoms at 0 and  $\pi$  that (RC) forbids. At  $c = \frac{1}{2}$  the body is the disc of radius  $\frac{1}{2}$  centred at  $(\frac{1}{2}, \frac{1}{2})$ , the incircle of the unit square, and every condition holds with  $H + H'' \equiv \frac{1}{2}$ . Its arms are

$$\alpha_1 \equiv \alpha_2 \equiv -\frac{1}{2},$$

so  $\{\alpha_1 < 0\} = [0, \pi/2)$  while  $\{\alpha_2 > 0\} = \emptyset$ , and the ordering fails identically. Moreover  $|N| = 0$  and  $|C_2| = \pi/4$  (both the cap formula and a direct polygon computation give 0.785398163), so the maximal ambidextrous sofa with this cap datum is the disc itself, and a disc of diameter

1 moves freely through the corridor and turns either corner. The datum is therefore realized by a connected ambidextrous moving sofa. Hence the ordered anchored sign structure follows neither from the gauge, mirror, convexity and (RC), nor from realizability, and the hypothesis of Theorem 98 is permanent rather than provisional.

**Remark 102** (A hypothesis of the niche formula, made explicit). The disc also isolates an assumption implicit in (20). The face-1 segment is  $s \in [\alpha_1^+, \sigma]$ , so its contribution is  $\frac{1}{2}((\sigma - \alpha_1^+)^+)^2$ , and writing it as  $\frac{1}{2}(\sigma - \alpha_1)^2$  presupposes

$$\sigma \geq \alpha_1^+ \quad \text{on } [0, \pi/2]. \quad (48)$$

For the disc  $\alpha_1^+ = 0$  while  $\sigma$  decreases to  $-\frac{1}{2}$ , and the unmodified expression returns  $V = -0.1427$  against  $|N| = 0$ , in apparent conflict with Proposition 40; with the segment truncated,  $V = 0 = |N|$  and the conflict disappears. For  $\Sigma$ , (48) holds:  $\sigma - \alpha_1 = \cos t x'(t)$  by Proposition 61 and  $x$  is strictly increasing by Proposition 42, while  $\sigma \geq 0$  throughout; the computed minimum of  $\sigma - \alpha_1^+$  is  $+2.5 \cdot 10^{-7}$ , attained as  $t \rightarrow \pi/2$ . All uses of (20) in this note are on data satisfying (48).

**Proposition 103** (An explicit ball inside the domain). *Let  $r_0 = \frac{1}{20}$ . Every  $\eta$  with  $\eta(0) = \eta(\pi/2) = 0$ , satisfying the mirror condition, and with  $\max(\|\eta\|_\infty, \|\eta'\|_\infty, \|\eta''\|_\infty) \leq r_0$ , gives  $H = H_\Sigma + \eta$  satisfying (RC),  $M(H) < \frac{1}{2}$ , and the sign hypothesis of Theorem 98 along the entire segment  $[H_\Sigma, H]$ . In particular the stability bound applies to every connected ambidextrous sofa with such cap data.*

*Proof.* (RC):  $\max(H_\Sigma + H''_\Sigma)_{ac} = \frac{3}{4}f_1\sqrt{4 - 2\sqrt{2}}\cos(\frac{\beta}{2} + \frac{\pi}{8}) = 0.8388253\dots$  in closed form, and  $\|\eta + \eta''\|_\infty \leq 2r_0 = 0.1 \leq 0.1611$ . Ceiling:  $M(\Sigma) = 0.38784\dots$  in closed form, and  $|\delta c_y| \leq \|\eta\|_\infty(\sin t + \cos t) \leq \sqrt{2}r_0 = 0.0708$ , so  $M(H) \leq 0.4586 < \frac{1}{2}$ . Signs: the perturbations obey  $|\delta\alpha_i| \leq \|\eta\|_\infty + \|\eta'\|_\infty \leq 2r_0 = 0.1$ , and six margins of the unperturbed arms are certified on grids of spacing  $h = 5 \cdot 10^{-4}$  with the Lipschitz constant  $L = 2$  supplied by Lemma 99: writing  $S := \sup(|\alpha_1| \vee |\alpha_2|)$ , the sandwich gives  $\sup|\alpha'_i| \leq S + 1$ , and the grid maximum 0.7506 bootstraps to  $S \leq 0.7506 + (S + 1)h/2$ , hence  $S \leq 0.7513$  and  $L \leq 1.76 \leq 2$ . The certified margins are:  $\alpha_1 \leq -g$  on  $[10^{-3}, \beta - d]$  and  $\alpha_1 \geq g$  on  $[\beta + d, \pi/2]$ ;  $\alpha_2 \geq g$  on  $[10^{-3}, \pi/2 - \beta - d]$  and  $\alpha_2 \leq -g$  on  $[\pi/2 - \beta + d, \pi/2]$ ; and  $\alpha_2 \geq 0.5975$ ,  $\alpha_1 \geq 0.5975$  on the windows  $[\beta - d, \beta + d]$ ,  $[\pi/2 - \beta - d, \pi/2 - \beta + d]$  respectively, with  $d = 0.15$  and  $g = 0.1224 > 2r_0$ . Outside the windows the perturbed arms keep their signs. Inside  $\alpha_1$ 's window, Lemma 99 gives  $\alpha'_1 \geq \alpha_2 \geq 0.5975 - 0.1 > 0$  for the perturbed data, so  $\alpha_1$  crosses zero exactly once there; likewise  $\alpha'_2 \leq -\alpha_1 \leq -(0.5975 - 0.1) < 0$  in its window. So  $\{\alpha_1 < 0\} = [0, \tau_1]$  with  $\tau_1 < \beta + d \leq \pi/4$  and  $\{\alpha_2 > 0\} = [0, \tau_2]$  with  $\tau_2 > \pi/2 - \beta - d > \beta + d$ , ordered and anchored. Replacing  $\eta$  by  $s\eta$ ,  $s \in [0, 1]$ , only shrinks the perturbation, so the same holds along the segment.  $\square$

The grid evaluations use double precision; the margins exceed the evaluation error by eleven orders of magnitude, and the Lipschitz constant is proved, not sampled. This replaces, in the load-bearing direction, the numerical statement that  $\mathcal{D}$  is a  $C^2$ -neighbourhood: the domain contains the explicit  $C^2$ -ball of radius  $\frac{1}{20}$  within the mirror class. The outer containment (that the radius along  $\sin(2k\theta)$  shrinks like  $k^{-2}$ ) remains a measurement.

The hypotheses of Theorem 98 are non-perturbative: no smallness of  $H - H_\Sigma$  is assumed, and the class contains data as far from  $\Sigma$  as the mirror caps with  $\tau_1$  anywhere in  $[0, \pi/4]$ . The binding smallness constraint that remains in  $\mathcal{D}$  is (RC) alone.

The sign cell is the least structural part of  $\mathcal{D}$ . On a natural linear subclass it collapses to a single scalar inequality.

**Lemma 104** (Mirror caps). *Suppose the cap is symmetric about some vertical line, equivalently*

$$H(\theta) - H(\pi - \theta) = A \cos \theta, \quad A := H(0) - H(\pi). \quad (49)$$

*Then  $\alpha_2(\pi/2 - t) = \alpha_1(t)$  for every  $t \in [0, \pi/2]$ . Consequently  $E_2 = \frac{\pi}{2} - E_1$  as sets, and if  $E_1 = [0, \tau_1)$  then  $E_2 = [0, \frac{\pi}{2} - \tau_1)$ , so*

$$\tau_1 + \tau_2 = \frac{\pi}{2}.$$

*Proof.* Condition (49) says  $h(u) - h(\sigma u) = \langle (A, 0), u \rangle$  for the reflection  $\sigma(x, y) = (-x, y)$ , that is, the cap coincides with its mirror image translated by  $(A, 0)$ ; the axis is  $x = A/2$ . Replacing  $\theta$  by  $\pi/2 + t$  in (49) gives  $H(\pi/2 - t) = H(\pi/2 + t) + A \sin t$ , and differentiating (49) gives  $H'(\pi - \theta) = -H'(\theta) - A \sin \theta$ . Hence

$$\alpha_2(\frac{\pi}{2} - t) = H(\frac{\pi}{2} - t) - 1 + H'(\pi - t) = [H(\frac{\pi}{2} + t) + A \sin t] - 1 + [-H'(t) - A \sin t] = H(\frac{\pi}{2} + t) - 1 - H'(t) = \alpha_1(t). \quad \square$$

Condition (49) is linear in  $H$  once  $A$  is treated as a free parameter, so the mirror caps form a linear subspace and intersecting  $\mathcal{D}$  with it preserves convexity. Both  $\Sigma$  and Gerver's cap satisfy it, to  $7 \cdot 10^{-16}$  in the computed data. On that subclass the anchored sign pattern is fixed by the single number  $\tau_1$ , and the order condition  $\tau_1 \leq \tau_2$  of Proposition 107 becomes

$$\tau_1 \leq \frac{\pi}{4},$$

which at  $\Sigma$  reads  $\beta = 0.289654 \leq 0.785398$ , a margin of 0.4957. Lemma 104 is also what makes  $L_1 + L_2 = \pi/2$  exact in (41), hence  $\Lambda(0) = 1$  exactly: the two lengths there are  $\tau_2$  and  $\tau_1$ .

**Remark 105** (Why Gerver's cap is excluded). Gerver's cap satisfies (49) and, by Theorem 97, the excess budget along the whole segment  $[H_\Sigma, H_G]$  at  $\theta = 0.0072$ ; and (RC) fails on that segment only at  $s = 0.398$ . None of these is what excludes it. Along the segment  $M = \max_t c_y(t)$  crosses  $\frac{1}{2}$  at  $s = 0.406$ , and at Gerver  $M = 0.6643$ , so by Theorem 11 any ambidextrous sofa with that corner path omits a vertical strip of width 0.1646 and a connected one lies entirely on one side of it, contradicting that it meets both arms. So *no connected ambidextrous moving sofa has Gerver's cap data*: the exclusion is a property of the competitor class, not a limitation of the curvature hypothesis. Weakening (RC) cannot reach Gerver, and should not.

**Remark 106** (Disconnected competitors). Connectedness is used in exactly one place, Remark 105, and it is load-bearing there: when  $M = \max_t c_y(t)$  exceeds  $\frac{1}{2}$  the two niches overlap, and inclusion-exclusion gives  $|S| = |C_2| - 2|N| + |U \cap \rho U| > Q(H)$ , so overlap *increases* area and cannot be assumed away on extremal grounds. A moving sofa is connected by definition, so the hypothesis is part of the problem rather than an added assumption, but that is a weak answer if the excluded competitors would win. They do not.

Rasterising the intersection directly, rather than through the cap and niche bookkeeping that is exactly what fails when the niches overlap,

$$S = \bigcap_t [\text{Hall}(t) \cap \rho \text{Hall}(t)],$$

$\Sigma$  returns 1.64548 against  $A_R^* = 1.6449552$ . Refining the spatial grid from  $400^2$  to  $2000^2$  and the rotation sampling from 120 to 900 angles moves the value by at most 0.0015, which is the error bar for everything below. Over admissible caps drawn with (RC) holding by construction

and the forced boundary data exact, 46 samples reached  $M \geq \frac{1}{2}$ , and the largest area among them was 1.6032, below  $A_R^*$  by 0.042, some 28 times the error bar. The overlap gain is real but the caps that produce it lose more elsewhere.

So dropping connectedness does not appear to raise the supremum, and Remark 105 is not smuggling in a restriction that does work. Rule 7: this is HEURISTIC, a rasterised measurement over 55 caps, and it is not a proof that no disconnected competitor beats  $A_R^*$ .

**Proposition 107** (Beyond  $\Sigma$ 's own cell).  *$Q$  is  $C^1$  and piecewise quadratic, so if every cell met by the segment  $[H_\Sigma, H]$  has  $\delta^2 Q \leq 0$ , then  $Q$  is concave along that segment and  $Q(H) \leq Q(\Sigma) + \delta Q(\Sigma)[H - H_\Sigma] = A_R^*$ . Convexity of the union of those cells is not needed. In particular this holds whenever the segment meets only cells with  $\{\alpha_1 < 0\} \subseteq \{\alpha_2 > 0\}$  and  $\{\alpha_2 > 0\}$  anchored, which is no longer a measured condition: Corollary 92 proves concavity on every such cell. The eleven sample values  $-0.8751, -0.1404, -0.3602, -0.5397, -0.4497, -0.3431, -0.2285, -0.1074, -0.0457, -0.7876, -0.042$  at  $(\tau_1, \tau_2) = (0, \frac{\pi}{2}), (0.1, 0.2), (0.3, 0.5), (0.5, 0.9), (0.8, 1.2), (1.0, 1.3), (1.2, 1.4), (1.4, 1.5), (1.5, \frac{\pi}{2}), (0.2, 1.5), (0.05, 0)$  all negative, stand as a numerical cross-check, with  $\Sigma$ 's own cell giving  $-0.7331$  at  $m = 128$  (converged value  $-0.7309566$  by Theorem 88).*

Since  $\Sigma$  sits at  $(\beta, \pi/2 - \beta)$  with  $\beta < \pi/2 - \beta$ , it lies in that family, and the class of competitors covered is strictly larger than  $\mathcal{D}$  and is checkable from the sign pattern alone. (The restriction to *anchored* intervals is essential: Remark 38 exhibits  $E_1 = E_2 = [0.4, 1.2]$ , unanchored, with  $\sup \frac{1}{2} \delta^2 Q / \|\eta\|_{L^2(0, \pi)}^2 = +0.4123$ .)

The constant in (45) is 99.9% of the sharp  $c^* = 0.7309566 \dots$ . What closed the earlier factor of 4.8 was giving up the pointwise splitting in two stages: Theorem 60 uses three separate Poincaré constants, of which  $P_3 = 1$  is marginal and  $P_2 = 1.5035$  nearly so; Theorem 87 replaces all three by two weighted eigenvalues that see the whole of each half at once; and Theorem 88 replaces those two by one system that sees both halves at once, losing nothing.

A correction this forced. Earlier drafts quoted the sharp constant as 0.7323, the P1 value at  $m = 256$ . Rayleigh–Ritz gives an upper bound on an eigenvalue and that sequence had not converged; it decreases through 0.731311, 0.731247, 0.731073 at  $m = 512, 1024, 2048$  towards  $c^* = 0.7309566$ . Every certificate in this note is a lower bound and every finite-element number an upper bound, so nothing crossed and no conclusion moves; the target was mis-stated by 0.2%.

**Proposition 108** ( $H^1$  stability, measured). *The second variation controls the derivative as well. Measured against assembled mass and stiffness matrices, the largest  $c_0$  with  $\frac{1}{2} \delta^2 Q[\eta] \leq -c_0 \|\eta\|_{L^2(0, \pi)}^2 - c_1 \|\eta'\|_{L^2(0, \pi)}^2$  is*

| $c_1$     | 0      | 0.1    | 0.3    | 0.5    |
|-----------|--------|--------|--------|--------|
| $m = 64$  | 0.7339 | 0.6132 | 0.3595 | 0.0796 |
| $m = 128$ | 0.7331 | 0.6123 | 0.3583 | 0.0779 |
| $m = 256$ | 0.7323 | 0.6114 | 0.3571 | 0.0760 |
| $m = 512$ | 0.7313 | 0.6103 | 0.3556 | 0.0739 |

*These are Rayleigh–Ritz values, hence upper bounds on the true constants, and every column is still decreasing at  $m = 512$ : the  $c_1 = 0$  column is the one whose limit is known independently, and it is descending towards the certified  $c^* = 0.7309566 \dots$ , not towards its own  $m = 256$  entry. So the table bounds the admissible constants from above and does not establish any of the inequalities. Only the  $c_1 = 0$  case is proved, by Theorem 88. Rule 7: the  $c_1 > 0$  entries are HEURISTIC, and the frontier  $c_0 \approx 0.731 - 1.31 c_1$  read off them, vanishing near  $c_1 = 0.56$ , is a measurement rather than a theorem.*



**Remark 109** (Why the earlier statement was wrong). An earlier version of this proposition printed the  $m = 256$  row alone, asserted that the entries were converged, and concluded  $\frac{1}{2}\delta^2 Q \leq -0.357\|\eta\|^2 - 0.3\|\eta'\|^2$ . The assertion was false in exactly the way the paragraph above this proposition already describes for 0.7323: a Rayleigh–Ritz sequence that has not converged, read as if it had. The same mesh value appears in both places, which is how the inconsistency survived. Reproducing a printed number at the mesh it was computed on confirms its provenance, not its correctness.

Two simplifications are worth recording. First, the  $\sigma$  term has no singularity: since  $-2 \int \eta \eta' \tan t = -\int (\eta^2)' \tan t = \int \eta^2 \sec^2 t$  when  $\eta(0) = \eta(\pi/2) = 0$ , and  $\sec^2 t - \tan^2 t = 1$ ,

$$-\int_0^{\pi/2} (\eta \tan t + \eta')^2 dt = \int_0^{\pi/2} (\eta^2 - \eta'^2) dt,$$

checked to  $10^{-12}$  on five test functions, so (24) may be written with bounded coefficients throughout. Second, the sharp  $L^2$  constant is 0.7309566, not the 0.7285 of an earlier draft; that figure came from a mass matrix that gave the endpoint hat at  $\theta = \pi$  the interior weight  $2h/3$  instead of  $h/3$ .

**Remark 110** (The size of  $\mathcal{D}$ , and what Theorem 84 is not). It is not a proof that  $\Sigma$  is optimal, and the reason is quantitative.

Measuring the radius of  $\mathcal{D}$  along  $H_\Sigma + \varepsilon \sin(2k\theta)$  gives  $\varepsilon_k = 0.0864, 0.0117, 0.0047, 0.0036, 0.0021$  at  $k = 2, 4, 6, 8, 10$ , so  $\varepsilon_k k^2$  stays between 0.17 and 0.35: the radius scales like  $k^{-2}$ .  $\mathcal{D}$  therefore contains a ball in the  $C^2$  norm and in no weaker one, which is exactly what a bound on  $H''$  should give. It is a genuine ball rather than a sliver, of 60 random eight-mode perturbations with coefficients of size  $0.02/k^2$ , 51 land but every member is a small  $C^2$ -perturbation of  $\Sigma$ . **Theorem 84 is thus a  $C^2$ -local statement**, with explicit convex hypotheses in place of a neighbourhood of unspecified size.

In every one of those probes the binding constraint was (b); condition (c) never became active. That is a statement about the probe and not about the geometry: (c) is *independent* of (b). Adding  $\delta \sin(\theta - \pi/2)$  on  $[\pi/2, \pi]$  and 0 on  $[0, \pi/2]$  leaves  $H + H''$  identically unchanged, so (b) and the gauge hold exactly, while the ceiling facet grows by  $\delta$  and  $c_y$  grows by  $\delta \sin t \cos t$ ; at  $\delta = 0.2243\dots$  the maximum  $M$  crosses  $\frac{1}{2}$ . So (c) cannot be derived and must remain a hypothesis.

$\mathcal{D}$  does not reach Gerver’s cap, and by Remark 111 injectivity genuinely fails once the density exceeds 1. The failure is sharp in the condition but gentle in magnitude: comparing the flux  $\frac{1}{2} \int (\alpha_2^+)^2$  with the area actually swept, the excess is  $5.4 \cdot 10^{-7}$  at density 1.047 and  $7.4 \cdot 10^{-5}$  at 1.267, and Gerver’s own excess is about 0.25%. Widening  $\mathcal{D}$  therefore does not need a weaker constant; it needs an upper bound on that excess which is *concave* in  $H$ , so that subtracting it preserves the concavity of  $Q$ . Truncating the sweep at  $\inf_{t'} s_{22}(t, t')$  fails because that infimum is concave in  $H$  and  $\frac{1}{2}\lambda^2$  then loses the convexity the Mamikon step supplies; the first-capture decomposition is exact but its parameter domain stops being of the form  $0 \leq s \leq \alpha_2$ , so its area is no longer quadratic in  $H$ . So the honest reading is that the one-corner architecture transfers to the ambidextrous problem on an explicit convex  $C^2$ -neighbourhood of  $\Sigma$ , and that enlarging that neighbourhood is where the remaining difficulty sits.

**Remark 111** ((RC) is sharp). The constant 1 in the forcing  $R = 1 - (G + G'')$  of Theorem 52 is the corridor width: it enters through  $c(t) = (F - 1)\mu_t + (G - 1)\nu_t$ , so  $\langle c(t), \nu_t \rangle = n + G - 1$



and  $n'' = -n + 1 - (G + G'')$ . The argument therefore cannot tolerate a density above 1. Nor can the conclusion: perturbing  $\Sigma$  so that the maximum density passes through 1, the face-2 sweep acquires self-intersections exactly there. At densities 0.838, 0.921, 0.976 there are none; at 1.004, 1.047, 1.119 there are 160, 1510, 3086 among  $\binom{301}{2}$ -many tested pairs, and a second mode gives 0 at 0.976 against 200 and 526 at 1.045 and 1.114. So (b) is sharp as stated.

## 22 Formalisation

Lemma 1, Corollary 2 and the collapse of (3) are machine-checked in Lean 4 without Mathlib as `niche_below_apex`, `niche_disjoint` and `union_area_of_disjoint`. The identity underlying  $\Sigma$ 's middle-phase equation is also formalised: writing  $D$  for the formal derivative in the half-angle,  $D(Dv) = -(v + 1)$  for every arc of constant term  $-1$ , which is  $4v'' = -(v + (1, 1))$  for the bracket of (4) (`Trig.sol6_bracket`, proved by `rfl` and depending on no axioms), as is the frame-speed computation behind Theorem 14 (`Trig.sol1_speed_mu`) and the covering criterion of Theorem 11 (`strip_covers_iff`).

Two steps of the unconditional bound are formalised, chosen as the ones where a slip would be fatal and where no analysis is needed to state them. The four-piece decomposition of Lemma 74 is a fact about a linear order, so  $\mathbb{Z}$  is a faithful model: `four_piece` states that a point avoids both removed open intervals exactly when it lies in one of the four pieces, `four_piece_no_cross` specialises to the case where both cross pieces are empty, and `cross_absorbed` is the containment that makes Lemma 78 need no new estimate. The maximum of the profile integral reduces to  $2x^2 + 2(\sqrt{2} - x)^2 = 2 + (2x - \sqrt{2})^2$ , an identity in  $\mathbb{Z}[\sqrt{2}]$ , which is built explicitly rather than posited: a hypothesis  $S^2 = 2$  over  $\mathbb{Z}$  has no witness and would make the statement vacuous, so  $\mathbb{Z}2$  carries the ring structure and `sqr2_sq` exhibits the model, after which `profile_identity` is the identity and `equality_case` locates the maximiser at  $2x = \sqrt{2}$ .

The profile integral itself is now verified, not merely its middle-range formula. The step is `integral_tent_le`: for every  $x$ ,  $\int_0^{\sqrt{2}} (2 - 2|s - x|)^+ ds \leq 2\sqrt{2} - 1$ , with `integral_tent_at_mid` giving equality at  $x = \sqrt{2}/2$ , so the pair says  $2\sqrt{2} - 1$  is the supremum. The Lean proof is shorter than the case analysis of Proposition 75 because it avoids it: the identity  $\max(0, y) = y + \max(0, -y)$  turns the truncation into a correction term,

$$(2 - 2|s - x|)^+ = 2 - 2|s - x| + 2(|s - x| - 1)^+,$$

valid with no hypothesis on  $s$ , and since  $|y| - (|y| - 1)^+ = \min(|y|, 1)$  the bound becomes

$$\int_0^{\sqrt{2}} \min(|s - x|, 1) ds \geq \frac{1}{2} \quad \text{for every } x \in \mathbb{R},$$

which is `half_le_integral_cap`. That has a one-line proof: split the band at its midpoint and pair  $s$  with  $s + \sqrt{2}/2$ . The triangle inequality gives  $|s - x| + |s + \sqrt{2}/2 - x| \geq \sqrt{2}/2$ , and  $\min(a, 1) + \min(b, 1) \geq \min(a + b, 1)$  with  $\sqrt{2}/2 < 1$  turns that into  $\min(|s - x|, 1) + \min(|s + \sqrt{2}/2 - x|, 1) \geq \sqrt{2}/2$ ; integrating over  $[0, \sqrt{2}/2]$  gives  $(\sqrt{2}/2)^2 = \frac{1}{2}$ . Equality throughout at  $x = \sqrt{2}/2$ . The  $-1$  in  $2\sqrt{2} - 1$  is exactly twice this  $\frac{1}{2}$ , which is the same  $-1$  that separates the constant from Hammersley's  $2\sqrt{2}$ .

`lake build` is clean with no `sorry` across the development, and `#print axioms` reports nothing beyond `propext`, `Quot.sound` and `Classical.choice`. The second variation is partially formalised. Both decoupling inequalities of Theorem 87 are verified outright, each as an exact

square over the field (`decouple_E2`, `decouple_E1`), and the assembly of Proposition 85 from its analytic inputs to the constant  $\frac{1}{12}$  is verified as pure arithmetic (`secondvar_assembly`); three of the five inputs are theorems: the absorption inequality, the subset monotonicity and the tail cut are verified for continuously differentiable data (`habs_verified`, `hsub_verified`, `hcut_verified`), all from one engine,  $(f(b) - f(a))^2 \leq (b - a) \int_a^b f'^2$  (`sq_sub_le`), proved by monotonicity of  $t \mapsto (t - a) \int_a^t f'^2 - (f(t) - f(a))^2$ , whose derivative is  $\int_a^t (f'(s) - f'(t))^2 ds \geq 0$ ; no Cauchy–Schwarz machinery is needed. The two Wirtinger inequalities are also verified, and by a route worth recording: the classical ground-state substitution divides by  $\sin$ , which vanishes at the Dirichlet end, and the resulting integration by parts has a removable singularity that is expensive to formalise. Shifting the Riccati supersolution avoids it entirely. The function  $v = k \cot(kt + c)$  with  $c > 0$  is regular on all of  $[0, T]$  when  $kT + c < \pi$  and satisfies  $v' + v^2 + k^2 = 0$  exactly, so the pointwise square  $(f' \sin - kf \cos)^2 \geq 0$  and the ordinary fundamental theorem of calculus give

$$k^2 \int_0^T f^2 \leq \int_0^T f'^2 - v(T)f(T)^2 + v(0)f(0)^2,$$

with no limits taken anywhere (`riccati_block`). At the Dirichlet end  $f(0) = 0$  absorbs the  $v(0)$  term however large  $\cot c$  is, and the loss  $k \tan(c) f(T)^2$  at the other end is controlled by the verified engine  $f(T)^2 \leq T \int f'^2$ . The result is  $\int f^2 \leq (1 + \pi c) \int f'^2$  for the Dirichlet–Neumann problem and  $4 \int f^2 \leq (1 + 2\pi c) \int f'^2$  for the Dirichlet–Dirichlet one (`wirtinger_DN`, `wirtinger_DD`, the latter glued from two shifted blocks by reflection), for every  $c > 0$ ; the elementary chain has slack  $1/1886$ , and  $c = 1/8000$  fits inside it (`secondvar_assembly`). The capstone `elementary_constant_verified` carries *no analytic hypothesis*: for continuously differentiable data with the boundary conditions and  $\beta \leq \frac{3}{10}$ , the decoupled forms dominate  $\frac{1}{12}$  of the mass, machine-checked end to end. The ball-arithmetic certificate of Theorem 88 for the sharp  $\frac{73}{100}$  remains outside Lean, and its label remains PROVED, not VERIFIED. The geometric reduction is verified, including the containment. The placed region is defined (`placedRegion`) and shown measurable; its fibres lie in the four-piece union (`fibre_subset_four_piece`); and, in the sign-restricted case that  $\Sigma$  satisfies, the slicing identity is *proved*: Fubini restricts the fibre integral to the band because the region is empty outside it (`placedRegion_volume_eq`), each fibre is bounded by the two tents (`fibre_volume_le`), and the verified profile bound evaluates the result (`placedRegion_volume_le`). The containment is supplied by `Containment.lean`: `AmbiSofa` records a sofa together with the two hallway placements it occupies at angle  $\pi/4$ , which is what “lies in the hallway at every instant” means, so `sofa_subset_pair` is definitional; and `toFibre_mem_placedRegion` carries the intersection into `placedRegion` by affine arithmetic, which is the content of Lemma 72. The hallways are *defined* there by their half-plane descriptions in the rotated frame rather than derived from a definition of rotation, so that identification is recorded rather than re-derived; it is the one modelling step of the chain, and it is not an estimate. Lemma 80 is verified end to end: `lemmaE_verified` carries no hypotheses, and `profile_bound_verified` is the lemma in its original  $g$ -form,  $g(m_1) + g(m_2) + 2 \int F \leq 2(2\sqrt{2} - 1)$  for every  $m_1 \leq m_2$ . The analytic inputs are discharged as follows: (L1) by `L1_triangle` (extend the domain, split at the midpoint, two ramp integrals); (L2) by pointwise domination; (L3) and (L4) by `L3_full` and `L4_full`, which rest on the splitting  $C = A - K$  of `Excess.lean`, the four ramp integrals `ramp_up_in` through `ramp_down_le`, the zone analysis `Kloss_le`, the closed forms `Cfun_right` and `Cfun_far` beyond the band, and the reflection `Cfun_symm`, which delivers the left side for free. No covering, no interval arithmetic and no numerics of any kind appear in the verified chain. Trigonometric quantities are carried as formal symbols with the

sign hypotheses  $\sin t \geq 0$ ,  $\cos t \geq 0$ , which is all Lemma 1 uses.

## References

- [1] J. Baek, *Optimality of Gerver's sofa*, arXiv:2411.19826, 2024.
- [2] J. L. Gerver, *On moving a sofa around a corner*, Geom. Dedicata **42** (1992), 267–283.
- [3] Y. Kallus and D. Romik, *Improved upper bounds in the moving sofa problem*, Adv. Math. **340** (2018), 960–982.
- [4] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, 2nd ed., Cambridge Univ. Press, 2014.
- [5] D. Romik, *Differential equations and exact solutions in the moving sofa problem*, Exp. Math. **27** (2018), 316–330.
- [6] J. M. Hammersley, *On the enfeeblement of mathematical skills by modern mathematics and by similar soft intellectual trash in schools and universities*, Bull. Inst. Math. Appl. **4** (1968), 66–85.
- [7] B. Romera-Paredes et al., *Mathematical exploration and discovery at scale*, arXiv:2511.02864, 2025.